

Entropy-Based Learning

Principles, Methods, and Scientific Applications

Yatao Bian

RELEASED

August 2, 2026

LAST UPDATED

October 6, 2026



Blue Whale Lab

Preface

Welcome to **Entropy-Based Learning**. This short book is an introductory, evolving companion to a research program on *modern entropy- and energy-based learning*, developed at the [Blue Whale Lab](#) led by [Yatao Bian](#).

Entropy is one of the deepest bridges between physics, information theory, and machine learning — a single quantity reinvented, again and again, for heat engines, telephone lines, quantum ensembles, and even theories of human anxiety. Three complementary principles organize the material in this book:

- **The Maximum Entropy Principle** — among all models consistent with the observed constraints, prefer the one that is maximally noncommittal (highest entropy). This principle underlies statistical physics, Boltzmann machines, maximum-entropy reinforcement learning, and a principled class of game-theoretic valuation methods.
- **The Minimum Entropy Principle** — when a model’s own predictions can serve as a training signal, reducing predictive uncertainty can elicit capabilities such as reasoning, *without* external supervision.
- **The Minimax Entropy Principle** — maximize entropy given the features a model includes, then choose the features that make that entropy as small as possible. This couples the two principles above and turns model selection itself into a well-posed problem ([Zhu et al. 1997](#); [Carcamo et al. 2025](#)).

Binding them together is the **free energy** $F = U - TS$, in which temperature T sets the exchange rate between energy and entropy. It is the equation that carried statistical physics into machine learning, and it makes the three principles above three settings of one dial rather than three separate doctrines.

How complete is this draft?

Work in Progress

This book is an ongoing project, released early so you can follow its development. Most chapters are incomplete. The table below shows the current status of each chapter. Please keep this in mind before quoting or citing any part of the text, or before judging drafts that are still being written.

Chapter	Status
1 — Introduction	early draft
2 — Entropy and the Maximum Entropy Principle	mostly complete
3 — Energy, Entropy, and Free Energy	mostly complete
4 — Energy-Based Models	mostly complete
5 — Maximum-Entropy Learning	mostly complete
6 — Minimum-Entropy Learning	mostly complete
7 — Minimax-Entropy Learning	early draft
8 — Energy-Based Guidance and Refinement	first full draft
9 — Maximum-Entropy Post-Training	first full draft
10 — Entropy-Based Learning for Science	early draft
11 — Summary	early draft

“Mostly complete” means the argument is in place and has been checked, though details may still move. “First full draft” means the chapter is written end to end but has not yet been checked with the same care. “Early draft” means little more than a scaffold. The table will be updated as chapters mature. Feedback and contributions are welcome at any stage.

A printable PDF of the current draft is available from the download button in the sidebar, or directly as [Entropy-Based-Learning.pdf](#).

To learn more about the underlying research, visit <https://yataobian.com/> and the Blue Whale Lab.

Citation

If you found this book useful, please cite it as:

```
@book{bian2026entropy,
  title    = {Entropy-Based Learning},
  author   = {Yatao Bian},
  year     = {2026},
  publisher = {Blue Whale Lab},
  url      = {https://yataobian.github.io/entropy-based-learning/}
}
```

Or, for attribution in text: Bian, Yatao. 2026. *Entropy-Based Learning*. Blue Whale Lab. <https://yataobian.github.io/entropy-based-learning/>.

Contents

- 1. Introduction
 - 1.1 Why entropy?
 - 1.2 Roadmap

Part I — Foundations

- 2. Entropy and the Maximum Entropy Principle
 - 2.1 The many faces of entropy
 - * 2.1.1 Information: Shannon entropy
 - * 2.1.2 Heat: thermodynamic entropy
 - * 2.1.3 Quantum states: von Neumann entropy
 - * 2.1.4 One knob, a family: Rényi and Tsallis
 - * 2.1.5 Minds: psychological entropy
 - * 2.1.6 One form, many readings
 - 2.2 Why these formulas and no others: the axioms
 - * 2.2.1 The load-bearing axiom
 - * 2.2.2 Loosening one axiom at a time
 - * 2.2.3 The other faces, axiomatized
 - * 2.2.4 From entropy to diversity
 - * 2.2.5 Similarity
 - * 2.2.6 What axioms cannot settle
 - 2.3 The maximum entropy principle
 - 2.4 Minimum description length and minimax entropy
 - 2.5 Why it matters for learning
- 3. Energy, Entropy, and Free Energy
 - 3.1 Maximum entropy with an energy constraint
 - 3.2 Free energy
 - 3.3 What entropy buys you
 - * 3.3.1 Isothermal expansion
 - * 3.3.2 The rubber band
 - * 3.3.3 Melting
 - 3.4 From spins to networks
 - 3.5 How this reached AI
 - 3.6 The temperature dial

- 3.7 Why it matters for learning
- 4. Energy-Based Models
 - 4.1 The energy view of inference
 - 4.2 Why learn an energy at all?
 - 4.3 Learning is shaping a landscape
 - 4.4 Strategy one: sample the pull-up
 - 4.5 Strategy two: match slopes, not heights
 - 4.6 Strategy three: learn by telling things apart
 - 4.7 One family, three estimators
 - 4.8 Latent variables, and the free energy again
 - 4.9 The modern landscape
 - 4.10 Where this sits in the book
 - 4.11 Takeaway

Part II — Methods

- 5. Maximum-Entropy Learning
 - 5.1 What this chapter means by maximum-entropy learning
 - * 5.1.1 Representation: maximum-entropy models in two forms
 - * 5.1.2 Inference: free-energy minimization over a tractable family
 - * 5.1.3 Learning: fitting the energy through the inference
 - * 5.1.4 The family, and a running example
 - 5.2 Graphical models as an explanatory family
 - 5.3 A worked example: energy-based cooperative games
 - * 5.3.1 Why this is useful
 - 5.4 Discrete maximum-entropy learning: set functions under the optimal subset oracle
 - * 5.4.1 The optimal subset oracle
 - * 5.4.2 Learning through mean-field inference: EquiVSet
 - * 5.4.3 Which statistics may the energy depend on? INSET
 - * 5.4.4 What the strand adds to the maximum-entropy programme
 - 5.5 Maximum-entropy reinforcement learning: energies over trajectories
 - * 5.5.1 A Gibbs distribution over trajectories
 - * 5.5.2 Soft Bellman equations as free energies
 - * 5.5.3 Inverse reinforcement learning: the optimal subset oracle on trajectories
 - 5.6 What the family adds, and where it stops
 - * 5.6.1 Maximum likelihood as a saddle-point problem
 - * 5.6.2 Maximum entropy inside, minimum entropy outside: the bridge to mini-max entropy
 - * 5.6.3 Where the family stops
 - 5.7 Takeaway

- 6. Minimum-Entropy Learning
 - 6.1 Two regimes, and a warning about the name
 - 6.2 From word-level to meaning-level uncertainty
 - 6.3 Entropy minimisation as an objective: a short history
 - 6.4 EMPO: reward agreement, not correctness
 - 6.5 What “reward agreement” actually optimises
 - 6.6 The free-energy view, and why entropy collapses
 - 6.7 The information ceiling
 - 6.8 Schrödinger, read carefully
 - 6.9 Prigogine, and an analogy that does not hold
 - 6.10 So why call it minimum-entropy learning?
 - 6.11 When agreement means truth, and when it does not
 - 6.12 Takeaway
- 7. Minimax-Entropy Learning
 - 7.1 Which features should a model include?
 - 7.2 The minimax entropy principle
 - 7.3 Three equivalent views
 - 7.4 Solving it
 - 7.5 Takeaway
- 8. Energy-Based Guidance and Refinement
 - 8.1 The tilted reference
 - 8.2 Continuous guidance
 - 8.3 Discrete guidance: residual energies over a proposal
 - 8.4 Verifiers as energies, best-of-N, and sequential Monte Carlo
 - 8.5 Refinement and reasoning as energy minimization
 - 8.6 Guidance on the three levels
 - 8.7 Takeaway
- 9. Maximum-Entropy Post-Training
 - 9.1 From the reference to its tilt
 - 9.2 Amortizing the tilt
 - 9.3 Learning the energy
 - 9.4 Self-referential energies
 - 9.5 The continuous case
 - 9.6 Post-training on the three levels, and where it stops
 - 9.7 Takeaway

Part III — Applications

- 10. Entropy-Based Learning for Science
 - 10.1 Two synergistic directions
 - 10.2 Molecular foundation models

- 10.3 Outlook
- 11. Summary
- References

1 Introduction

1.1 Why entropy?

Modern machine learning has long been nurtured by scientific disciplines. A canonical example is the **Boltzmann machine**, rooted in statistical physics and motivated by the *Maximum Entropy Principle*. This book takes entropy — and its close relative, energy — as the organizing thread for a family of learning methods that are both principled and practically effective.

That lineage is unusually direct. In 1982 John Hopfield borrowed the energy function of a magnetic spin system to build an associative memory (Hopfield 1982); within a year, simulated annealing had turned the competition between energy and entropy into an optimization schedule (Kirkpatrick et al. 1983); and by 1985 the Boltzmann machine placed Hopfield’s network at a finite temperature and learned its couplings from data (Ackley et al. 1985). The same trade-off resurfaced as the evidence lower bound (Dayan et al. 1995; Neal and Hinton 1998), as the contrastive objectives of energy-based learning (LeCun et al. 2006), and as the temperature parameter of every language-model decoder. In 2024 the Nobel Prize in Physics was awarded to Hopfield and Hinton for this line of work (The Royal Swedish Academy of Sciences 2024). Chapter 3 develops the single equation, $F = U - TS$, that runs beneath all of it.

We study three directions that are connected:

1. **Maximizing entropy** to obtain the least-biased model consistent with given constraints. This yields exponential-family / Gibbs distributions and, as we will see, a principled framework for valuation problems in machine learning (Bian et al. 2022). When the energy is itself learned rather than given, the same principle becomes a learning method on discrete domains — neural set functions trained from their optimal subsets alone (Ou et al. 2022; Xie, Bian, et al. 2024) — and, on trajectories, the inverse reinforcement learning that infers a reward from demonstrations (Ziebart et al. 2008).
2. **Minimizing entropy** to sharpen a model’s own predictions. When applied to large language models in a latent semantic space, this idea enables fully *unsupervised* elicitation of reasoning capabilities (EMPO) (Zhang et al. 2025).
3. **Minimaxing entropy** to decide which features a model should include at all. The optimal features are those whose maximum-entropy model has the *minimum* entropy, which makes model selection a well-posed problem (Carcamo et al. 2025).

1.2 Roadmap

- **Part I — Foundations** surveys the thermodynamic, information-theoretic, quantum, and psychological notions of entropy, develops the maximum-entropy principle together with the free-energy relationship $F = U - TS$ that connects entropy to energy, and introduces energy-based models and the minimum-description-length view that leads to minimax entropy.
- **Part II — Methods** develops maximum-entropy, minimum-entropy, and minimax-entropy learning, and then applies the maximum-entropy machinery to pretrained models in two ways: energy-based guidance and refinement at inference time, and maximum-entropy post-training, which reads RLHF, GRPO, DPO, and their relatives as one Gibbs tilt of a pretrained reference.
- **Part III — Applications** connects these ideas to scientific foundation models and AI-for-science.

Part I

Part I — Foundations

2 Entropy and the Maximum Entropy Principle

2.1 The many faces of entropy

Entropy was invented three separate times — for heat engines in the 1860s, for quantum ensembles in the 1920s, and for telephone lines in the 1940s — and has been rediscovered many times since. The names, the units, and the motivating problems all differ, but each version measures the same thing: given what we have specified about a system, how much remains unspecified. This section walks through those notions before the rest of the book settles on one of them.

2.1.1 Information: Shannon entropy

For a discrete random variable X with probability mass function $p(x)$, the **Shannon entropy** is

$$H(p) = - \sum_x p(x) \log p(x),$$

which quantifies the average uncertainty (in nats or bits) of X ([Shannon 1948](#)). Entropy is maximized by the uniform distribution, where it equals $\log |\mathcal{X}|$, and minimized (equal to zero) by a deterministic one.

This functional form is not an arbitrary choice. Shannon showed that three requirements — continuity in the probabilities, monotonicity in the number of equally likely outcomes, and consistency when outcomes are grouped — determine H uniquely up to the base of the logarithm; the requirements, their reformulation as the Shannon–Khinchin axioms, and the uniqueness theorem are stated in [Section 2.2](#). It also carries a concrete operational meaning: $H(p)$ is the expected code length of an optimal lossless encoding of samples from p — exactly, in the limit of coding long blocks, and to within one bit if each symbol must be coded on its own. Uncertainty and description length are one quantity seen from two sides, a duality that returns in [Chapter 7](#).

2.1.2 Heat: thermodynamic entropy

The original entropy is older and, on its face, unrelated to information. Studying the irreversibility of heat flow, Clausius defined a state function whose differential along a reversible path is

$$dS = \frac{\delta Q_{\text{rev}}}{T},$$

and named it from the Greek $\tau\rho\omicron\tau\tau$, transformation, deliberately shaping the word to rhyme with *energy* because the two quantities seemed to him so closely related (Clausius 1865). That 1865 paper closes with the two sentences that became the standard statement of the laws of thermodynamics: the energy of the universe is constant, and its entropy tends towards a maximum.

Boltzmann supplied the microscopic reading. A macrostate — a temperature, a pressure — is compatible with an enormous number W of microstates, and entropy simply counts them (Boltzmann 1877):

$$S = k_B \log W.$$

Gibbs then generalized the count to microstates that are not equally likely (Gibbs 1902):

$$S = -k_B \sum_i p_i \log p_i.$$

The Gibbs expression and the Shannon expression are the same formula, separated only by the constant k_B that carries the physical units. The coincidence is not accidental, and Jaynes turned it into the argument developed in Section 2.3: thermodynamic entropy is the Shannon entropy of our ignorance about the microstate, which makes equilibrium statistical mechanics a problem of inference rather than of mechanics (Jaynes 1957).

2.1.3 Quantum states: von Neumann entropy

Quantum mechanics replaces a probability distribution with a density matrix ρ , a positive semidefinite operator of unit trace. Von Neumann extended entropy to this setting in 1927, two decades before Shannon (Neumann 1927):

$$S(\rho) = -\text{Tr}(\rho \log \rho).$$

Written in the eigenbasis of ρ , this reduces to the Shannon entropy of its eigenvalues, so the same functional form reappears with a different argument. It vanishes exactly for pure states, which are completely specified, and is maximal for the maximally mixed state.

The quantum case does break the analogy in one instructive place. Classically, conditional entropy is never negative: learning Y cannot leave us knowing more than everything about X . Quantum mechanically it can be — the joint state may be more sharply specified than either of its parts. A negative value therefore certifies entanglement, though the implication runs one way only: entangled states with non-negative conditional entropy exist.

2.1.4 One knob, a family: Rényi and Tsallis

Shannon’s requirements can be relaxed, and each relaxation produces a one-parameter family. Rényi kept additivity over independent systems but dropped the requirement that entropy be a linear average of $-\log p$ (Rényi 1961):

$$H_\alpha(p) = \frac{1}{1-\alpha} \log \sum_x p(x)^\alpha, \quad \alpha > 0, \alpha \neq 1.$$

The order α is a knob controlling how much weight rare outcomes receive. The limit $\alpha \rightarrow 1$ recovers Shannon entropy; $\alpha = 0$ gives the Hartley entropy $\log |\text{supp } p|$, which counts possibilities and ignores their probabilities; $\alpha = 2$ gives the **collision entropy**

$$H_2(p) = -\log \sum_x p(x)^2,$$

whose argument $\sum_x p(x)^2$ is the probability that two independent draws coincide — a quantity that returns as a training objective in Section 6.5; and $\alpha \rightarrow \infty$ gives the **min-entropy**

$$H_\infty(p) = -\log \max_x p(x),$$

the worst-case measure used in cryptography, where what matters is not average uncertainty but the chance that an adversary guesses right on the first try.

A warning about that last name, since both readings appear later in this book. The min-entropy H_∞ is a *specific functional*, the most pessimistic member of the Rényi family. It should not be confused with the **minimum-entropy principle** of Chapter 6, which is not a functional at all but a class of learning *objectives* that reduce the entropy — usually the Shannon or the collision entropy — of a model’s predictive distribution. The two are unrelated, and the shared prefix is an accident of terminology in two literatures that developed apart.

Tsallis relaxed the other requirement, replacing the chain rule by a q -deformed version (Section 2.2), so that additivity itself is lost (Tsallis 1988):

$$S_q(p) = \frac{1}{q-1} \left(1 - \sum_x p(x)^q \right).$$

Here $q \rightarrow 1$ again recovers the Boltzmann–Gibbs form, but for $q \neq 1$ two independent systems combine as $S_q(A, B) = S_q(A) + S_q(B) + (1-q) S_q(A) S_q(B)$. This *non-extensivity* was introduced to describe systems with long-range interactions and multifractal structure, where the usual assumption that entropy scales with system size fails.

Both families are useful in their own domains, and both make the same point by contrast: Shannon entropy is the unique member satisfying the full set of Shannon–Khinchin axioms, including the chain rule that makes conditional entropy and mutual information well behaved, as Section 2.2 spells out. That is why it is the form used throughout the remainder of this book.

2.1.5 Minds: psychological entropy

The most surprising reappearance is in psychology. Hirsh, Mar, and Peterson propose the **entropy model of uncertainty**, which applies Shannon’s formula directly to the human information system (Hirsh et al. 2012). At any moment an organism faces a set of competing *affordances*: possible interpretations of the sensory input, and possible actions to take. The model treats these as a subjectively weighted probability distribution and defines **psychological entropy** as its Shannon entropy.

The reading is unusually literal. A familiar situation places nearly all the weight on one dominant affordance, giving low entropy and a fast habitual response. An unfamiliar or ambiguous one spreads the weight across many competing frames, giving high entropy and sustained neural competition — which, the authors argue, is what anxiety *is* subjectively, with anterior cingulate activity and noradrenaline release as its physiological correlates.

Two features of the account matter for what follows. First, goals act as constraints: adopting a goal reweights the distribution towards the actions that serve it, and thereby lowers entropy. This is structurally the same move as imposing moment constraints in Section 2.3, since a constraint is whatever narrows the space of live possibilities. Second, the model makes entropy reduction a *drive* rather than a calculation. That idea runs from Schrödinger’s argument that organisms persist by exporting entropy to their environment (Schrödinger 1944) through to the free-energy principle (Friston 2010), and it is the biological precedent for the minimum-entropy learning objectives of Chapter 6.

2.1.6 One form, many readings

Notion	Probabilities over	Definition	Origin
Thermodynamic	no explicit distribution	$dS = \delta Q_{\text{rev}}/T$	Clausius, 1865
Statistical	microstates, equally likely	$k_B \log W$	Boltzmann, 1877
Gibbs	microstates, unequally likely	$-k_B \sum_i p_i \log p_i$	Gibbs, 1902
Information	a random variable or message source	$-\sum_x p(x) \log p(x)$	Shannon, 1948
Quantum	eigenvalues of a density matrix	$-\text{Tr}(\rho \log \rho)$	von Neumann, 1927
Rényi	any distribution, at order α	$\frac{1}{1-\alpha} \log \sum_x p(x)^\alpha$	Rényi, 1961
Tsallis	non-extensive systems, at order q	$\frac{1}{q-1} (1 - \sum_x p(x)^q)$	Tsallis, 1988
Psychological	competing perceptions and actions	$-\sum_x p(x) \log p(x)$	Hirsh et al., 2012

Two observations follow. The first is mathematical: almost every row is the same functional $-\sum p \log p$, differing only in what the probabilities range over and what constant multiplies

the result. Entropy is less a family of analogies than a single quantity applied to different objects.

The second is interpretive, and it is the one that does work in this book. The same number reads as dispersal of energy to a physicist, as expected code length to an engineer, and as felt uncertainty to a psychologist. Those readings are what make the two principles that follow feel necessary rather than arbitrary: maximizing entropy is being honest about what we have not specified, and minimizing it is committing to what we have.

2.2 Why these formulas and no others: the axioms

In two places so far, this chapter has referenced the uniqueness of Shannon entropy without stating the basis for that claim: first, in attributing the form of H to Shannon's trio of requirements (which specify H up to the logarithm's base), and again at the end of Section 2.1.4, which asserts that Shannon entropy alone satisfies the full set of Shannon–Khinchin axioms. In this section we state the axioms, give the uniqueness theorem in its sharpest known form, and then loosen one axiom at a time; H_α and S_q are as defined in Section 2.1.4.

The same method is then applied to a second quantity, which this book needs from Section 5.4 onward: the diversity of a set of objects, with or without a notion of similarity between them. Diversity has its own axiomatic literature, in ecology, economics and recently machine learning, and it is related to entropy by an exponential; we give the correspondence axiom by axiom and record what the axiomatic method cannot decide.

2.2.1 The load-bearing axiom

Let us start with a numerical example. Consider a distribution on five outcomes,

$$p = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6} \right),$$

and think of drawing from it in two stages: first a coin toss $w = (\frac{1}{2}, \frac{1}{2})$ decides between a left block and a right block, then within the left block one draws from $p_1 = (\frac{1}{2}, \frac{1}{2})$ and within the right block from $p_2 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Written as a composite, $p = w \circ (p_1, p_2)$. Computed directly, the Shannon entropy of p is $\frac{1}{2} \log 4 + \frac{1}{2} \log 6 = \frac{3}{2} \log 2 + \frac{1}{2} \log 3$, about 2.29 bits. The two-stage computation gives $H(w) + \frac{1}{2} H(p_1) + \frac{1}{2} H(p_2) = \log 2 + \frac{1}{2} \log 2 + \frac{1}{2} \log 3$, which is the same number. The equality is an instance of the **chain rule**,

$$H(w \circ (p_1, \dots, p_n)) = H(w) + \sum_{i=1}^n w_i H(p_i),$$

valid for every $w \in \Delta_n$ (the simplex of distributions on n outcomes) and every choice of $p_i \in \Delta_{k_i}$. With X the block and Y the outcome within the block, it reads $H(X, Y) = H(X) + H(Y | X)$; for independent X and Y (all p_i equal to one p , a composite written $w \otimes p$) it reduces to additivity, $H(X, Y) = H(X) + H(Y)$, which is a strictly weaker statement.

Shannon's requirements. Shannon's own theorem (Shannon 1948, Theorem 2) assumes i)

continuity in the p_i ; ii) that $A(n) = H(1/n, \dots, 1/n)$ increases in n ; iii) consistency under grouping, which is the two-stage computation above, and concludes that $H = -c \sum_i p_i \log p_i$ for a constant $c > 0$.

Khinchin's axioms. Most later work uses Khinchin's reformulation ([Khinchin 1957](#)) (the 1957 translation of his 1953 Uspekhi paper). Khinchin himself isolates three properties, namely maximality at the uniform distribution, the chain rule $H(AB) = H(A) + H_A(B)$ and expansibility, and proves that any H continuous in its arguments and satisfying them is $-\lambda \sum_i p_i \log p_i$ with $\lambda > 0$; later authors package continuity together with these three as the four **Shannon–Khinchin axioms**, with labels SK1–SK4. For a sequence of functions $H : \Delta_n \rightarrow \mathbb{R}, n \geq 1$, they are:

- SK1 (continuity): H is continuous in p ;
- SK2 (maximality): $H(p) \leq H(u_n)$ for all $p \in \Delta_n$, where u_n is the uniform distribution;
- SK3 (expansibility): $H(p_1, \dots, p_n, 0) = H(p_1, \dots, p_n)$, i.e., adding an outcome of probability zero changes nothing;
- SK4 (strong additivity): $H(X, Y) = H(X) + \sum_x p(x) H(Y \mid X = x)$, i.e., the chain rule.

Any sequence satisfying SK1–SK4 is $-c \sum_i p_i \log p_i$ with $c \geq 0$, where $c = 0$ is the trivial $H \equiv 0$ that Khinchin's $\lambda > 0$ tacitly excludes ([Khinchin 1957](#)). Note that SK4 is strictly stronger than additivity for independent systems, which is the special case in which every conditional distribution $p(y \mid x)$ equals the marginal $p(y)$. SK4 is the load-bearing axiom in the following sense: every generalized entropy discussed below keeps SK1–SK3 and modifies SK4. This is what the phrase “unique member satisfying the full set of Shannon–Khinchin axioms” in Section 2.1.4 means.

Faddeev's theorem. Once SK4 is assumed, only continuity is needed from SK1–SK3, as Leinster's variant of Faddeev's theorem shows ([Faddeev 1956](#); [Leinster 2021](#), Theorem 2.5.1). Let $(I : \Delta_n \rightarrow \mathbb{R})_{n \geq 1}$ be a sequence of functions. Then the following are equivalent: i) the functions I are continuous and satisfy the chain rule $I(w \circ (p_1, \dots, p_n)) = I(w) + \sum_i w_i I(p_i)$ for all $n, k_1, \dots, k_n \geq 1, w \in \Delta_n$ and $p_i \in \Delta_{k_i}$; ii) $I = cH$ for some $c \in \mathbb{R}$.

Proof sketch. Firstly, the chain rule applied to $u_{mn} = u_m \circ (u_n, \dots, u_n)$ gives $I(u_{mn}) = I(u_m) + I(u_n)$; together with $I(u_{n+1}) - I(u_n) \rightarrow 0$, which follows from continuity, this gives $I(u_n) = c \log n$. Secondly, a distribution with non-zero rational entries $p = (k_1/N, \dots, k_n/N)$ satisfies $p \circ (u_{k_1}, \dots, u_{k_n}) = u_N$, so the chain rule gives $c \log N = I(p) + \sum_i p_i c \log k_i$, i.e., $I(p) = cH(p)$. Lastly, continuity extends the identity to all distributions ([Leinster 2021, sec. 2.5](#)).

We add some remarks on the hypotheses ([Leinster 2021](#), Remarks 2.5.2; [Faddeev 1956](#)). Firstly, continuity is the only regularity condition; SK3 is not assumed, it follows from the conclusion $I = cH$ for every c , and SK2 follows as soon as the sign of c is fixed, e.g., by a normalization such as $I(u_2) = \log 2$. That Khinchin's maximality and expansibility axioms can be dispensed with is Faddeev's own observation: his system replaces the former by positivity of I at a single point and the latter, together with strong additivity, by a single recursion. Secondly, Faddeev's original 1956 theorem assumed in addition that I is symmetric, and assumed only the two-block

special case $I(tw_1, (1-t)w_1, w_2, \dots, w_n) = I(w) + w_1 I(t, 1-t)$ (Faddeev split the last block, which is equivalent under symmetry), while requiring continuity only of the binary function $t \mapsto I(t, 1-t)$; under symmetry the two forms of the chain rule are equivalent by induction, so the general form allows symmetry to be removed (if symmetry is kept, continuity can instead be weakened to measurability, which is a 1964 theorem of Lee). Lastly, some regularity condition is necessary: for an additive but nonlinear $f : \mathbb{R} \rightarrow \mathbb{R}$ (which exists under the axiom of choice), $p \mapsto -\sum_{i \in \text{supp}(p)} p_i f(\log p_i)$ satisfies the chain rule without being a multiple of H .

2.2.2 Loosening one axiom at a time

The five-outcome example makes the failure of SK4 concrete. For the collision entropy H_2 of Section 2.1.4, one computes directly $H_2(p) = -\log(2 \cdot \frac{1}{16} + 3 \cdot \frac{1}{36}) = \log \frac{24}{5} \approx 1.569$ nats, whereas the chain-rule prediction is $H_2(w) + \frac{1}{2}H_2(p_1) + \frac{1}{2}H_2(p_2) = \frac{3}{2} \log 2 + \frac{1}{2} \log 3 \approx 1.589$ nats. The two disagree, so H_2 violates SK4, although one can verify that it is still additive for independent X, Y . Now take the Tsallis entropy $S_2(p) = 1 - \sum_i p_i^2 = \frac{19}{24}$. The chain-rule prediction fails again, but the modified rule

$$S_q(w \circ (p_1, \dots, p_n)) = S_q(w) + \sum_{i=1}^n w_i^q S_q(p_i)$$

gives $S_2(w) + \frac{1}{4}S_2(p_1) + \frac{1}{4}S_2(p_2) = \frac{1}{2} + \frac{1}{8} + \frac{1}{6} = \frac{19}{24}$, exactly. We now describe the two modifications one at a time.

Rényi: a different mean. Rényi kept additivity on independent systems (his Postulate 4) and replaced the arithmetic mean of $-\log p_i$ under p by a Kolmogorov–Nagumo quasi-arithmetic mean $g^{-1}(\sum_i p_i g(-\log p_i))$ with g continuous and strictly monotone (Rényi 1961). Two choices of g are compatible with additivity: the linear functions, which return Shannon entropy, and the exponential functions $g(x) \propto 2^{(1-\alpha)x}$, which return H_α (Rényi’s Theorem 2; his printed formula has the exponent $(\alpha - 1)x$, but with his convention $H(\{p\}) = \log_2(1/p)$ the generator for H_α is $2^{(1-\alpha)x}$). Rényi himself called it an open question whether any other g is admissible for entropy, and proved the linear-or-exponential dichotomy only for the information gain $I(Q \| P)$ (his Theorem 3); for the entropy of incomplete distributions the dichotomy was proved by Aczél and Daróczy (Aczél and Daróczy 1963). Therefore the Rényi entropies satisfy SK1–SK3 and additivity on independent systems, and fail SK4 for $\alpha \neq 1$. Jizba and Arimitsu derive the family from SK-type axioms with a quasi-linear conditional entropy in place of SK4 (Jizba and Arimitsu 2004). Lazarev revisits Rényi’s question without assuming additivity (Lazarev 2026) (a preprint at the time of writing): under the admissibility condition that a measure never has larger entropy than a reference measure dominating it, a continuous strictly monotone g is admissible exactly when $t \mapsto g(1/t)$ is strictly convex for increasing g , or strictly concave for decreasing g , and additivity on products then singles out the Rényi family.

Tsallis: a different chain rule. The functional S_q was introduced by Havrda and Charvát and by Daróczy in information theory, and independently by Tsallis in statistical physics (Havrda and Charvát 1967; Daróczy 1970; Tsallis 1988). Its defining property is the q -chain rule displayed above, in which the weights w_i of SK4 are replaced by w_i^q ; for independent systems, where all p_i are equal, it reduces to the pseudo-additivity $S_q(A, B) = S_q(A) + S_q(B) + (1-q) S_q(A) S_q(B)$

stated in Section 2.1.4. The wording used there can accordingly be made more precise: rather than “giving up additivity itself”, Tsallis replaced the chain rule by its q -deformation, and non-additivity on independent systems is the consequence. Uniqueness theorems with generalized SK axioms were given by Santos and by Abe (Santos 1997; Abe 2000). The shortest statement is Leinster’s (Leinster 2021, Theorem 4.1.5): for real $q \neq 1$ and a sequence of functions $(I : \Delta_n \rightarrow \mathbb{R})_{n \geq 1}$, the following are equivalent: i) I is symmetric and satisfies

$$I(w \otimes p) = I(w) + \left(\sum_{i \in \text{supp}(w)} w_i^q \right) I(p)$$

for all $n, k \geq 1$, $w \in \Delta_n$ and $p \in \Delta_k$; ii) $I = cS_q$ for some $c \in \mathbb{R}$. Notice that no regularity condition appears, in contrast to the Shannon case.

Hanel–Thurner: no SK4. One can also keep SK1–SK3 only and ask what freedom remains in a trace-form entropy $S = \sum_i g(p_i)$. Hanel and Thurner showed that the admissible g are classified by two asymptotic scaling exponents (c, d) , with a representative

$$S_{c,d}(p) \propto \sum_i \Gamma(1 + d, 1 - c \log p_i)$$

for each class, up to constants, where $\Gamma(\cdot, \cdot)$ is the incomplete gamma function (Hanel and Thurner 2011). Shannon entropy is the point $(1, 1)$, the Tsallis family is the line $(c, 0)$, and the stretched-exponential entropies are the line $(1, d)$; the associated maximum-entropy distributions are Lambert- W exponentials. Tempesta’s composability axiom requires, for statistically independent subsystems A and B , that $S(A \cup B) = \Phi(S(A), S(B))$ with Φ symmetric, associative and satisfying $\Phi(x, 0) = x$, i.e., a commutative formal group law over the reals; the axiom replaces SK4 and generalizes the group entropies that Tempesta had introduced in 2011 (Tempesta 2016). Korbel’s review surveys these frameworks (Shannon–Khinchin, Tempesta’s group composability, Hanel–Thurner scaling, Shore–Johnson, Lieb–Yngvason) (Korbel 2026) (also a preprint). Once SK4 is dropped, choosing a trace-form entropy amounts, up to its asymptotic class, to choosing a point (c, d) , i.e., a modelling decision, to which Chapter 7 returns.

The following table records which axiom each family keeps (Khinchin 1957; Faddeev 1956; Rényi 1961; Aczél and Daróczy 1963; Daróczy 1970; Abe 2000; Leinster 2021; Hanel and Thurner 2011).

axiom	Shannon H	Rényi H_α	Tsallis S_q	Hanel–Thurner $S_{c,d}$
SK1 continuity	yes	yes	yes	yes
SK2 maximal at uniform	yes	yes	yes	yes
SK3 expansibility	yes	yes	yes	yes
SK4 chain rule	yes	no	no (q -chain rule)	no
additive on independent systems	yes	yes	no (pseudo-additive)	no in general

axiom	Shannon H	Rényi H_α	Tsallis S_q	Hanel–Thurner $S_{c,d}$
arithmetic mean of $-\log p$	yes	no (quasi-arithmetic)	—	—
uniqueness theorem	Khinchin; Faddeev	Rényi, 1961 (given a Kolmogorov–Nagumo mean); Aczél–Daróczy, 1963	Daróczy, 1970; Abe, 2000; Leinster, Theorem 4.1.5	Hanel–Thurner, 2011 (classification into asymptotic (c, d) -classes; $S_{c,d}$ is a representative)

The rows SK1–SK3 hold for Rényi and Tsallis when $\alpha, q > 0$; for negative orders the uniform distribution is not the maximizer (Leinster 2021, Remarks 4.4.4(ii)). Hanel and Thurner assume the trace form $S = \sum_i g(p_i)$ with g continuous and concave, $g(0) = 0$, and $c \in (0, 1]$.

2.2.3 The other faces, axiomatized

For three readings in the table closing the preceding section, the axiomatic method gives an independent derivation.

Entropy as information loss. Return to the five-outcome distribution and merge the two outcomes of probability $\frac{1}{4}$ into one. The result is $q = (\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ with $H(q) = \log 2 + \frac{1}{2} \log 3$, so the merge loses $H(p) - H(q) = \frac{1}{2} \log 2$, exactly the term $w_1 H(p_1)$ that the chain rule attributes to the left block. Baez, Fritz and Leinster take this loss as the object of the axioms (Baez et al. 2011). On the category FinProb , whose objects are finite probability spaces (X, p) and whose morphisms $f : (X, p) \rightarrow (Y, q)$ are measure-preserving maps, an information loss assigns a number $F(f) \in [0, \infty)$ to each morphism and is i) functorial, $F(g \circ f) = F(g) + F(f)$; ii) convex-linear, $F(\lambda f \oplus (1 - \lambda)g) = \lambda F(f) + (1 - \lambda)F(g)$; iii) continuous. Their theorem states that every such F equals $c [H(p) - H(q)]$ for a constant $c \geq 0$; replacing the weights $\lambda, 1 - \lambda$ in convex-linearity by $\lambda^\beta, (1 - \lambda)^\beta, \beta > 0$, gives $c [S_\beta(p) - S_\beta(q)]$ with the Tsallis entropy S_β instead (Baez et al. 2011, Theorem 7). Baez and Fritz characterize relative entropy in the same style on a category FinStat (Baez and Fritz 2014).

Quantum states. Ochs gave a list of postulates for a quantity measuring the intrinsic dispersion of a quantum state, including invariance under unitaries and partial isometries, additivity on product states, subadditivity, continuity, and some technical conditions, and proved that these single out $S(\rho) = -k \text{Tr } \rho \log \rho$ up to the scale k (Ochs 1975). Section 2.2.5 applies this functional to a normalized similarity kernel.

Heat. Earlier in this chapter, the argument connecting the thermodynamic row to the rest was Jaynes’s interpretive one (Jaynes 1957); Lieb and Yngvason give an axiomatic one, which assumes nothing about heat, temperature or microscopic constituents (Lieb and Yngvason 1999). The primitive is the relation of adiabatic accessibility $X \prec Y$ between equilibrium states, meaning that Y can be reached from X by a process whose only net effect on the surroundings is that a weight has been raised or lowered. The axioms are properties of this relation: reflexivity, transitivity, consistency under composition of systems, scaling invariance,

splitting and recombination, and stability, together with a comparison hypothesis (any two states of the same system are comparable under $\prec$). From these they prove that there is a function S on states, unique up to affine transformations, with $X \prec Y$ if and only if $S(X) \leq S(Y)$, additive under composition and extensive under scaling. Weilenmann, Krämer, Faist and Renner apply the framework, including its 2013 extension to non-equilibrium states, to finite-dimensional quantum systems ordered by majorization; the resulting entropy coincides with the von Neumann entropy on the equilibrium states, in this setting those with flat spectrum, and with the min- and max-entropies of one-shot information theory, $H_{\min}(\rho) = -\log \|\rho\|_{\infty}$ and $H_{\max}(\rho) = \log \text{rank } \rho$, away from them (Weilenmann et al. 2016). The result gives an axiomatic identification of the Clausius and von Neumann rows of the table, where the chapter so far had only Jaynes’s interpretive argument; the thermodynamic side returns in Chapter 3.

The inference rule. Shore and Johnson axiomatize the rule that takes a prior and a set of constraints on expected values to a posterior, rather than the entropy functional. They require of such a rule, realized by minimizing a functional $H(q, p)$, uniqueness, invariance under coordinate changes, system independence and subset independence, and prove that every rule satisfying these is equivalent to minimization of relative entropy (their “cross-entropy”, $\int q \log(q/p)$), which reduces to the maximum-entropy principle of Section 2.3 for a uniform prior (Shore and Johnson 1980). Chapter 5 uses this result, which axiomatizes a procedure rather than H . Csiszár’s survey (Csiszár 2008) treats the characterizations of functionals, of entropic vectors and of inference rules in one place.

2.2.4 From entropy to diversity

The following example is due to Jost (Jost 2009; Jost et al. 2010), in the form given by Leinster (Leinster 2021, Example 2.4.11); Jost’s original has 20 islands with Barro Colorado abundance data, and the oil company, the lawyers and the figures below are Leinster’s re-casting. An oil company plans work on 16 equally sized islands which will destroy all wildlife on 8 of them. The islands share no species and each has diversity 4, in the sense of the effective number defined below, so the group has diversity $16 \times 4 = 64$ before the work and 32 after, a loss of 50%. With Shannon entropy as the index, as ecologists have long used it, the company’s lawyers compute $\log 64$ before and $\log 32$ after, hence $\log 32 / \log 64 = 5/6 \approx 83\%$ of the diversity preserved; the environmentalists’ lawyers compute the entropy of the islands to be destroyed, $\log 32$ out of $\log 64$, and claim that 83% is lost. The two conclusions contradict each other, and both are wrong, since half of the diversity is lost; the reason is that H fails the replication principle stated below.

Effective numbers. The remedy, argued by Jost under the slogan that entropies are not diversities (Jost 2006), is to exponentiate. For an abundance distribution p on n species and the Rényi entropy H_q of Section 2.1.4, the **Hill number** of order q is

$$D_q(p) = \left(\sum_{i=1}^n p_i^q \right)^{1/(1-q)} = \exp H_q(p),$$

introduced into ecology by Hill and, for $q = 1$, by MacArthur (Hill 1973; MacArthur 1965). $D_q(p)$ is the **effective number** of species: a community with $D_q = m$ counts as diverse as one of

m equally abundant, completely distinct species. Specifically, D_0 is species richness (the Hartley count of possibilities of Section 2.1.4), $D_1 = e^H$, $D_2 = 1/\sum_i p_i^2$ is the inverse of Simpson's concentration (Simpson 1949), and $D_\infty = 1/\max_i p_i$. Patil and Taillie read a diversity index as an average rarity $\sum_i p_i R(p_i)$, with R a rarity function (Patil and Taillie 1982); the choice $R(p) = (1 - p^{q-1})/(q - 1)$ gives the q -logarithmic entropies S_q , whose numbers equivalents are the Hill numbers. D_q itself is an average rarity, namely the power mean $M_{1-q}(p, 1/p)$ of the rarities $1/p_i$ weighted by p (Leinster 2021, Definition 4.3.4), which is Hill's own reading of D_q as the reciprocal of a mean proportional abundance (Hill 1973). The property on which the oil-company example depends is the **replication principle**: if n communities of equal size, sharing no species and each of diversity D , are pooled, the pooled community has diversity nD . Formally, $D(u_n \otimes p) = n D(p)$, whereas $H(u_n \otimes p) = \log n + H(p)$.

Leinster's characterization. The Hill numbers are characterized by this property plus a few elementary ones (Leinster 2021, Theorem 7.4.3). One condition needs a definition (Leinster 2021, Definition 7.4.1): a sequence $D : \Delta_n \rightarrow (0, \infty)$ is *modular-monotone* if $D(p_i) \leq D(\tilde{p}_i)$ for all i implies $D(w \circ (p_1, \dots, p_n)) \leq D(w \circ (\tilde{p}_1, \dots, \tilde{p}_n))$, i.e., if no subcommunity's diversity decreases then neither does that of the whole. It implies modularity, i.e., dependence on the p_i only through the $D(p_i)$. The theorem states that for a sequence of functions $(D : \Delta_n \rightarrow (0, \infty))_{n \geq 1}$ the following are equivalent: i) D is symmetric, absence-invariant (deleting a species of abundance zero does not change D), continuous in the positive probabilities, normalized ($D(u_1) = 1$), modular-monotone, and satisfies the replication principle; ii) $D = D_q$ for some $q \in [-\infty, \infty]$. A refinement in the same section excludes negative q under an additional condition.

Compared with the SK axioms, the correspondence holds exactly for SK3 and for additivity, and up to a deliberate weakening for SK4. Expansibility (SK3) corresponds to absence-invariance. Additivity on independent systems corresponds to multiplicativity $D(w \otimes p) = D(w) D(p)$, of which the replication principle is the special case $w = u_n$; the theorem assumes only this special case and derives multiplicativity from it (Leinster 2021, Lemma 7.4.7). The chain rule (SK4), whose exact diversity form is $D(w \circ (p_1, \dots, p_n)) = D(w) \prod_i D(p_i)^{w_i}$ (Leinster 2021, Corollary 2.5.8), corresponds to the strictly weaker modular-monotonicity, i.e., the diversity of the whole depends monotonically only on w and the diversities of the parts; this weakening is the reason the theorem produces the whole family D_q instead of D_1 alone. Finally, the effective-number property $D(u_n) = n$, which is the exponential-scale counterpart of the normalization $H(u_n) = \log n$ rather than of the maximality inequality SK2, is not assumed at all, because it follows from normalization and replication: $D(u_n) = D(u_n \otimes u_1) = n D(u_1) = n$ (Leinster 2021, Lemma 7.4.4). Maximality at the uniform distribution itself is not a consequence of the six hypotheses: the theorem admits Hill numbers of negative order, which are not maximized at u_n (Leinster 2021, Remarks 4.4.4(ii)); it holds for the refinement $q \geq 0$. Since $H_q = \log D_q$, the theorem is also a new characterization of the Rényi entropies, without Rényi's quasi-arithmetic mean among the hypotheses (Leinster 2021, Remark 7.4.15).

2.2.5 Similarity

Hill numbers treat every pair of distinct species as equally distinct. A similarity matrix removes this restriction, and the two constructions below use it differently. Let two objects have similarity $s \in [0, 1]$ and equal weight, so that $Z = K = \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix}$. The eigenvalues of $K/2$ are $(1 \pm s)/2$, and the exponential of their Shannon entropy is $\exp H(\frac{1+s}{2}, \frac{1-s}{2})$, about 1.75 for $s = 0.5$. Alternatively, the expected similarity of a random individual to either object is $(Zp)_i = (1 + s)/2$, and $1/(Zp)_i = 2/(1 + s)$, which is 1.33 for $s = 0.5$. The two functionals coincide when K is block-diagonal with all-ones blocks, in which case the non-zero eigenvalues of K/n are the block masses (as noted in Section 6.5), and they differ in general.

Similarity-sensitive diversity. Rao's quadratic entropy $\sum_{i,j} p_i p_j d_{ij}$, the expected dissimilarity between two randomly drawn individuals, is the best-known early index that admits inter-species distances (the same form appears as Nei and Li's 1979 nucleotide diversity) (Rao 1982); Stirling decomposes diversity into variety, balance and disparity and proposes the heuristic $\sum_{i \neq j} d_{ij}^\alpha (p_i p_j)^\beta$, whose $\alpha = \beta = 1$ case is Rao's index (Stirling 2007). The family that contains the Hill numbers is Leinster and Cobbold's **similarity-sensitive diversity** of order q (Leinster and Cobbold 2012): for a similarity matrix Z with $Z_{ii} = 1$ and $0 \leq Z_{ij} \leq 1$, and $p \in \Delta_n$,

$$D_q^Z(p) = \left(\sum_{i: p_i > 0} p_i ((Zp)_i)^{q-1} \right)^{1/(1-q)}, \quad q \neq 1,$$

where $(Zp)_i = \sum_j Z_{ij} p_j$ is the expected similarity between species i and an individual chosen at random, and $q = 1, \infty$ are taken as limits. $Z = I$ recovers D_q . The properties they establish are of three kinds: i) elementary ones (symmetry, absent species, identical species, effective number); ii) the effect of similarity (monotonicity in Z , the naive model $Z = I$ as an upper bound, and the range $1 \leq D_q^Z(p) \leq n$); iii) partitioning (modularity and replication for mutually dissimilar subcommunities).

Maximum diversity. For a symmetric Z , i) there exists a distribution that maximizes D_q^Z for all $q \in [0, \infty]$ simultaneously, and ii) the maximum value $\sup_p D_q^Z(p)$ is independent of q (Leinster and Meckes 2016; Leinster 2021, Theorem 6.3.2). The maximum equals the magnitude $|Z_B|$ of some principal submatrix Z_B of Z (for invertible Z_B , the sum of all entries of Z_B^{-1}) (Leinster 2021, Theorem 6.3.13), and in important classes of cases the magnitude of Z itself. The same quantity had appeared in 1994, when Solow and Polasky proposed $V(S) = \mathbf{1}^\top F^{-1} \mathbf{1}$ with $F_{ij} = f(d_{ij})$ for a positive-definite, decreasing f with $f(0) = 1$ and $f(\infty) = 0$, their leading example being $f(d) = \exp(-\theta d)$, and called it the effective number of species (Solow and Polasky 1994); for the exponential choice it is exactly the magnitude of the scaled metric space, a coincidence that Leinster records as a historical surprise (Leinster 2021). Leinster's earlier note on the result is titled "A maximum entropy theorem with applications to the measurement of biodiversity" (Leinster 2009). The title is meant in a broader sense than the principle of Section 2.3: there are no moment constraints, the maximized quantity is the similarity-sensitive entropy $\log D_q^Z$, and the maximizer does not depend on q .

Diversity of a set: the economics tradition. Economists measure the diversity of a plain set,

without abundances. Weitzman defines a diversity function recursively from pairwise distances, $V(S) = \max_{i \in S} [V(S \setminus \{i\}) + d(i, S \setminus \{i\})]$ with $d(i, Q) = \min_{j \in Q} d_{ij}$, and shows that it has monotonicity in species, the twin property, and continuity and monotonicity in distances (Weitzman 1992). Solow and Polasky set out three natural requirements: monotonicity in species (adding a new distinct species does not decrease diversity), twinning (adding an exact copy of a present species does not increase it; the term is Weitzman’s), and monotonicity in distance; their $V(S)$ satisfies the first two, but they note that it is not monotone in distance in general, and conjecture that it is for exponential f when the distances satisfy the triangle inequality (Solow and Polasky 1994; as read by Emmerich et al. 2026b). Nehring and Puppe model diversity as the total weight of the attributes realized by some member of a set, $v(S) = \sum_{A: A \cap S \neq \emptyset} \lambda_A$ with $\lambda_A \geq 0$, and show that these are exactly the set functions with non-negative conjugate Möbius inverse, equivalently the monotone, totally submodular set functions, which are the duals of the totally monotone belief functions (Nehring and Puppe 2002). In particular such functions are submodular, which places set diversity among the set functions learned in Section 5.4.

Kernels in machine learning. The Vendi Score of Friedman and Dieng is the eigenvalue-based construction of the opening example (Friedman and Dieng 2023). Given a positive semidefinite kernel k with $k(x, x) = 1$ and samples $x_1, \dots, x_n$, let $K_{ij} = k(x_i, x_j)$ and $\rho = K/n$, a positive semidefinite matrix of unit trace, i.e., a density matrix. The **Vendi Score** is

$$\text{VS}_k(x_1, \dots, x_n) = \exp \left(- \sum_i \lambda_i \log \lambda_i \right) = \exp (- \text{Tr } \rho \log \rho),$$

with λ_i the eigenvalues of ρ , i.e., the exponential of the von Neumann entropy of ρ , hence the name. Their Theorem 3.1 lists four properties: 1) effective number: if $k(x_i, x_j) = 0$ for all $i \neq j$ then VS is maximal and equals n , and if $k(x_i, x_j) = 1$ for all i, j it is minimal and equals 1; 2) identical elements: if $k(x_i, x_j) = 1$ for some $i \neq j$, then merging the weights, $p'_i = p_i + p_j$ and $p'_j = 0$ in the weighted form of the score, leaves VS unchanged; 3) partitioning: if $S_1, \dots, S_m$ are mutually dissimilar collections ($k = 0$ across groups) of relative sizes p_i , then $\text{VS}(S_1, \dots, S_m) = \exp (H(p_1, \dots, p_m)) \prod_i \text{VS}(S_i)^{p_i}$; 4) symmetry under permutation. No reference dataset or labels are required. Property 3) is the chain rule in exponential form. Pasarkar and Dieng replace the Shannon entropy of the spectrum by H_q , giving $\text{VS}_q = \exp H_q(\lambda)$ (Pasarkar and Dieng 2024); this is the matrix-based Rényi entropy of Giraldo et al. (Sánchez Giraldo et al. 2015), as Section 6.5 notes. At $q = 2$ no eigendecomposition is needed, since $\text{Tr } \rho^2 = n^{-2} \sum_{i,j} K_{ij}^2$; the entropy $-\log (n^{-2} \sum_{i,j} K_{ij}^2)$ is the RKE of Jalali, Li and Farnia, computable in $O(n^2)$ (Jalali et al. 2023). FKEA accelerates the general case with random features (Ospanov et al. 2024), and quality-weighted variants exist (Nguyen and Dieng 2024). A ninth row could be added to the table of readings in the preceding section: diversity, over the eigenvalues of a normalized similarity kernel, $\exp(-\text{Tr } \rho \log \rho)$, Friedman and Dieng, 2023.

DCScore. A cheaper summarization of the same kernel reads diversity as a classification problem (Zhu et al. 2025). With embeddings h_i , kernel matrix K and temperature $\tau > 0$, form $P = \text{softmax}(K/\tau)$ row-wise, $P_{ij} = \exp(K_{ij}/\tau) / \sum_{j'} \exp(K_{ij'}/\tau)$, and set $\text{DCScore} = \text{tr}(P) = \sum_i P_{ii}$, the total probability that each sample is classified back into its own class. Zhu et al. prove four properties, named after those of Leinster and Cobbold; stated precisely, they are

the following. i) Effective number: for a kernel whose rows are maximal on the diagonal (cosine similarity of normalized embeddings, or an RBF kernel), $P_{ii} \geq 1/n$, so $1 \leq \text{DCScore} < n$; the value is 1 exactly when all samples coincide, and the value n for distinct samples is reached only in the limit, which the proof states as “tending to n ”, i.e., as $\tau \rightarrow 0$. ii) Identical samples: merging a dataset with an identical copy of itself leaves the score unchanged. This is a dataset-level property, not Leinster and Cobbold’s merging of two identical species, and it holds exactly for every kernel and every τ , since duplication halves each P_{ii} and doubles their number. iii) Symmetry under permutation. iv) Monotonicity, a comparative statement: two datasets of n mutually distinct samples, both with DCScore n , receive the same new sample, which is more similar to the samples of the second; then the first augmented dataset scores higher. Since $\text{DCScore} = n$ holds only as $\tau \rightarrow 0$, the version at a fixed τ is the elementwise one: for symmetric K and $i \neq j$,

$$\frac{\partial \text{DCScore}}{\partial K_{ij}} = -\frac{1}{\tau}(P_{ii}P_{ij} + P_{jj}P_{ji}) < 0,$$

where $K_{ij} = K_{ji}$ varies jointly, so the score strictly decreases whenever an off-diagonal similarity increases, and iv) holds whenever the two augmented kernel matrices differ only in the row and column of the new sample, with the entries of the second at least as large and one of them larger. Given K , the summarization costs $O(n^2)$ for DCScore against $O(n^3)$ for the eigendecomposition in the Vendi Score; with an inner-product kernel and $n \gg d$, the Vendi Score can use the $d \times d$ matrix $X^\top X$ instead, at total cost $O(d^2n)$, against $O(n^2d)$ for DCScore (Zhu et al. 2025, Table 2). The following table collects the properties of these measures and of the constructions of the next subsection, where the rows A1–A3 are defined.

property	Hill D_q	Leinster–Cobbold D_q^Z (equal weights, $q \in [0, \infty)$, $q \neq 1$)	Vendi / q -Vendi	DCScore	Mironov–Prokhorenkova constructions
symmetry	yes	yes	yes	yes	yes
absence-invariance / identical elements	yes	yes	yes (weighted form)	dataset duplication only	—
effective number	yes	yes	yes	1 for identical samples; n only as $\tau \rightarrow 0$	no (not normalized; inferred from the definitions)
modularity / partitioning	yes	yes	yes	no	—
replication	yes	yes	yes	no	—

property	Hill D_q	Leinster– Cobbald D_q^Z (equal weights, $q \in [0, \infty)$, $q \neq 1$)	Vendi / q -Vendi	DCScore	Mironov– Prokhorenkova constructions
strict monotonicity in distances (A1)	n/a	yes	no for $q = 1$; yes for RKE ($q = 2$, non-negative similarities)	yes, if $K_{ij} = f(d_{ij})$ with f strictly decreasing	yes
uniqueness (A2)	n/a	no	no	no	yes
continuity (A3)	yes	yes	yes	yes	yes
cost	$O(n)$	$O(n^2)$	$O(n^3)$ given $K, O(d^2n)$ with a linear kernel on d -dimensional embeddings; $O(n^2)$ for RKE	$O(n^2)$ given $K, O(n^2d)$ with a linear kernel	NP-hard

Several cells carry qualifications. The A2 failure of D_q^Z is established in Mironov and Prokhorenkova (2025) numerically, on one three-point example for $q \in [0, 100]$, and not by a proof for all q ; Leinster and Cobbald’s own monotonicity in Z (Leinster 2021, Lemma 6.2.2) is non-strict, and the strict A1 is Mironov and Prokhorenkova’s statement for the equal-weight case. “Not normalized” is not stated in Mironov and Prokhorenkova (2025) but follows from the definitions: MultiDimVolume equals $n - 1$ for n points at unit distance and 0 for coincident points, so it is not an effective number in $[1, n]$. The DCScore entries for A1–A3, partitioning and replication are not stated in Zhu et al. (2025) and follow from the definition. A1 holds whenever the similarity is a strictly decreasing function of the distance, e.g., cosine similarity of normalized embeddings, $1 - \|h_i - h_j\|^2/2$, by the derivative above; the argument is unaffected by duplicates and, unlike that for RKE, needs no sign condition on the similarities. A2 fails: on the three points of Mironov and Prokhorenkova (2025) used there against RKE (on a circle with cosine similarity, arcs 1.1 and 0.4 radians from x_1 to x_2 and from x_2 to x_3), making x_2 a copy of x_3 raises DCScore from 1.337 to 1.394 at $\tau = 1$. A smaller τ does not restore A2: by Proposition 6.1 of Mironov and Prokhorenkova (2025), a continuous measure satisfying A2 cannot depend on which element is duplicated, whereas for points at angles 0, 0.3, 1.0, 2.2 and $\tau = 0.1$, adding a copy of x_2 or of x_3 gives 2.998 or 3.095. Partitioning and replication fail at every $\tau > 0$, since a similarity of 0 still contributes $e^0 = 1$ to the softmax denominator: two samples with $K_{11} = K_{22} = 1$ and $K_{12} = 0$ score $2e^{1/\tau}/(e^{1/\tau} + 1) < 2$, e.g., 1.462 at $\tau = 1$. The two limits are degenerate: as $\tau \rightarrow \infty$ every dataset scores 1, and as $\tau \rightarrow 0$ DCScore tends to the number of distinct samples, provided $K_{ij} < K_{ii}$ whenever $x_j \neq x_i$, i.e., to the measure Unique of Mironov and Prokhorenkova (2025), which satisfies A2 only.

2.2.6 What axioms cannot settle

Consider three points on a circle with cosine similarity, with arc 0.6 radians from x_1 to x_2 and 1.4 radians from x_2 to x_3 . Move x_3 a further 0.1 radians away from the other two. No pairwise distance decreases, and the Vendi Score falls from 1.941 to 1.916. Now let the arcs be 0.2 and 0.3 radians and replace x_2 by a copy of x_1 . The set now contains a duplicate, and the Vendi Score rises from 1.187 to 1.233. Both computations are from (Mironov and Prokhorenkova 2025, sec. 3 and Appendix B).

Three axioms for the diversity of a set. Mironov and Prokhorenkova formalize the intuitions violated here (Mironov and Prokhorenkova 2025). The setting is n objects, possibly repeated, with pairwise distances d_{ij} satisfying $d_{ij} \geq 0$, $d_{ii} = 0$, symmetry, and the consistency condition that $d_{ij} = 0$ implies $d_{ik} = d_{jk}$ for all k ; the triangle inequality is not required. A diversity function is a permutation-invariant map from this space $\mathcal{D}_n$ to $\mathbb{R}$, with n fixed. The axioms are: A1 (monotonicity), the function is strictly increasing in every pairwise distance, and this must hold also when the set contains duplicates; A2 (uniqueness), replacing an element that coincides with some other element by any element that coincides with none strictly increases diversity; A3 (continuity) in the distances. Their Table 2 checks the existing measures. Average pairwise distance and its sum, RKE, and Species(q), the latter being Leinster–Cobbold diversity with equal weights, satisfy A1 and A3 and fail A2, at cost $O(n^2)$ (for RKE, which depends on the similarities only through s_{ij}^2 , A1 presupposes $s_{ij} \geq 0$); Diameter, Bottleneck and the Energy measures fail A1 and A2; the count of unique elements satisfies A2 only; the Vendi Score and the DPP determinant fail A1 and A2 and satisfy A3, at cost $O(n^3)$; #Circles(t) fails all three and HamDiv fails A1 and A2, and both are NP-hard. Two new constructions, MultiDimVolume and IntegralMaxClique, satisfy all three axioms, and both are NP-hard to compute. The paper leaves open whether a polynomial-time measure satisfying A1–A3 exists. Velikonitsev et al. state slightly weaker variants of A1 and A2 (they assume all objects pairwise distinct), and the two-object “dissimilarity” axiom of Xie et al., i.e., non-strict monotonicity of the diversity of a pair in its distance, is implied by A1 (Velikonitsev et al. 2024; Xie et al. 2023, Axiom 4.3; cf. Mironov and Prokhorenkova 2025, sec. 4.4). Emmerich et al. show, in two preprints, that selecting a fixed-cardinality subset of maximum Solow–Polasky diversity is NP-hard in general metric spaces, and remains so for points in the Euclidean plane (Emmerich et al. 2026b, 2026a).

It is important to read the Vendi row correctly. That the Vendi Score fails A2 does not contradict its own identical-elements property. The latter is an invariance of the weighted representation (two identical samples may be merged into one of double weight without changing the score), whereas A2 asks for a strict increase when a duplicate is replaced by a new element; the two axiom sets are about different properties. Propositions 6.1 and 6.2 of Mironov and Prokhorenkova (2025) explain the difficulty. Proposition 6.1 states that under A2 and A3 the diversity of a set does not depend on which of its elements are duplicated. Proposition 6.2 states that no measure satisfying A1–A3 decomposes as $F(d_{12}, \dots, d_{1n}) + G(\text{distances among } x_2, \dots, x_n)$, so no additive split into one element’s contribution plus the diversity of the rest is available.

Which similarity. The axiomatic method fixes a functional only after the similarity, or the embedding that induces it, has been chosen. Friedman and Dieng note the dependence themselves:

a too sensitive similarity makes every set look diverse, and one that is not sensitive enough gives every set a low score (Friedman and Dieng 2023). Zhao et al., reviewing 135 image and text datasets presented as diverse, borrow the vocabulary of measurement theory (D. Zhao et al. 2024): conceptualization, operationalization, and evaluation of the indicators for reliability (inter-annotator agreement, test–retest) and validity (convergent and discriminant); scale, they note, is not diversity. In these terms the choice of k or Z is a construct-validity question rather than a theorem, and, since diversity metrics depend on the embedding space, they recommend benchmarking a dataset across several embedding spaces suited to the chosen definition of diversity.

⚠ Terminology

Several words in this section have different meanings in different literatures. i) “Uniqueness” is a theorem in the axiomatics of entropy (the functional is unique given the axioms) and an axiom in Mironov and Prokhorenkova (a duplicate is worth less than a new element). ii) “Monotonicity” has at least four senses: Shannon’s increasing $A(n)$; Solow and Polasky’s monotonicity in species and in distance, both non-strict; Mironov and Prokhorenkova’s strict monotonicity in every distance, duplicates included; and the comparative property proved for DCScore. iii) “Effective number”, $D(u_n) = n$ for all n , is stronger than “normalized”, $D(u_1) = 1$; Leinster’s theorem assumes the latter and derives the former from replication. Compare the warning on min-entropy in Section 2.1.4.

In summary, this section supplies the following for learning. The chain rule is the reason the remainder of this book works with H by default; when a later chapter writes a Rényi or Tsallis quantity instead, one named axiom has been traded for a computational or modelling advantage, and Chapter 7 treats that trade as part of model selection. The kernel relaxation of collision entropy in Section 6.5 is the $q = 2$ member of the Vendi family, i.e., RKE up to a monotone transformation, and the table above records that it keeps A1 for non-negative similarities and gives up A2. Finally, the drop of output diversity after RLHF (Kirk et al. 2024) and the diversity collapse under RLVR (Yuan et al. 2026) (a preprint) are statements about the effective number of a model’s outputs under some similarity, and Chapter 9 discusses them in these terms.

2.3 The maximum entropy principle

Given constraints — typically moment constraints of the form $\mathbb{E}_p[f_k(X)] = \mu_k$ — the **maximum entropy** distribution solves

$$\max_p H(p) \quad \text{s.t.} \quad \sum_x p(x) = 1, \quad \mathbb{E}_p[f_k(X)] = \mu_k.$$

Its solution is the **Gibbs / Boltzmann** distribution

$$p(x) = \frac{1}{Z(\lambda)} \exp\left(\sum_k \lambda_k f_k(x)\right),$$

where λ are Lagrange multipliers and Z is the partition function. This is the least-biased distribution consistent with what we know (Jaynes 1957).

The single most important instance of this framework is the one in which the constrained feature is the energy of a physical system. That case, worked out in Chapter 3, is what turns the multipliers above into a temperature and connects the whole construction to thermodynamics.

2.4 Minimum description length and minimax entropy

The maximum-entropy principle can be *derived* rather than postulated. Under the **minimum description length** (MDL) view (Grünwald 2007), a model P encodes each state with a code word of length $-\log P(x)$, and its description length is the expected code length across all datasets consistent with the measured features. Among models matching those features, description length is minimized by the maximum-entropy model (Feder 1986).

MDL also poses a question that maximum entropy alone cannot answer: which features should be measured in the first place? For maximum-entropy models the description length coincides with the model's own entropy,

$$L(P_F) = S(P_F),$$

so the best feature set F is the one whose maximum-entropy model has the *smallest* entropy. This is the **minimax entropy** principle, developed in Chapter 7.

2.5 Why it matters for learning

The maximum-entropy principle explains why exponential families are ubiquitous, grounds the Boltzmann machine in statistical physics, and — as developed in Bian et al. (2022) — provides a principled basis for defining valuations in machine learning. The next chapter supplies the missing half of the picture: what happens when entropy is traded off against energy, and what the exchange rate between them turns out to be.

3 Energy, Entropy, and Free Energy

The previous chapter maximized entropy subject to abstract moment constraints and obtained a Gibbs distribution written in terms of Lagrange multipliers. The next chapter writes probabilities as $p(x) \propto e^{-E(x)}$ and calls the exponent an energy. Something has to connect the two, and that something is one of the most useful equations in science:

$$F = U - TS.$$

At a fixed temperature, nature neither minimizes energy nor maximizes entropy. It minimizes a combination of the two, and the temperature is the exchange rate between them. This chapter derives that statement, illustrates it with three classical examples, and follows it into artificial intelligence.

3.1 Maximum entropy with an energy constraint

The maximum-entropy problem of Section 2.3 was stated for arbitrary features f_k . Historically the original problem had exactly one constraint: the average energy of the system is known to be U . Written out,

$$\max_p H(p) \quad \text{s.t.} \quad \sum_x p(x) = 1, \quad \sum_x p(x) E(x) = U,$$

and solved with Lagrange multipliers, it yields the **Boltzmann distribution**

$$p(x) = \frac{1}{Z} e^{-\beta E(x)}, \quad Z(\beta) = \sum_x e^{-\beta E(x)},$$

where the multiplier enforcing the energy constraint is the inverse temperature, $\beta = 1/k_B T$.

That identification is worth pausing on, because it is what gives temperature a meaning outside physics. A Lagrange multiplier is a *price*: it measures how much of the objective must be surrendered to buy one unit of the constrained quantity. Here it measures how much entropy must be given up to raise the expected energy by one unit. Temperature is that price. A system at high temperature buys energy cheaply and stays spread out; a system at low temperature pays dearly for it and concentrates on low-energy states.

This also settles a notational discrepancy the attentive reader will have noticed. In Section 2.3 the solution appeared as $p(x) \propto \exp(\sum_k \lambda_k f_k(x))$, with a plus sign; here it appears as $\exp(-\beta E(x))$, with a minus. The two are the same object under the correspondence $E = -\sum_k \lambda_k f_k$. Energy is negative log-probability up to a constant, and the sign convention comes

from physics, where stable configurations should sit at *low* energy. Chapter 4 follows the physics convention and absorbs β into E by setting $k_B T = 1$.

3.2 Free energy

Given the Boltzmann distribution, the two quantities of interest follow at once: the average energy $U = \langle E \rangle$ and the entropy $S = -k_B \sum_x p(x) \log p(x)$. The combination that governs equilibrium is the **Helmholtz free energy** $F = U - TS$, and it has a compact closed form (Callen 1985):

$$F = -k_B T \log Z.$$

Free energy is the log partition function. What Chapter 4 will present as the central computational obstacle of energy-based modelling is, read from the other side, the central thermodynamic potential of the system.

The relationship has a third face, and it is the one that matters most for learning. Let q be any distribution over the same states, and define its **variational free energy**

$$F(q) = \langle E \rangle_q - T S(q).$$

A short calculation relates this to the equilibrium value:

$$F(q) = F + k_B T \cdot \text{KL}(q \parallel p),$$

where p is the Boltzmann distribution. Because the Kullback–Leibler divergence is non-negative, $F(q) \geq F$ with equality exactly when $q = p$. Equilibrium is therefore not merely *described* by the Boltzmann distribution, it is *characterized* by it, as the unique minimizer of a variational objective. Nearly everything in later chapters that goes by the name of a variational bound is this one inequality in disguise.

3.3 What entropy buys you

The trade-off is easiest to believe after seeing cases in which entropy alone decides the outcome.

3.3.1 Isothermal expansion: a process with no energy change

An ideal gas of N particles expands from volume V_1 to V_2 while held at temperature T . The internal energy of an ideal gas depends only on temperature, so $\Delta U = 0$: not one joule of stored energy is released. The gas expands regardless, because

$$\Delta S = Nk_B \log \frac{V_2}{V_1} > 0 \implies \Delta F = -T\Delta S < 0.$$

The work extracted, $W = Nk_B T \log(V_2/V_1) = T\Delta S$, is drawn entirely from the heat bath. An account of this process in terms of energy alone cannot even say which direction it runs.

3.3.2 The rubber band: a force made of entropy

Stretch a rubber band and it pulls back. The restoring force is not stored in stretched chemical bonds. A polymer chain is a random walk of N segments of length a , and there are vastly more coiled configurations than extended ones, so extending the chain to end-to-end length L costs entropy. The force is the gradient of that cost:

$$f = -T \frac{\partial S}{\partial L} \approx \frac{3k_B T}{Na^2} L.$$

The energy landscape here is essentially flat, and yet there is a real force, one measurable with a spring scale. The formula also makes two predictions that run against intuition, and both hold. The force is *proportional to temperature*, so heating a stretched rubber band makes it contract harder — the opposite of a metal rod, which expands. And stretching a band quickly warms it, an effect readily confirmed by holding one against your lip.

3.3.3 Melting: where the competition changes hands

Ice has lower energy than liquid water, since melting must pay $\Delta H > 0$ to break hydrogen bonds; water has the higher entropy. At the melting point the two contributions balance exactly, $\Delta H - T_m \Delta S = 0$, so

$$T_m = \frac{\Delta H}{\Delta S}.$$

Below T_m the energy term dominates and ice is the stable phase; above it the entropy term dominates and water is. A phase transition is precisely the temperature at which the exchange rate tips the competition from one side to the other.

3.4 From spins to networks

The simplest system with an energy is a single two-state variable $s = \pm 1$ with energy $\mp \varepsilon$. Its partition function and average value are

$$Z = 2 \cosh(\beta \varepsilon), \quad \langle s \rangle = \tanh(\beta \varepsilon),$$

and the temperature limits are exactly what the trade-off predicts: as $T \rightarrow 0$ the variable freezes into its low-energy state, and as $T \rightarrow \infty$ it becomes a fair coin with entropy $k_B \log 2$.

Now couple many such variables together. The **Ising model** assigns to a configuration $s = (s_1, \dots, s_N)$ the energy

$$E(s) = - \sum_{i < j} J_{ij} s_i s_j - \sum_i h_i s_i,$$

with J_{ij} the interaction between units i and j and h_i a local bias. When the couplings are disordered, the landscape acquires a large number of local minima and the system is called a *spin glass*.

That equation is the hinge of this chapter. It is a model of magnetism, and it is also, term for term, the energy function of a Hopfield network and of a Boltzmann machine. What follows is what happened when physicists noticed.

3.5 How this reached AI

Year	Contribution	Role of the entropy–energy trade-off
1982	Hopfield network (Hopfield 1982)	The Ising energy becomes an associative memory: stored patterns are local minima and recall is descent. The $T = 0$ limit — energy only
1983	Simulated annealing (Kirkpatrick et al. 1983)	Turns the trade-off into a <i>schedule</i> : start hot so entropy drives exploration, then cool so energy drives commitment
1985	Boltzmann machine (Ackley et al. 1985)	Places the Hopfield network at finite temperature, $p(s) \propto e^{-E(s)/T}$; learning compares correlations in a clamped and a free phase
1995–98	Helmholtz machine (Dayan et al. 1995); free-energy view of EM (Neal and Hinton 1998)	The evidence lower bound is identified as a negative variational free energy
2002	Contrastive divergence (Hinton 2002)	Makes the intractable $\log Z$ gradient practical and revives deep generative models
2006	Energy-based learning (LeCun et al. 2006)	Names the log-partition term “the Helmholtz free energy of the ensemble” and unifies probabilistic and non-probabilistic learning

Year	Contribution	Role of the entropy–energy trade-off
2024	Nobel Prize in Physics (The Royal Swedish Academy of Sciences 2024)	Awarded to Hopfield and Hinton; the citation rests on the correspondence between neural networks and spin models in statistical physics

Two threads in that table deserve to be stated plainly, because the rest of the book leans on them.

The first is that **the evidence lower bound is a free energy**. Take a latent variable z , set $E(z) = -\log p(x, z)$ and $T = 1$, and the variational inequality of the previous section reads

$$-\log p(x) \leq \langle -\log p(x, z) \rangle_q - H(q) = F(q),$$

with the gap equal to $\text{KL}(q \| p(\cdot | x))$. Variational autoencoders, variational inference, and the training objectives of diffusion models are all instances of minimizing $U - TS$ at unit temperature. The physics is not an analogy here; it is the same algebra.

The second is that **annealing is the trade-off used as a control signal**. Simulated annealing, the noise schedules of Langevin and diffusion samplers, and the practice of decoding a language model at a temperature that falls during generation are one idea: begin where entropy dominates and the search is broad, end where energy dominates and the answer is definite.

3.6 The temperature dial

The modern instance is worth making explicit, because it is the version most readers use daily. A language model produces logits $\ell(x)$ and samples

$$p(x) \propto \exp(\ell(x)/T),$$

which is the Boltzmann distribution with energy $E = -\ell$. Greedy decoding is $T \rightarrow 0$; sampling uniformly from the vocabulary is $T \rightarrow \infty$. The temperature field in every inference API is the same T that appears in $F = U - TS$.

Read this way, the three principles that organize this book are three settings of a single dial.

Principle	Where the dial sits	What it buys
Maximum entropy (Chapter 5)	High: entropy dominates	Honesty about what the constraints leave undetermined
Minimum entropy (Chapter 6)	Low: energy dominates	Commitment — sharpened predictions and elicited reasoning

Principle	Where the dial sits	What it buys
Minimax entropy (Chapter 7)	Both, nested	Maximize on the inside to model the world, minimize on the outside to choose what to model

The principles are not competing claims about the right amount of uncertainty. They are claims about *when* to be uncertain, and the free energy is what makes that question well posed. Read at the level of algorithms rather than slogans, the three also share one *inner* problem — the minimization of a free energy over distributions, which Chapter 5 calls the representation and inference levels — and differ only in the *outer* criterion: fit the data (maximum entropy), sharpen the model’s own output (minimum entropy), or choose the features whose maximum-entropy model has the least entropy (minimax entropy).

3.7 Why it matters for learning

The equation $F = U - TS$ supplies three things the rest of this book uses. It explains where the Gibbs form of Chapter 4 comes from and why its exponent carries a minus sign. It identifies the partition function — the central computational difficulty of energy-based modelling — as a free energy, which is why variational bounds are the natural way around it. And it gives *temperature* a precise meaning, so that annealing schedules, decoding parameters, and the coalition temperature τ of Chapter 5 can all be recognized as one knob rather than three coincidences.

The next chapter takes the Gibbs distribution as its starting point and asks what happens when the energy function is learned from data.

4 Energy-Based Models

The previous chapter left us with an open question: what changes when the Gibbs distribution’s energy function is no longer prescribed by physics, but instead *learned from data*? This chapter answers that question.

Two things change the moment the energy becomes learnable. First, the energy stops being handed to us by physics — the Ising couplings, the hydrogen bonds of Chapter 3 — and becomes a neural network $E_\theta(x)$ whose parameters we must fit. Second, the partition function, which for a physicist is a bookkeeping device, becomes for us the central computational villain of the story: nearly every algorithm in this chapter is best understood as a different strategy for *not computing it*.

An **energy-based model (EBM)** associates to each configuration x a scalar energy $E_\theta(x)$ and defines the Gibbs distribution

$$p_\theta(x) = \frac{e^{-E_\theta(x)}}{Z(\theta)}, \quad Z(\theta) = \int e^{-E_\theta(x)} dx, \quad (4.1)$$

with a sum in place of the integral when x is discrete. This is the Boltzmann distribution of Chapter 3 at unit temperature, with a learnable energy in place of a physical one. Low energy means high probability; the sign convention and the absorbed temperature were settled at the end of that chapter.

The material here draws chiefly on two tutorials, written fifteen years apart and complementary in spirit. The first ([LeCun et al. 2006](#)) treats the energy as a *decision surface*: inference is energy minimization, learning is shaping the landscape, and probabilities are an optional purchase. The second ([Song and Kingma 2021](#)) treats the energy as a *density*: the model is Equation 4.1 taken literally, and the question is how to estimate it when $Z(\theta)$ cannot be computed. Reading them together — decision surface first, density second — is the plan of this chapter. A regularly updated companion list of papers, tutorials, and software is maintained at *Awesome-EBM* ([Bian 2020](#)).

4.1 The energy view of inference

Start with the decision surface. In the energy-based view, a model is a function $E_\theta(x, y)$ that measures the *compatibility* between an observed input x and a candidate answer y : small energy means “these two belong together,” large energy means “they do not.” Inference is then not a forward pass but an *optimization*:

$$y^* = \arg \min_{y \in \mathcal{Y}} E_\theta(x, y). \quad (4.2)$$

When $\mathcal{Y}$ is a handful of class labels, the arg min is a table lookup and Equation 4.2 is just classification wearing unfamiliar clothes. The formulation earns its keep when $\mathcal{Y}$ is large or structured: all consistent segmentations of an image, all sentences of English, all conformations of a protein. In those cases no model can afford to enumerate answers; what it can afford is to *score* any proposed answer, and to search — by dynamic programming, by gradient descent on a continuous y , by annealing — for a good one. The energy function is precisely the object that makes such a search well posed.

This view immediately clarifies what a model owes us, and it is less than one might think. LeCun et al. distinguish the questions a model may be asked (LeCun et al. 2006):

1. **Decision.** “Which y is best for this x ?” Only the *ordering* of energies matters. Any strictly increasing transform of E_θ gives the same answers.
2. **Ranking.** “Is y_1 better than y_2 ?” Again only comparisons matter.
3. **Detection.** “Is this y good enough?” Now energies are compared to a threshold, so their *scale* begins to matter.
4. **Probability.** “How confident are we in each y ?” Only here must energies be converted into calibrated numbers that sum to one.

Only the fourth question requires the Gibbs conversion

$$p_\theta(y \mid x) = \frac{e^{-\beta E_\theta(x, y)}}{\int_{y'} e^{-\beta E_\theta(x, y')} dy'}, \quad (4.3)$$

(a sum over y' when $\mathcal{Y}$ is discrete), restoring the inverse temperature β of Chapter 3 as an explicit dial. The rest of the chapter works at $\beta = 1$ unless a limit in β is at issue, matching the unit-temperature convention of Equation 4.1. The denominator is the partition function — the free-energy bill of Chapter 3. This is the first strategic lesson of energy-based learning: *probabilities are expensive, and you should buy them only when the application actually needs them* (LeCun et al. 2006). A robot that must steer left or right needs the arg min, not a posterior. Much of this chapter is about what becomes possible when we decline to pay, and what it costs to pay when we must.

4.2 Why learn an energy at all?

If unnormalized models create such trouble, why prefer them? Three reasons, in increasing order of depth.

Freedom of form. A normalized density must integrate to one, and maintaining that constraint dictates the architecture: autoregressive models must factor into conditionals, normalizing flows must be invertible with tractable Jacobians, variational autoencoders must route everything through a directed latent-variable structure. An energy function must merely be a scalar-valued

function, bounded below well enough for Equation 4.1 to make sense. Any regression architecture qualifies — a convolutional network for images, a graph neural network for molecules, a transformer for text. In the words of Song and Kingma, density estimation is thereby “reduced to a nonlinear regression problem” (Song and Kingma 2021). The modeling question stops being “what distributions can my architecture normalize?” and becomes “what shape should the landscape have?”

Compatibility is often what we actually know. Domain knowledge usually arrives as statements about what configurations are good — physical constraints, grammatical constraints, consistency conditions — not as recipes for sampling. An energy function absorbs such knowledge directly, as terms that raise the energy of violating configurations.

Energies add. If E_1 scores grammaticality and E_2 scores factual accuracy, then $E_1 + E_2$ scores both at once, and at the level of distributions the sum of energies is the *product* of the corresponding Gibbs factors — a product of experts (Hinton 2002), in which each expert can veto a candidate by assigning it high energy. Normalized models do not compose this way: the product of two normalized densities is not normalized, and renormalizing it is exactly the partition-function problem again. This additivity is what makes energies the natural currency for *inference-time* composition — steering a generator with extra constraints — a theme that returns in force in Chapter 8.

The price of this freedom is stated by Equation 4.1 itself: $Z(\theta)$ is an integral over every configuration the model can represent, and for any interesting architecture it is hopelessly intractable. Neither exact likelihoods nor exact samples are available. Everything that follows is about living well anyway.

4.3 Learning is shaping a landscape

What should a learned energy landscape look like? Configurations resembling the training data should sit in valleys; everything else should sit high. The subtlety — and it is *the* central subtlety of EBM training — is that a learning rule which only digs valleys will destroy the landscape.

Consider the most innocent possible objective, the **energy loss**: for each training pair (x^i, y^i) , minimize $E_\theta(x^i, y^i)$. Push down on the data; done. For some architectures this works — in least-squares regression, $E_\theta(x, y) = \|y - G_\theta(x)\|^2$, pushing the correct answer’s energy toward zero automatically raises every other answer’s, because the architecture allows only one minimum per input. But give the model more freedom and the innocent objective finds the cheat: make the energy *low everywhere*. A constant landscape assigns the data the lowest possible energy and is perfectly useless — it has **collapsed** (LeCun et al. 2006). Flat landscapes answer every question with “everything is equally fine.”

The cure is as old as the perceptron: every push down on the correct answer must be paired with a **pull up** on incorrect ones. Losses that implement this pairing are called *contrastive*, and the zoo of them is organized by one question — *which* incorrect answers get pulled up, and how hard?

The most offending incorrect answer. The sharpest choice is to pull up only on the incorrect

answer the model currently likes best, $\bar{y}^i = \arg \min_{y \neq y^i} E_\theta(x^i, y)$ — the *most offending incorrect answer* (LeCun et al. 2006). The **perceptron loss** $E_\theta(x^i, y^i) - \min_y E_\theta(x^i, y)$ does this but demands no gap, and so can still be satisfied by a nearly flat landscape. Margin losses fix that. The **hinge loss** $\max(0, m + E_\theta(x^i, y^i) - E_\theta(x^i, \bar{y}^i))$ insists the correct answer sit at least m below its closest rival — the loss of support vector machines, re-read as landscape shaping. The **log loss** $\log(1 + e^{E_\theta(x^i, y^i) - E_\theta(x^i, \bar{y}^i)})$ is its soft, infinite-margin sibling. The **square-square loss** $E_\theta(x^i, y^i)^2 + \max(0, m - E_\theta(x^i, \bar{y}^i))^2$ pins correct answers near zero and rivals above m , which is natural when the energy is a distance and hence bounded below.

All incorrect answers at once. The other extreme is the **negative log-likelihood (NLL) loss**. Write the Gibbs conversion Equation 4.3, take $-\log$, and per sample:

$$\mathcal{L}_{\text{nll}} = E_\theta(x^i, y^i) + \frac{1}{\beta} \log \int_y e^{-\beta E_\theta(x^i, y)} dy. \quad (4.4)$$

(Replace the integral by a sum when $\mathcal{Y}$ is finite.) The second term is *minus* the Helmholtz free energy $F = -(1/\beta) \log \int e^{-\beta E} dy$ of Chapter 3 — at $\beta = 1$ it is $\log Z$, not F itself. The first term pushes down on the data; the $\log Z$ term pulls up on *every* answer. How hard? Differentiate (at $\beta = 1$):

$$\frac{\partial \mathcal{L}_{\text{nll}}}{\partial \theta} = \frac{\partial E_\theta(x^i, y^i)}{\partial \theta} - \int_y p_\theta(y | x^i) \frac{\partial E_\theta(x^i, y)}{\partial \theta} dy. \quad (4.5)$$

Each answer is pulled up with force proportional to *the probability the model currently assigns it*. The landscape’s own valleys attract the correction: wherever the model is confidently wrong, the pull is strongest. This is a beautiful rule and an expensive one — the integral in Equation 4.5 is an expectation over the model distribution, which is the partition-function problem in gradient form. Two limits knit the zoo together (LeCun et al. 2006). As $\beta \rightarrow \infty$, $(1/\beta) \log \int e^{-\beta E} dy \rightarrow -\min_y E_\theta(x^i, y)$, so Equation 4.4 becomes the perceptron loss (only the minimum-energy rival matters). And when $\mathcal{Y}$ has just two elements $\{y^i, \bar{y}^i\}$, the $\beta = 1$ case collapses algebraically onto the log loss: $E(y^i) + \log(e^{-E(y^i)} + e^{-E(\bar{y}^i)}) = \log(1 + e^{E(y^i) - E(\bar{y}^i)})$.

i Collapse is a recurring character

Hold on to the failure mode, because this book meets it twice more. In Section 6.6, a language model trained on its own agreement signal can collapse to confident nonsense — entropy minimization with the pull-up term missing. And in modern self-supervised representation learning, joint-embedding architectures collapse to constant embeddings unless a contrastive or regularization term props the landscape up (LeCun 2022; Dawid and LeCun 2024). The lesson is always the same: *an objective that only rewards fitting what you see will be gamed by a model that makes everything look equally good*. The interesting design space is in how — and how cheaply — you pull up.

The remainder of the chapter follows the density view (Song and Kingma 2021): three families of estimators, distinguished by how they approximate or evade the pull-up integral of Equation 4.5. In caricature: **sample it** (maximum likelihood with MCMC), **differentiate it away** (score

matching), or **turn it into a classification problem** (noise contrastive estimation).

4.4 Strategy one: sample the pull-up

Take maximum likelihood seriously and estimate the intractable term by Monte Carlo. For the unconditional model Equation 4.1, the log-likelihood gradient splits into the same two forces as Equation 4.5:

$$\nabla_{\theta} \log p_{\theta}(x) = -\nabla_{\theta} E_{\theta}(x) + \mathbb{E}_{\tilde{x} \sim p_{\theta}}[\nabla_{\theta} E_{\theta}(\tilde{x})]. \quad (4.6)$$

The derivation is worth doing once by hand, because the punchline is neat. Start from $\nabla_{\theta} \log Z(\theta)$, use the chain rule, and watch the model distribution assemble itself:

$$\nabla_{\theta} \log Z = \frac{1}{Z} \nabla_{\theta} \int e^{-E_{\theta}(x)} dx = \int \frac{e^{-E_{\theta}(x)}}{Z} (-\nabla_{\theta} E_{\theta}(x)) dx = -\mathbb{E}_{x \sim p_{\theta}}[\nabla_{\theta} E_{\theta}(x)]. \quad (4.7)$$

The intractable normalizer has become an *expectation under the model* — not computable in closed form, but estimable from a single model sample, without bias. Substituting into $\nabla_{\theta} \log p_{\theta} = -\nabla_{\theta} E_{\theta} - \nabla_{\theta} \log Z$ gives Equation 4.6.

Read the two terms as the two phases of learning. The **positive phase** lowers the energy of real data. The **negative phase** raises the energy of whatever the model itself likes to produce — its *fantasies*, in the old Boltzmann-machine vocabulary, where the phases were called *clamped* and *free* (Ackley et al. 1985) (the 1985 row of the table in Chapter 3). Learning stops exactly when the fantasies become statistically indistinguishable from the data, at which point the two forces cancel in expectation. Confabulate, compare, correct.

The entire difficulty hides in four words: *sample from the model*. The standard tool is **Langevin dynamics**, which needs only the gradient of the log-density in x — and here EBMs catch a genuine break, because the normalizer does not depend on x :

$$\nabla_x \log p_{\theta}(x) = -\nabla_x E_{\theta}(x) - \underbrace{\nabla_x \log Z(\theta)}_{=0} = -\nabla_x E_{\theta}(x). \quad (4.8)$$

This quantity — the gradient of the log-density with respect to the *input*, not the parameters — is called the **score**, and Equation 4.8 says an EBM hands it to us free of charge. Langevin MCMC then iterates

$$x_{k+1} = x_k - \frac{\epsilon^2}{2} \nabla_x E_{\theta}(x_k) + \epsilon z_k, \quad z_k \sim \mathcal{N}(0, I) : \quad (4.9)$$

roll downhill, but shake while doing so. The reader will recognize finite-temperature descent — energy pulling toward minima, injected noise buying the entropy that keeps the walker exploring (Chapter 3). The *ratio* of noise amplitude to drift sets the temperature; Equation 4.9 is the unit-temperature case already built into Equation 4.1. The step size ϵ is a discretization

parameter, not a temperature: as $\epsilon \rightarrow 0$ and the number of steps grows, the iterates distribute as p_θ (Song and Kingma 2021).

“Long enough” is the problem: fresh chains per gradient step are ruinously slow, so practice truncates. **Contrastive divergence (CD)** starts chains at data points and runs a handful of steps (Hinton 2002); **persistent CD** never restarts, carrying chains across parameter updates (Tieleman 2008); replay buffers reinitialize chains from a reservoir of past samples (Du and Mordatch 2019); short-run MCMC accepts non-convergence and reinterprets the truncated sampler itself as the generative model (Nijkamp et al. 2019). These approximations power most large-scale EBM image models, but the bias of truncated chains is real, and taming it remains a research area of its own (Song and Kingma 2021).

4.5 Strategy two: match slopes, not heights

Here is a different idea, and it begins with an observation about Equation 4.8. The score $\nabla_x \log p_\theta(x)$ does not contain $Z(\theta)$ at all. Two normalized densities with the same score everywhere on a connected domain are the same density — if their log-densities have equal gradients, they differ by a constant, and normalization forces that constant to zero. (When the domain falls into well-separated pieces the argument fails, and that failure is the mode-weight blind spot below.) So instead of matching *heights* of the log-density (which requires knowing sea level, i.e. Z), match *slopes*. Picture two hikers comparing terrain maps: they need not agree on altitude above sea level to verify they hold maps of the same mountain — agreeing on the steepness and direction of the slope at every point is enough.

Score matching (Hyvärinen 2005) makes this precise by minimizing the **Fisher divergence** between data and model:

$$D_F(p_{\text{data}} \parallel p_\theta) = \mathbb{E}_{p_{\text{data}}} \left[\frac{1}{2} \|\nabla_x \log p_{\text{data}}(x) - \nabla_x \log p_\theta(x)\|^2 \right]. \quad (4.10)$$

An objection leaps out: Equation 4.10 contains the score of the *data distribution*, which nobody knows. The resolution is a small miracle of integration by parts, and it is worth seeing in one dimension. Expand the square in Equation 4.10; the term $\frac{1}{2} \mathbb{E}[(\partial_x \log p_{\text{data}})^2]$ is a constant in θ and can be dropped, and the term $\frac{1}{2} \mathbb{E}[(\partial_x \log p_\theta)^2]$ is computable. The troublesome cross-term is where the trick lands:

$$-\mathbb{E}_{p_{\text{data}}} [\partial_x \log p_{\text{data}} \cdot \partial_x \log p_\theta] = - \int p_{\text{data}}(x) \frac{\partial_x p_{\text{data}}(x)}{p_{\text{data}}(x)} \partial_x \log p_\theta(x) dx = - \int \partial_x p_{\text{data}}(x) \cdot \partial_x \log p_\theta(x) dx.$$

The unknown density has been reduced to a plain factor, and one integration by parts (boundary terms vanishing for well-behaved densities) moves the derivative off it entirely:

$$= \int p_{\text{data}}(x) \partial_x^2 \log p_\theta(x) dx = \mathbb{E}_{p_{\text{data}}} [\partial_x^2 \log p_\theta(x)].$$

The unknown score has become a second derivative of the *model* — which for an EBM is a second

derivative of the energy — averaged over data samples we have. In d dimensions (Hyvärinen 2005):

$$D_F = \mathbb{E}_{p_{\text{data}}} \left[\sum_{i=1}^d \frac{1}{2} \left(\frac{\partial E_{\theta}(x)}{\partial x_i} \right)^2 - \frac{\partial^2 E_{\theta}(x)}{\partial x_i^2} \right] + \text{const.} \quad (4.11)$$

Signs deserve a moment of care here, because they are easy to slip on. Since $\log p_{\theta} = -E_{\theta} - \log Z$, second derivatives of the log-density are *negatives* of second derivatives of the energy, which is where the minus sign in Equation 4.11 comes from; the squared first-derivative term is immune because squaring erases the sign. A one-line sanity check: fit a Gaussian $E_{\theta}(x) = x^2/2\sigma^2$ to standard-normal data. Equation 4.11 gives $\frac{1}{2\sigma^4} - \frac{1}{\sigma^2}$, minimized at $\sigma^2 = 1$ as it must be; with the sign flipped, the objective would be minimized by sending $\sigma^2 \rightarrow \infty$ — a model dissolving into uniformity. (For *location* families the second-derivative term is constant in the parameter, so both signs give the same fit; that is exactly why the slip is easy to miss.)

No sampling, no partition function, and a consistent estimator. The catch is the Hessian trace: for deep networks and high-dimensional x , exact second derivatives cost roughly d backpropagations. Two descendants remove the bottleneck, one by adding noise and one by throwing dice.

Denoising score matching (DSM). Perturb each data point with Gaussian noise, $\tilde{x} = x + \sigma z$, and match the model’s score to the score of the *noisy* data distribution. Vincent’s identity (Vincent 2011) shows this equals (up to a constant) a beautifully simple objective:

$$\mathbb{E}_{x \sim p_{\text{data}}, z \sim \mathcal{N}(0, I)} \left[\frac{1}{2} \left\| \nabla_{\tilde{x}} \log p_{\theta}(\tilde{x}) + \frac{z}{\sigma} \right\|^2 \right], \quad \tilde{x} = x + \sigma z. \quad (4.12)$$

Intuitively: given a noisy point, the score of the noised distribution points back toward the clean data that produced it, so *learning the score is learning to denoise*. Second derivatives are gone. The costs are a bias — the optimum matches the noisy distribution, not the data — and, if one shrinks σ to reduce that bias, a variance in the z/σ term that blows up as $\sigma \rightarrow 0$ (Song and Kingma 2021). Noise is a knob with no free setting.

Sliced score matching (SSM). Keep the clean data, and check the score match only along random directions: project both scores onto a random vector v and match the resulting scalars (Song et al. 2019). The Hessian-trace term of Equation 4.11 is replaced by the directional second derivative $v^{\top} (\nabla_x^2 E_{\theta}) v$ — the Skilling–Hutchinson estimator — *keeping the minus sign*. Both the squared directional derivative and that Hessian–vector product are computable with two backpropagations at any dimension. SSM is consistent, like exact score matching, at the price of extra estimator variance from the random projections.

The blind spot, and how it launched diffusion. Score matching has one structural weakness worth internalizing. Suppose the data has two well-separated modes — a mixture $\pi p_0 + (1 - \pi) p_1$ with essentially disjoint supports. *Inside* each mode’s region, the mixture’s score is just that component’s own score: the weight π enters the density as a constant factor, and gradients of log-densities kill constant factors. The relative weight of the modes lives precisely in the no-man’s-land *between* them, where there is no data to enforce it. A score-matched model can therefore reproduce each island perfectly and still apportion probability between the islands

arbitrarily (Song and Ermon 2019; Song and Kingma 2021).

The fix is to flood the landscape: perturb the data with noise large enough to bridge the modes — making the weights visible in the score — and, since heavy noise distorts fine detail, do it at *many scales*, learning a noise-conditional score for each. Sampling then anneals from coarse to fine (Song and Ermon 2019). The reader of Chapter 3 will recognize the annealing schedule — start where entropy dominates, finish where energy does — and the reader of the wider literature will recognize the result: this multi-scale, score-based recipe, together with its diffusion-process formulation (Sohl-Dickstein et al. 2015; Ho et al. 2020; Song, Sohl-Dickstein, et al. 2021), became the generative-modeling workhorse of the 2020s. Diffusion models are the energy-based lineage’s most successful child: what they inherit is the score Equation 4.8, the Langevin walk Equation 4.9, and the annealed temperature dial of Chapter 3.

4.6 Strategy three: learn by telling things apart

The third strategy is the slyest. If estimating a density is hard, compare it to one you know.

Noise contrastive estimation (NCE) (Gutmann and Hyvärinen 2010) introduces a noise distribution p_n with tractable density and easy sampling — a Gaussian, say — and mixes noise samples with data samples. Now play a game: given a point, was it data or noise? By Bayes’ rule, the ideal referee computes

$$P(\text{data} \mid x) = \frac{\nu p_{\text{data}}(x)}{\nu p_{\text{data}}(x) + p_n(x)}, \quad (4.13)$$

with ν the data-to-noise mixing ratio. NCE builds a classifier of exactly this functional form, but with the model density in the data slot,

$$P_{\theta}(\text{data} \mid x) = \frac{\nu p_{\theta}(x)}{\nu p_{\theta}(x) + p_n(x)}, \quad (4.14)$$

and trains it by ordinary logistic regression on the mixed samples. If the classifier family is rich enough, training drives Equation 4.14 to the true posterior Equation 4.13 for every x , and matching posteriors at fixed p_n and ν forces $p_{\theta} = p_{\text{data}}$. Density estimation has been smuggled inside a classification problem — the one problem deep learning solves in its sleep.

Two features distinguish NCE in the EBM toolkit (Song and Kingma 2021). First, the normalizer Z_{θ} can be treated as an explicit learnable scalar — the classification game is only won when the model’s *absolute* density matches the data’s, so the game itself calibrates Z . Alternatively, fix $Z = 1$ and obtain a *self-normalized* model, a trick beloved of language modeling, where NCE-style objectives trained word embeddings and early neural language models at vocabulary scales where softmax normalization was unaffordable (Mnih and Teh 2012). Second, the choice of p_n is the whole game in practice: the classification task must be neither trivial nor impossible, which means the noise should resemble the data — a requirement that becomes brutal in high dimension, and has spawned a cottage industry of learned, adaptive noise distributions (Song and Kingma 2021).

One more connection ties the room together. Choose the noise to be the data itself, shifted by a small vector v , and the NCE objective Taylor-expands — as $\|v\| \rightarrow 0$ — into the sliced score matching objective with projection direction v (Song and Kingma 2021). Telling the data apart from a nudged copy of itself means, in the limit, matching directional slopes.

4.7 One family, three estimators

The three strategies look like three separate inventions. They are better seen as one family, distinguished by which divergence they minimize and what they demand of us.

Table 4.1: Three roads around the partition function. Each row answers: how does this method avoid computing $Z(\theta)$, and what does it pay instead?

	MCMC-based MLE (Section 4.4)	Score matching (Section 4.5)	NCE (Section 4.6)
Minimizes	KL divergence	Fisher divergence	KL of classification posteriors
Pull-up term handled by	sampling it	differentiating it away	canceling it against a reference
Requires	model samples (MCMC)	input derivatives of E_θ	a tractable noise density
Access to Z ?	no	no (energy determined up to a constant)	yes (Z learned, or fixed to 1)
Characteristic failure	biased truncated chains	invisible mode weights	noise too far from data

And the roads interconnect. Contrastive divergence run with a single infinitesimal Langevin step *is* score matching, up to rescaling (Hyvärinen 2007). NCE against infinitesimally shifted data is sliced score matching, as we just saw. Underneath both coincidences sits a single identity — a relative form of de Bruijn’s identity — relating the two divergences. Perturb data and model with the *same* Gaussian noise of variance t , write q_t and $p_{\theta,t}$ for the smoothed densities, and

$$\frac{d}{dt} D_{\text{KL}}(q_t \| p_{\theta,t}) = -\frac{1}{2} D_F(q_t \| p_{\theta,t}) : \quad (4.15)$$

the Fisher divergence is the *rate* at which the KL divergence dissolves under noise (Song and Kingma 2021). (The classical de Bruijn identity is the special case that tracks entropy rather than KL.) KL is the integral view, Fisher the differential view, of one and the same discrepancy — which is why an MCMC shortcut and a derivative trick keep converging on the same estimator.

There are further roads — minimizing Stein discrepancies, adversarially training a sampler network against the energy, differences of KL divergences — for which Song and Kingma’s survey (Song and Kingma 2021) and the running list at *Awesome-EBM* (Bian 2020) are the entry points. The taxonomy above is enough to read that literature: whenever a new EBM training method appears, ask first *how it implements the pull-up*.

4.8 Latent variables, and the free energy again

Often the energy needs helper variables that are observed never: the pose of a face, the segmentation of a sentence, the rotation of a molecule. Give the energy an extra argument $E_\theta(x, y, z)$ and let inference minimize over z as well. But minimizing over z is a $T = 0$ policy — it credits y with only its single best explanation. The finite-temperature policy sums over explanations:

$$F_\beta(x, y) = -\frac{1}{\beta} \log \int_z e^{-\beta E_\theta(x, y, z)} dz, \quad (4.16)$$

which the reader will recognize as a free energy — Equation 4.16 is $F = -\frac{1}{\beta} \log Z$ with the latent variable playing the role of microstate, and it interpolates between marginalization ($\beta = 1$: an answer is good if its explanations are *collectively* probable) and minimization ($\beta \rightarrow \infty$: an answer is as good as its best explanation) (LeCun et al. 2006). Latent-variable EBMs thereby fold the machinery of Chapter 3 inside the energy function itself, and the variational bound of that chapter — minimize $F(q) = \langle E \rangle_q - T S(q)$ over tractable q , with $T = 1/\beta$ — becomes the standard training route when the integral is intractable. The evidence lower bound of variational autoencoders is this construction at $T = 1$, as Chapter 3 already made explicit.

This is also where the energy-based view connects to the present frontier of self-supervised learning. LeCun’s proposal for world-model architectures — JEPA, the joint-embedding predictive architecture — is at bottom a latent-variable EBM over *representation* space: energy measures prediction error between embeddings, latent variables absorb what cannot be predicted, and the central design problem is once again preventing collapse, either by contrastive pull-up or by regularizers that keep the embedding distribution spread out (LeCun 2022; Dawid and LeCun 2024). Decades of this subject can be compressed into one sentence: *dig valleys where the data is, and find an affordable way to keep the rest of the landscape high.*

4.9 The modern landscape

A brief tour of where energies appear in current practice, with pointers rather than treatments — the companion list (Bian 2020) tracks the full literature.

Your classifier was an EBM all along. A softmax classifier computes logits $f_\theta(x)[y]$ and normalizes over labels only. Reuse those logits as a joint energy, $E_\theta(x, y) = -f_\theta(x)[y]$, and the *marginal* energy of an input is $E_\theta(x) = -\log \sum_y e^{f_\theta(x)[y]}$ — the free energy of the label ensemble. Training this hidden generative model alongside the classifier improves calibration and robustness (Grathwohl et al. 2020). Even without retraining, the same marginal energy is a useful *score* for out-of-distribution detection, with higher energy read as more unfamiliar (Liu et al. 2020). (A classifier that was never trained as a generative model does not automatically place OOD inputs high; the energy score is still often a better OOD detector than the raw softmax confidence.) This is the “energy dial” of Chapter 3 read in reverse — the logits every deep classifier already emits are energies in thin disguise.

Generation and its guidance. Diffusion and score-based models (Section 4.5) carry the EBM inheritance in generation, and the additivity of energies (Section 4.2) becomes *composability*

at inference time: adding a classifier’s gradient to a sampler’s score steers generation toward a class (Dhariwal and Nichol 2021); adding hand-built constraint energies composes concepts (Du et al. 2020); in text, a lightweight energy can re-rank or steer a powerful autoregressive proposal — residual EBMs (Deng et al. 2020), constraint-guided decoding (Qin et al. 2022). Discrete spaces lack Langevin’s gradient walk, and a lively literature builds gradient-informed proposals for them (Grathwohl et al. 2021). This inference-time story is the subject of Chapter 8; its training-time counterpart, in which the tilt is absorbed into the weights of the pretrained model, is the subject of Chapter 9.

Science. Molecules are the native habitat of energy functions — the word itself came from physics, and here it goes home. Denoising-style score matching over molecular geometries drives conformation generation (Shi et al. 2021); SE(3)-equivariant energies score protein structures end-to-end (Wu et al. 2021). The learned energy (or its score) sits in the same seat as the physical force field it approximates or replaces — Chapter 10 takes up this thread.

4.10 Where this sits in the book

This chapter closes Part I, and it is the load-bearing wall for Part II. Backward: Chapter 2 supplied entropy and the maximum-entropy principle; Chapter 3 derived the Gibbs distribution and identified $-\log Z$ as a free energy; this chapter made the energy learnable and catalogued the ways around the normalizer. Forward, each methods chapter picks up a specific thread:

- **Maximum-entropy learning** (Chapter 5) is the family in which Gibbs distributions are *used* as models and the entropy term is explicit in the inference, whereas this chapter supplied *estimators* for a Gibbs model, in which the entropy is implicit in $\log Z$ (maximum likelihood, contrastive divergence) or replaced by a generalized entropy (score matching, NCE; see Section 5.6.3). The family is defined through the three levels of representation, inference, and learning (Section 5.1), and it contains classical MaxEnt, probabilistic graphical models as an explanatory family (Section 5.2), maximum-entropy and inverse reinforcement learning (Section 5.5), energy-based guidance, and two strands of our own — a Gibbs distribution over coalitions in a *given* cooperative game, from which game-theoretic valuations are derived (Bian et al. 2022) (Section 5.3), and a Gibbs distribution over subsets whose energy is a neural set function *learned* from optimal subsets alone (Ou et al. 2022), a fourth road around the partition function that Table 4.1 does not list (Section 5.4).
- **Minimum-entropy learning** (Chapter 6) trains a model on its own agreement signal; its collapse mode (Section 6.6) is the energy loss’s flat-landscape failure in a new costume, and the free-energy decomposition used to analyze it is Equation 4.4 seen from the other side.
- **Minimax-entropy learning** (Chapter 7) asks which *features* an exponential-family model should include — which is precisely the architecture-selection problem for the EBMs of this chapter, posed variationally.
- **Energy-based guidance and refinement** (Chapter 8) keep a pretrained generator frozen and spend inference-time compute tilting it by an energy: guidance is gradient descent on

a composed energy, refinement is repeated local descent, and both inherit the annealing wisdom of Chapter 3. The energy may itself have been trained — by the NCE of this chapter, for instance — but it enters only through the sampler.

- **Maximum-entropy post-training** (Chapter 9) performs the same tilt at the learning level: the pretrained model is fitted to absorb the tilt, or the energy is learned through the model’s log-ratio to its reference, which is how RLHF, GRPO, and DPO enter this book.

4.11 Takeaway

An energy-based model is a learnable landscape: inference rolls downhill on it, probability — when needed — is bought from it at the price of a partition function. Training is landscape-shaping, and its non-negotiable rule is that digging valleys under the data must be paired with pulling up elsewhere, lest the landscape collapse into a useless plain. The three classical training strategies are three ways to afford the pull-up: sample it (MCMC-based maximum likelihood), sidestep it by matching slopes (score matching, the road that led to diffusion models), or fold it into a classification game against known noise (NCE). One identity — a relative form of de Bruijn’s — reveals them as views of a single discrepancy, and one design question organizes the whole field, from Boltzmann machines to JEPA: *how do you keep the rest of the world high?*

Part II

Part II — Methods

5 Maximum-Entropy Learning

5.1 What this chapter means by maximum-entropy learning

Throughout this book the term *maximum-entropy learning* is used in a broad sense: it refers to a family of methods rather than to one algorithm. The family contains methods in which nothing is learned at all, e.g., the cooperative games of Section 5.3, as well as methods in which an energy function is fitted from data, e.g., the set functions of Section 5.4. We therefore need a vocabulary that keeps the two apart. We borrow it from probabilistic graphical models, whose standard textbooks are organized into three parts, *representation*, *inference*, and *learning* (Koller and Friedman 2009), and we read every method through these three levels in turn. In this section we make each level precise, state the definition of the family, and fix the notation used in the rest of the chapter.

Notation. We write $\mathsf{T}(x) = (f_1(x), \dots, f_K(x))$ for the vector of features whose expectations are constrained, i.e., the *sufficient statistics* of the resulting exponential family. The upright sans-serif T is used on purpose: the italic T is reserved for temperature throughout the book, and the two must not be confused. $h(x)$ denotes a *base measure*, i.e., a reference distribution, taken to be uniform unless stated otherwise; λ denotes the vector of Lagrange multipliers (the natural parameters), Z the partition function, and $\text{KL}(q \parallel p)$ the Kullback–Leibler divergence.

5.1.1 Representation: maximum-entropy models in two forms

The first level asks which model class a method uses. In maximum-entropy learning the answer is the Gibbs distribution, equivalently the exponential family, and it can be written in two forms.

The **constrained form** is the one of Section 2.3, stated here with a base measure (Jaynes 1957; Kullback 1959; Csiszár 1975):

$$\begin{aligned} p^* &= \arg \min_q \text{KL}(q \parallel h) \quad \text{s.t.} \quad \mathbb{E}_q[\mathsf{T}(x)] = \mu, \\ p^*(x) &= \frac{h(x) \exp(\lambda^\top \mathsf{T}(x))}{Z(\lambda)}. \end{aligned} \tag{5.1}$$

When h is uniform, minimizing $\text{KL}(q \parallel h)$ is the same as maximizing the entropy $H(q)$, and Equation 5.1 is exactly the Jaynes problem of Section 2.3. For a general h the principle is that of *minimum relative entropy*, known as minimum discrimination information in Kullback’s work (Kullback 1959) and as minimum cross-entropy in the axiomatic treatment of Shore and Johnson (1980); the solution p^* is the *I-projection* of h onto the constraint set (Csiszár 1975).

The **free-energy form** is the one introduced in Chapter 3. Given an energy $E(x)$ and a temperature $T > 0$, define the variational free energy

$$\begin{aligned}\mathcal{F}_T(q; E) &= \mathbb{E}_q[E] + T \text{KL}(q \| h), \\ p_{E,T} &= \arg \min_q \mathcal{F}_T(q; E) = \frac{h(x) e^{-E(x)/T}}{Z}, \quad \min_q \mathcal{F}_T(q; E) = -T \log Z.\end{aligned}\tag{5.2}$$

For uniform h this is $\mathbb{E}_q[E] - T H(q)$ up to a constant, i.e., the $U - TS$ of Chapter 3, and the identity $\min_q \mathcal{F}_T = -T \log Z$ is the variational representation of the log-partition function (Wainwright and Jordan 2008, sec. 3.6), known in statistical physics as the Gibbs variational principle.

The two forms are equivalent by Lagrangian duality. Setting $E = -\lambda^\top \mathbf{T}$ and $T = 1$ in Equation 5.2 recovers Equation 5.1, and the constraint values and the multipliers are in one-to-one correspondence through $\mu = \nabla_\lambda \log Z(\lambda)$, provided that μ lies in the interior of the marginal polytope (the regularity condition discussed in Section 5.2). One can also read the equivalence geometrically. For the empirical distribution $\hat{p}$ and the maximum-likelihood member p^* of the exponential family, Csiszár’s Pythagorean identity $\text{KL}(\hat{p} \| h) = \text{KL}(\hat{p} \| p^*) + \text{KL}(p^* \| h)$ holds (Csiszár 1975), so the maximum-entropy solution and the maximum-likelihood solution are one point approached from two directions (Della Pietra et al. 1997).

Both forms will be used in this chapter, and the choice is a matter of emphasis. We state a model in the constrained form when the point is *what is known*, i.e., the form of $\mathbf{T}$ declared by the modeler: clique indicators in a Markov random field, n -gram features in a maximum-entropy classifier (Berger et al. 1996), the invariant statistics of INSET (Section 5.4.3), reward features together with the dynamics in reinforcement learning (Section 5.5). We state it in the free-energy form when the point is *what is computed*: an ELBO, a mean-field iteration, a soft Bellman equation, an annealing schedule. It is worth noting where the knowledge sits. It does not sit in the value of μ , which the data supply, but in the form of $\mathbf{T}$: the content of the theorem behind Equation 5.1 is that the energy is *confined to the span of the declared features*, $E = -\lambda^\top \mathbf{T}$, not that it has an exponential shape, since any strictly positive p can be written as e^{-E} with $E := -\log p$.

A related point is that the two forms constrain different variables. The constrained form restricts the distribution q ; an architecture for E_θ restricts the energy, i.e., the dual variable, and the two descriptions coincide exactly only when the admissible energies form a linear space (equality constraints, with free multipliers) or a convex cone (inequality constraints, with nonnegative multipliers). For a neural energy the correspondence holds slice by slice through the last linear layer, and at every stationary point through the likelihood equations of Section 5.1.3, but not for the family as a whole, which is a curved rather than an exponential family (Efron 1975). Symmetry is the exception that lives on both sides at once: invariance of the distribution under a group G is a set of homogeneous linear constraints $\mathbb{E}_q[f \circ g - f] = 0$, the maximum-entropy solution inherits it (Jaynes 1968), and it is equivalent to restricting the features to functions of the maximal invariant of the group (Bloem-Reddy and Teh 2020), which is the route taken in Section 5.4.3.

5.1.2 Inference: free-energy minimization over a tractable family

The second level asks how the distribution is computed or approximated. In maximum-entropy learning, inference is the minimization of the free energy of Equation 5.2 over a tractable family $\mathcal{Q}$ of distributions,

$$q_Q^*(E, T) = \arg \min_{q \in \mathcal{Q}} \mathcal{F}_T(q; E). \quad (5.3)$$

The choice of $\mathcal{Q}$ is what distinguishes the algorithms. If $\mathcal{Q}$ is the set of all distributions, the inference is exact and returns $p_{E,T}$. If $\mathcal{Q}$ is the set of product distributions, Equation 5.3 is mean-field variational inference (Jordan et al. 1999; Wainwright and Jordan 2008), the workhorse of Section 5.3 and Section 5.4. If $\mathcal{Q}$ is the set of trajectory distributions induced by a policy under the true dynamics, it is maximum-entropy reinforcement learning (Section 5.5). Markov chain Monte Carlo replaces the minimization by sampling, and a path in T is an annealing schedule (Chapter 8). The two limits are instructive: as $T \rightarrow 0$, Equation 5.3 becomes energy minimization, i.e., the inference of energy-based models in the sense of Section 4.1; as $T \rightarrow \infty$, it returns the base measure.

A remark on terminology is in order. We deliberately avoid the phrase “maximum-entropy inference”. In Jaynes’ writing, *inference* means choosing a distribution from constraints, which is our representation level; in the graphical-model literature it means computing marginals or posteriors, which is the present level. For the latter we say *variational inference* or *free-energy minimization*.

5.1.3 Learning: fitting the energy through the inference

The third level is active when the energy itself carries parameters, E_θ , which are fitted from data $\mathcal{D}$ through the inference of the second level:

$$\min_{\theta} \mathcal{L}(\mathcal{D}; q_Q^*(E_\theta, T)). \quad (5.4)$$

The supervision in $\mathcal{D}$ can take several forms: i.i.d. samples, as in maximum likelihood; prescribed moments, as in the classical setting; observed *optimal solutions*, as in the optimal subset oracle of Section 5.4; demonstrated trajectories, as in inverse reinforcement learning (Section 5.5); or preferences. In every case the division of labour is the same. The learner chooses the form of T , i.e., an architecture or a symmetry; the data supply the constraint content; and the maximum-entropy principle fixes the distribution. The inner problem Equation 5.3 is strictly convex in q , whereas the outer problem Equation 5.4 is in general non-convex in θ .

Maximum likelihood deserves a separate comment, because it is where the two levels meet. For the Gibbs model $p_\theta \propto h e^{-E_\theta}$ (we set $T = 1$ and absorb the scale into E_θ), the gradient of the negative log-likelihood is $\mathbb{E}_{\hat{p}}[\nabla_\theta E_\theta] - \mathbb{E}_{p_\theta}[\nabla_\theta E_\theta]$ (Equation 4.5), so at any stationary point

$$\mathbb{E}_{\hat{p}}[\nabla_\theta E_\theta] = \mathbb{E}_{p_\theta}[\nabla_\theta E_\theta]. \quad (5.5)$$

These are the classical *likelihood equations*, and they say that the model matches the data on the moments of the features $\nabla_{\theta} E_{\theta}$. For a linear energy $E = -\lambda^{\top} \mathbf{T}$ they are exactly the constraints of Equation 5.1 with μ the empirical moments — this is the maximum-entropy / maximum-likelihood duality — and the Boltzmann machine learning rule, clamped correlations minus free correlations (Ackley et al. 1985), is their best-known early instance in machine learning. For a neural energy the features move with θ , so the constrained form holds only at each stationary point; the general case is better written in the free-energy form, as a saddle-point problem (Section 5.6).

5.1.4 Definition of maximum-entropy learning, and a running example

We can now state the definition used in this chapter.

i Definition: maximum-entropy learning

A method belongs to **maximum-entropy learning** if i) its model class is the Gibbs distribution of Equation 5.1 or Equation 5.2, determined by a declared form of constraint; ii) its inference is a free-energy minimization Equation 5.3 in which the entropy term is explicit and load-bearing, i.e., $T > 0$; and, when the energy is fitted from data, iii) the fitting is carried out through that inference, as in Equation 5.4. A method satisfying i) and ii) is a *maximum-entropy model* with variational inference; a method satisfying i)–iii) is termed a **maximum-entropy learner**. When the Kullback–Leibler divergence in i)–ii) is replaced by the Bregman divergence of another proper scoring rule, the same structure defines a *generalized entropy* (Grünwald and Dawid 2004); we then speak of maximum-entropy learning *in the generalized sense*, and reserve the unqualified term for the Shannon case.

Two consequences of the definition are worth spelling out. First, the energy-based models of Chapter 4 are all maximum-entropy models in the sense of i), and their training criteria differ in how they treat the inner problem that carries the entropy. Maximum likelihood and its approximations — sampled, as in persistent contrastive divergence (Tieleman 2008); truncated, as in contrastive divergence (Hinton 2002) and short-run MCMC (Nijkamp et al. 2019); or amortized, as when a generator is trained to maximize its own entropy (Kumar et al. 2019) — are maximum-entropy learners in the strict sense of iii). Score matching and noise-contrastive estimation replace the log score by another proper scoring rule; each such rule carries its own generalized entropy and Bregman divergence (Grünwald and Dawid 2004), and we call these methods maximum-entropy learners in the generalized sense, naming the entropy in each case (Section 5.6.3). The zero-temperature losses of energy-based learning, e.g., the perceptron and hinge losses, lie on the boundary of the family. Second, methods in which the energy is *given* — a cooperative game (Section 5.3), a reward function in the soft actor–critic (Section 5.5), a composed energy used for guidance (Chapter 8) — are maximum-entropy models with variational inference and no learner. Nothing is learned in them, and yet they belong to the family, because the entropy term does the work. Table 5.1 places the strands of this chapter, and two neighbouring ones, on the three levels.

Table 5.1: The strands of maximum-entropy learning read through the three levels. A dash means that the level is not active: the energy is given rather than learned.

Strand	Representation	Inference	Learning
Classical MaxEnt, log-linear models, CRFs (Section 5.2)	Gibbs with declared features	exact, or message passing	multipliers by maximum likelihood
Undirected graphical models (Section 5.2)	Gibbs over cliques	variational or sampling	parameters by maximum likelihood
Energy-based models by maximum likelihood or CD (Chapter 4)	Gibbs with learned features	MCMC, sampled or truncated	energy by maximum likelihood
Energy-based models by score matching or NCE (Chapter 4)	Gibbs with learned features	none (local divergence) or a fixed noise proposal	energy by a generalized (Bregman) divergence; see Section 5.6.3
Energy-based cooperative games (Section 5.3)	Gibbs over coalitions, one constraint	mean field	— (game given)
Optimal subset oracle (Section 5.4)	Gibbs over 2^V , neural set function	mean field, unrolled	energy from optimal subsets
Maximum-entropy RL with a given reward (Section 5.5)	Gibbs over trajectories	soft Bellman recursion	— (reward given)
Inverse RL (Section 5.5)	Gibbs over trajectories	soft Bellman recursion	reward from demonstrations
Guidance and refinement (Chapter 8)	Gibbs tilt of a pretrained generator	guided or annealed sampling, reranking	— (energy given to the sampler; if trained, by a separate learner)
Maximum-entropy post-training (Chapter 9)	Gibbs tilt of a pretrained model	amortized into the weights	reward from preferences, or through the closed form (DPO)

The status of the temperature follows the same pattern. When the energy is given and no constraint level is prescribed, T is a genuine hyperparameter: the coalition temperature τ of Section 5.3, the entropy coefficient α of the soft actor–critic, the KL coefficient β of policy optimization (Equation 6.2). When the energy is learned and its family is closed under rescaling, T is absorbed into the scale of E_θ and is not identifiable; this is the case of Section 5.4. When a constraint level is prescribed, T is the reciprocal of a multiplier and is determined by that level, as in the canonical ensemble of Chapter 3.

A running example. The simplest maximum-entropy learner is the maximum-entropy classifier of classical natural language processing (Berger et al. 1996), i.e., multinomial logistic regression. The features are indicator functions $f_k(x, y)$, e.g., “word k occurs in document x and the label y is *sports*”. The constrained form asks for the conditional distribution closest to uniform that matches the empirical expectations of all f_k ; the solution is $p(y | x) \propto \exp(\lambda^\top \mathbf{T}(x, y))$, and fitting λ by maximum likelihood is the dual problem, so the likelihood equations Equation 5.5 are precisely the moment constraints. In the free-energy form, the negative log-likelihood of

one example is $-\lambda^\top \mathbf{T}(x, y) + \log \sum_{y'} \exp(\lambda^\top \mathbf{T}(x, y'))$, i.e., the energy of the observed label plus the free energy of the label ensemble. Inference is exact because the label set is small, and the temperature is absorbed into the scale of λ . Levels i)–iii) are all present and all three are trivial; the rest of this chapter is about what happens when the space is the power set 2^V or the set of trajectories, and the inference is no longer exact.

5.2 Graphical models as an explanatory family

Probabilistic graphical models (PGMs) organize a joint $p(x)$ by a graph that *encodes* a set of conditional independencies (Pearl 1988; Koller and Friedman 2009). The encoding is not the same in the two standard realizations. On an undirected graph, a missing edge is pairwise independence given all other variables (and, when p is strictly positive, the pairwise, local, and global Markov properties coincide). On a directed acyclic graph the criterion is *d-separation*. A missing arrow still carries an independence — the local Markov property says each variable is independent of its non-descendants given its parents — but not the *same* independence as a missing undirected edge: conditioning on all the remaining variables can *open* a path through a collider, so the MRF’s pairwise reading has no directed analogue (Pearl 1988). The field is older than the modern energy-based revival of Chapter 4, larger than this chapter, and still the working language for structured uncertainty in statistics and in much of classical machine learning — vision, speech, computational biology, and expert systems among them.

An **undirected** model, or Markov random field (MRF), scores *compatibilities*. On a finite graph G , a strictly positive distribution is Markov with respect to G if and only if it is a Gibbs distribution that factors over the cliques of G (Hammersley–Clifford; the published proof usually cited is Besag (1974)). In energy language that is already the definition of Chapter 4, with the extra discipline that E is a sum of local terms:

$$p(x) = \frac{1}{Z} \exp\left(- \sum_{C \in \text{cliques}(G)} E_C(x_C)\right).$$

(The sum may equally be taken over maximal cliques: smaller clique terms can be absorbed.) The Gibbs $\Rightarrow$ Markov direction is elementary and needs no positivity; it is Markov $\Rightarrow$ Gibbs that does. Without it the implication fails: Moussouris’ four-cycle example is a distribution that is Markov with respect to G but admits no Gibbs factorization over the cliques of G (Moussouris 1974).

A **directed** model, or Bayesian network, tells a *generative* story. If pa_i are the parents of coordinate i in a directed acyclic graph,

$$p(x) = \prod_i p(x_i \mid \text{pa}_i) \tag{5.6}$$

(Pearl 1988). The arrows are often read as explanation or cause; the undirected edges of an MRF are not. Pearl’s own contrast is that a Bayes net is a knowledge base of directed influences, while a Markov network is a system of symmetric associations. That reading is a stretch beyond what the joint distribution alone licenses: from observational data a DAG is identifiable only up to its

Markov equivalence class — every DAG with the same skeleton and the same v-structures — so “the graph is the explanation” should be heard as a claim about which independencies the graph asserts, not about which arrow points which way.

This book treats the pair as an **explanatory family**, in a local sense that is not a standard name in the literature. The graph is the explanation: it says which independencies the joint is required to satisfy, and the factorization is what makes the model intelligible rather than a mere scoring rule. That reading is closest to Pearl’s directed case. It is more of a stretch for MRFs, whose native language is compatibility rather than cause. Both, however, explain the joint by a sparse independence hypothesis, and that is the sense in which they belong together here.

How much of this is classical maximum entropy? Enough to be worth saying, not enough to be a slogan. The maximum-entropy distribution of Section 2.3 that matches prescribed expectations of clique-supported features is an exponential-family MRF: the Lagrange multipliers are the feature weights, and the solution is the Gibbs form above (with $E_C = -\sum_k \lambda_k f_k$ over the features supported on C ; the sign is the usual energy convention) (Jaynes 1957; Della Pietra et al. 1997; Wainwright and Jordan 2008). On a finite discrete state space every strictly positive MRF arises this way, because the clique-configuration indicators are a sufficient statistic. The modern synthesis of Wainwright and Jordan writes undirected graphical models, exponential families, and entropy duality as one subject. Feature induction for random fields (Della Pietra et al. 1997) and the maximum-entropy models of classical NLP (Berger et al. 1996) are the same *exponential-family* construction; they are MRFs only when the features have local graphical support — Stephen Della Pietra, Vincent Della Pietra, and John Lafferty are explicit that their fields may be non-Markovian. Conditional random fields (Lafferty et al. 2001) take the construction over to $p(y \mid x)$ and became, for a decade, the default of structured prediction. None of that is controversial.

The duality itself has a real regularity condition, not just a name: the program $\max_p H(p)$ subject to $\mathbb{E}_p[f] = \mu$ is dual to maximum likelihood in the exponential family exactly when μ sits in the *interior* of the marginal polytope; on the boundary the MLE fails to exist, and the maximum-entropy optimum is attained only in the closure of the exponential family (Wainwright and Jordan 2008, sec. 3.4).

What is *not* settled — and should not be written as if it were — is the claim that “PGM is classical maximum entropy.” Three qualifications matter.

1. The PGM literature organizes itself around *representation, inference, and learning* — the triad we borrowed in Section 5.1 — not around Jaynes. Standard treatments (Pearl 1988; Koller and Friedman 2009) may discuss log-linear Markov networks, whose maximum-likelihood dual *is* a maximum-entropy problem, without taking that dual as the field’s founding slogan. The maxent identification is a precise reading of the exponential-family parameterization, not the community’s self-description.
2. The tight theorem is undirected, finite, and strictly positive. Directed models are a different organizing principle: Equation 5.6 is not a Jaynes problem. A Bayes net can be *moralized* into an MRF on the graph obtained by marrying co-parents and dropping directions; the resulting undirected model is an **I-map** (every independence it asserts by missing edges does hold in p), but it need not be a *perfect* map — moralization can lose

independencies that the DAG had (v-structures are the usual example, since marrying the parents removes exactly the edge whose absence encoded that independence). The rewriting $E(x) = -\sum_i \log p(x_i \mid \text{pa}_i)$ is an energy in the sense of Chapter 4, and in this case $Z = 1$ already, so the network sits inside the energy-based tent as a *normalized* model. That does not make it a maximum-entropy model, and the precise reason is the dimensional cousin of the loss of independencies above: a Jaynes problem with *linear* constraints has a solution confined to a *linear* exponential family. Fully observed discrete **undirected** graphical models are exactly such linear exponential families; fully observed discrete **directed** (DAG) models are in general only a *curved* exponential family (Geiger et al. 2001), with equality precisely when the DAG has no immoralities (v-structures) — that is, when its Markov equivalence class contains a decomposable (chordal) undirected model. Away from finite discrete variables the identification loses its grip for a further, dimensional reason: the clique potentials now range over an infinite-dimensional function space, so the MRFs on a fixed graph no longer form a *finite-dimensional* exponential family. (A single Gibbs distribution is always trivially exponential-family — the claim has content only for the model class.) Gaussian MRFs remain a finite-dimensional exponential family; potentials given by mixtures do not.

3. Even on the undirected side, maxent fixes the *distribution given the constraints*. Choosing the graph, or which clique features to include, is a *feature-selection* question with more than one answer — the greedy KL-gain feature induction of Della Pietra, Della Pietra, and Lafferty (Della Pietra et al. 1997), and the minimax entropy of Zhu, Wu, and Mumford (Zhu et al. 1997), the subject of Chapter 7 — and neither is part of Jaynes’ original programme.

So the accurate statement, and the one this chapter will use, is narrower than a slogan and stronger than a metaphor: **a strictly positive exponential-family MRF on a finite graph is a classical maximum-entropy model with structured constraints**; Bayesian networks are an explanatory sibling that can be rewritten as (already normalized) energies but should not be renamed maxent; the PGM field as a whole is larger than either identification, and still doing work that this book will not reproduce. A future revision should include full derivations of both the Hammersley–Clifford theorem and the moralization construction. For now, this chapter simply points to them as essential references.

5.3 A worked example: energy-based cooperative games

The variational machinery of the previous section is not only for graphs. The same mean-field relaxation that Wainwright and Jordan use to approximate a graphical model’s log-partition function applies verbatim to a Gibbs distribution over coalitions, and what it produces there turns out to be a familiar object.

As one concrete instance of the programme above, consider a **valuation problem** in machine learning: how much does a feature, data point, or player contribute to an outcome? The classical answer is the **Shapley value** from cooperative game theory, but it is one particular choice among many.

In *Energy-Based Learning for Cooperative Games* (Bian et al. 2022), the valuation problem is recast through an energy / maximum-entropy lens. A cooperative game with characteristic function $v(S)$ over coalitions $S \subseteq N$ is turned into a distribution over coalitions by a single Jaynes constraint: match a prescribed expected payoff and nothing else. The solution is the Gibbs distribution

$$p(S) \propto \exp(v(S)/\tau),$$

with energy $-v(S)$. Here τ is a temperature in exactly the sense of Chapter 3: as $\tau \rightarrow 0$ the mass concentrates on the highest-value coalitions, while as $\tau \rightarrow \infty$ it flattens towards uniform. In the strict Jaynes reading $\tau = 1/\lambda$ is not a free knob — it is pinned by whichever expected payoff the constraint matches — so sweeping τ below is the same move as sweeping that matched level.

Player valuations are then read from a mean-field approximation to p . The payoff term of that mean-field ELBO is exactly Owen’s **multilinear extension** of the game (Owen 1972, 1975). What the variational reading adds is the entropy term — which turns the multilinear extension from an interpolation device into an ELBO with a temperature — and the trajectory beyond one step: further mean-field iterations define a family of variational valuations whose fixed point — the minimizer of the mean-field KL divergence, i.e. the member with the best conceivable decoupling error — is the **Variational Index**. Shapley is therefore not a special case of the maximum-entropy *distribution*, but a one-step mean-field reading of it.

5.3.1 Why this is useful

- **Principled interpretation.** Valuations become variational parameters of a mean-field approximation to a maximum-entropy distribution over coalitions, rather than isolated axiomatic scores.
- **A family, not a point.** The temperature τ and the number of mean-field steps trace a spectrum of valuations, with Shapley and Banzhaf as one-step readings rather than the only options.
- **Bridges to physics.** The same Gibbs machinery is the one that carried the Ising energy into neural networks in the first place, by way of the Hopfield network (Hopfield 1982) and the Boltzmann machine (Ackley et al. 1985); see Chapter 3.

5.4 Discrete maximum-entropy learning: set functions under the optimal subset oracle

Everything in Section 5.3 presupposed the game. The characteristic function $v(S)$ was given, the Gibbs distribution over coalitions was *derived* from it by one Jaynes constraint, and mean-field inference on that distribution was an *interpretation* device — a way to read valuations off a distribution nobody had to fit. Nothing was learned. This section drops that assumption. The set function is unknown, the only supervision is which subset turned out to be optimal, and the same Gibbs-plus-mean-field machinery becomes a *training loop* on the power set 2^V — an

exponentially large discrete space on which none of the Langevin machinery of Chapter 4 applies. This section most concretely embodies the concept of maximum-entropy *learning* as defined in Section 5.1: here, the energy function is not given a priori but is instead learned directly on a discrete set space, so the strand is a maximum-entropy *learner* in the sense of Section 5.1.4. The published line runs through EquiVSet (Ou et al. 2022) and INSET (Xie, Bian, et al. 2024), and has since been taken up by others (Xie, Wang, et al. 2024; Özcan et al. 2025); a further step of our own is still a preprint and is noted only at the end. We present the line as a paradigm of **discrete maximum-entropy learning**, and in Section 5.4.4 say what it adds to the programme and what it leaves open.

5.4.1 The optimal subset oracle

A **set function** $F(S; V)$ assigns a utility to every subset S of a ground set V : the products a customer would want among those displayed, the anomalous images in a batch, the compounds worth advancing from a screening library. The object of interest is the maximizer,

$$S^* = \arg \max_{S \subseteq V} F(S; V), \quad (5.7)$$

and the question is how to learn F_θ from data. The classical setting is the **function-value (FV) oracle**: pairs $(S_i, F(S_i))$ on a fixed ground set, from which F_θ is fit by regression. It is expensive — one utility label per subset, on a space of $2^{|V|}$ subsets — and the learnability theory built on it carries strong inapproximability results even for submodular functions (Balcan and Harvey 2018). The **optimal subset (OS) oracle** is weaker and far more common: pairs (V_i, S_i^*) , a ground set together with the subset that was actually chosen, and no utility values at all. The two are not comparable in general. Max-cut has a trivial FV oracle and an NP-hard OS oracle; a recommender log has a natural OS oracle — what the customer selected from what was shown — and no FV oracle, because nobody knows the customer’s utility.

Learning under the OS oracle was posed by Ou et al. (2022) as maximum likelihood over a distribution on subsets whose mass grows with the utility:

$$\max_{\theta} \mathbb{E}_{(V, S^*)} [\log p_{\theta}(S^* | V)], \quad p_{\theta}(S | V) = \frac{\exp(F_{\theta}(S; V))}{\sum_{S' \subseteq V} \exp(F_{\theta}(S'; V))}. \quad (5.8)$$

This is a conditional energy-based model in the sense of Section 4.1 — input $x = V$, answer $y = S$, answer space $\mathcal{Y} = 2^V$, energy $E_{\theta}(S; V) = -F_{\theta}(S; V)$ — and the Gibbs form is not a convenience but the maximum-entropy choice. Among all distributions on 2^V that match a prescribed expected utility $\mathbb{E}_p[F(S)] = \mu$, the entropy maximizer is unique and is $p(S) \propto \exp(\lambda F(S))$ (Ou et al. 2022, Thm. 1); the argument is Jaynes’ own (Jaynes 1957). It is the single-constraint problem of Chapter 3 and exactly the coalition distribution of Section 5.3 with v replaced by F_{θ} ; the multiplier $\lambda = 1/\tau$ is absorbed into the scale of F_{θ} , as β was absorbed into E in Chapter 4. The paper calls the property *minimum prior*: the model assumes nothing about the subset beyond what its utility says. The contrast is with the probabilistic greedy model (Tschitschek et al. 2018), which builds $p(S)$ by summing an order-dependent,

autoregressive greedy-sampling distribution over all orderings of S — importing an inductive bias the data never asked for, and at factorial cost.

It is worth seeing what the likelihood asks for. In the normalization of Equation 4.4, with an explicit inverse temperature, the negative log-likelihood of one pair is $-F_\theta(S^*; V) + \frac{1}{\beta} \log \sum_S e^{\beta F_\theta(S; V)}$. As $\beta \rightarrow \infty$ the second term tends to $\max_S F_\theta(S; V)$, and the loss to $\max_S F_\theta(S; V) - F_\theta(S^*; V)$: the perceptron loss of Section 4.3 on the answer space 2^V , which vanishes exactly when S^* is a maximizer — Equation 5.7 read as a loss. Maximum likelihood under the OS oracle is therefore the finite-temperature relaxation of “make the observed subset the arg max.” It buys, at the price of a partition function, a smooth objective whose pull-up (Equation 4.5) is spread over every rival subset in proportion to the probability the model currently assigns it. In the taxonomy of Section 4.1 this is the *decision* question answered through the *probability* question, and the detour is deliberate.

The same objective can be read in the constrained form of Section 5.1.1. What the OS oracle tells us about the unknown set function is a collection of *optimality constraints*, $F(S^*; V) \geq F(S; V)$ for all $S \subseteq V$, i.e., one polyhedral cone per instance, and the set functions satisfying them are exactly those under which the observed subset is optimal. Inferring an objective from observed optimal solutions is the classical problem of *inverse optimization* (Burton and Toint 1992; Ahuja and Orlin 2001; Heuberger 2004), and the zero-temperature loss above is the structured perceptron of structured prediction (Collins 2002; Taskar et al. 2003; Tsochantaridis et al. 2005). Seen this way, the setting is not new in kind, and it is worth being explicit about its nearest relatives. Max-margin learning of submodular set functions from reference summaries (Lin and Bilmes 2012; Sipos et al. 2012) is the $T \rightarrow 0$ version on a fixed function class; structured prediction energy networks (SPENs) place a deep energy on a structured output and train it through unrolled gradient-based inference (Belanger and McCallum 2016; Belanger et al. 2017). What EquiVSet adds is the combination: a permutation-invariant neural set function on a variable ground set, the maximum-entropy justification of the set mass function, amortized mean-field inference on the power set, and a copula that restores the correlations which mean field discards. The finite temperature is what turns the optimality constraints into a likelihood, and the likelihood into an objective that a gradient method can train.

Two further requirements shape the architecture. The set mass function must be **permutation invariant**, and it must accept a **varying ground set**, since every instance arrives with its own V . Both are met by parameterizing F_θ as a DeepSets network, $F(S) = \rho(\sum_{s \in S} \kappa(s))$, a form that is universal for permutation-invariant set functions under mild conditions (Zaheer et al. 2017). With minimum prior and **scalability** — training in polynomial rather than factorial time — these are the four desiderata of Ou et al. (2022), and scalability is where the difficulty lives.

5.4.2 Learning through mean-field inference: EquiVSet

The difficulty is the one this book keeps meeting. The pull-up term of Equation 5.8 is an expectation under $p_\theta(\cdot \mid V)$ over $2^{|V|}$ subsets; there is no continuous S on which to run Langevin dynamics, and running a discrete sampler (Grathwohl et al. 2021) or contrastive divergence inside the training loop is costly and, on large ground sets, unstable (Ou et al. 2022). EquiVSet takes a road absent from the table of Section 4.7: *approximate the model by a tractable*

family, and train through the approximation.

The tractable family is mean field — the product of independent Bernoullis $q(S; \psi) = \prod_{i \in S} \psi_i \prod_{i \notin S} (1 - \psi_i)$ with $\psi \in [0, 1]^{|V|}$, ψ_i the probability that element i belongs to the optimal subset. Minimizing $\text{KL}(q_\psi \parallel p_\theta)$ is maximizing the evidence lower bound,

$$\text{ELBO}(\psi) = f_{\text{mt}}^{F_\theta}(\psi) + H(q_\psi), \quad f_{\text{mt}}^{F_\theta}(\psi) := \sum_{S \subseteq V} F_\theta(S; V) \prod_{i \in S} \psi_i \prod_{i \notin S} (1 - \psi_i), \quad (5.9)$$

and the payoff term is again the **multilinear extension** (Owen 1972; Calinescu et al. 2007) — the object that carried the valuations of Section 5.3, now built from a learned F_θ rather than a given v . Read through Chapter 3, Equation 5.9 is minus a variational free energy at unit temperature, $\mathcal{F}(\psi) = \langle E_\theta \rangle_{q_\psi} - H(q_\psi)$, with the multilinear extension of the energy as the expected energy. The stationarity condition of the bound is the mean-field fixed point $\psi_i = \sigma(\partial_i f_{\text{mt}}(\psi))$, and the partial derivative has a form that makes it estimable:

$$\partial_i f_{\text{mt}}^{F_\theta}(\psi) = \mathbb{E}_{S \sim q(\cdot; \psi_{-i})} [F_\theta(S \cup \{i\}; V) - F_\theta(S; V)], \quad (5.10)$$

the expected *marginal gain* of i over a random subset of the other elements, drawn from q with ψ_i set to zero. That is the quantity read off as a player’s value in Section 5.3; here it is estimated by Monte Carlo and pushed through a sigmoid, and K such updates, run in parallel over coordinates, define a differentiable map $\psi^* = \text{MFVI}(\psi^{(0)}, V, K)$.

Learning then never touches $\log Z$. The parameters are trained by the cross-entropy of the observed optimal subset under the marginals the inference algorithm produces,

$$\mathcal{L}(\theta) = - \sum_{j \in S^*} \log \psi_j^*(\theta) - \sum_{j \in V \setminus S^*} \log (1 - \psi_j^*(\theta)), \quad (5.11)$$

with gradients back-propagated through the K fixed-point steps. This is Domke’s *marginal-based loss* (Domke 2013) — the training rule behind differentiable mean field for dense random fields and CRF-as-RNN (Krähenbühl and Koltun 2013; Zheng et al. 2015) — carried over to a deep energy on a variable ground set. In the terminology of the broader literature it is end-to-end training through *unrolled inference*, also termed deep unfolding: the iterations of an inference algorithm are treated as the layers of a network, and the loss is back-propagated through them (Gregor and LeCun 2010; Stoyanov et al. 2011; Domke 2012; Hershey et al. 2014; Belanger et al. 2017). Two properties should be stated plainly. The loss is not a bound on the log-likelihood of Equation 5.8: what is fit is the output of an approximate inference algorithm, which is a virtue (inference error is accounted for at training time) and a limitation (the link to the maximum-entropy distribution runs only through the fixed point). And the mean-field family cannot represent interactions between elements — the decoupling error of Section 5.3 returns as a modelling limit.

EquiVSet addresses cost and decoupling together. A permutation-*equivariant* recognition network $\psi^{(0)} = \text{EquiNet}(V; \phi)$ amortizes inference by proposing the initial marginals; it is trained to maximize Equation 5.9 (through a Gumbel-softmax relaxation of the discrete

samples), optionally with a Gaussian copula that correlates the Bernoullis. A single mean-field step from that initialization then carries the gradient to θ , and the two networks are trained cooperatively (Xie et al. 2018). At test time the same pipeline — proposal, one step, top- k rounding — returns a subset. A telling ablation: taking *more* mean-field steps from the copula initialization hurts, because the iteration converges to the plain mean-field fixed point and washes out the correlations the copula supplied.

On product recommendation (Amazon baby-registry categories), set anomaly detection (Double MNIST, CelebA), and compound selection from PDDBind and BindingDB, EquiVSet outperformed the probabilistic greedy model by large margins and, more tellingly, outperformed a DeepSets network trained directly on membership labels — the comparison that isolates the value of learning an explicit set function. The last application is the one that matters for Part III. A virtual-screening pipeline filters by bioactivity, diversity, and ADME in stages whose intermediate labels are costly or proprietary, but the final selected library is observed. That is an OS oracle and nothing else (Chapter 10).

5.4.3 Which statistics may the energy depend on? INSET

EquiVSet’s set function looked only at S . But a subset’s utility depends on the ground set it was drawn from: an image is anomalous *relative to* its batch, a bundle is chosen *from* what was displayed. INSET (Xie, Bian, et al. 2024) asks what an energy $F_\theta(S, V)$ is permitted to depend on, and derives the answer from symmetry. The conditional law the network must represent — that of the membership probabilities $Y \in [0, 1]^{|V|}$ given (S, V) — should be invariant under permutations of S and under the nested permutations of V viewed as a set of disjoint subsets (a wreath-product group). The theorem, proved through the noise-outsourcing lemma of probabilistic symmetry (Bloem-Reddy and Teh 2020), is that $P(Y \mid S, V)$ has this invariance if and only if $Y \stackrel{\text{a.s.}}{=} f(\xi, M(S, V))$ for independent noise ξ and a *maximal invariant* M — a statistic that is constant on the orbits of the group and separates them. Such an M is then an **adequate statistic**: conditioning on it loses nothing about Y . The invariant sufficient representations of S and of V may be built separately and combined, and each is approximated by a DeepSets sum, which gives the architecture

$$F_\theta(S, V) = \sigma\left(\theta_1 \sum_{x \in S} \phi(x) + \theta_2 \sum_{x \in V} \phi(x)\right). \quad (5.12)$$

The move is the one Section 5.2 attributed to Jaynes, with invariance in place of moment constraints: the constraints fix *which statistics* the distribution may depend on — there, clique indicators; here, the maximal invariant of the symmetry group — and leave the function of those statistics free for the data to determine. In the vocabulary of Section 5.1.1, invariance is the one structural constraint that can be stated on either side of the duality, as linear equality constraints on the distribution or as the requirement that the features factor through the maximal invariant, and INSET takes the second route. The two-branch design improved on EquiVSet across every benchmark (on Double MNIST the mean Jaccard index rose from 0.575 to 0.707), converged in fewer epochs, and an ablation shows the gain is not bought with parameter count.

5.4.4 What the strand adds to the maximum-entropy programme

Five points, and then what is open.

1. **The feature is learned, not the multipliers.** Classical maximum entropy fixes the distribution given the features; Section 5.3 read valuations off that distribution for a given game. Here the confinement of Section 5.1.1 collapses to a single feature — the utility itself — and that feature is a neural network trained from data. In the vocabulary of Section 5.1.4 this is a maximum-entropy *learner* on a discrete domain, with the coalition temperature τ of Section 5.3 reappearing as the learned scale of F_θ .
2. **Learning from observed optimal solutions.** The OS oracle is neither function values nor samples from the model; it is arg-max points of the energy, i.e., the supervision of inverse optimization. Maximum likelihood is the finite-temperature relaxation of the perceptron condition “the observed subset is a maximizer,” and the pull-up over 2^V is the price of smoothing it. The data constrain F only through its maximizers, so what is learned is a landscape with the right valleys, not a calibrated utility: with noiseless optimal subsets the likelihood alone would send the scale of F_θ to infinity, as in separable logistic regression, and regularization and early stopping are what pin it.
3. **A fourth road around the partition function.** Table 4.1 listed three ways to afford the pull-up: sample it, differentiate it away, cancel it against noise. EquiVSet’s road — relax the discrete distribution to a tractable family, unroll the inference, and train the marginals it outputs — is a fourth, with its own characteristic failure: the decoupling error of the family, and a loss that is not a likelihood bound. The bridge between 2^V and $[0, 1]^{|V|}$ is always the multilinear extension, which is why continuous DR-submodular maximization (Bian et al. 2019; Hassani et al. 2017) is the right theory for the guarantees. In Section 5.3 mean-field inference was an interpretation device; here it is the training loop.
4. **Constraints from symmetry.** INSET shows that the statistics an energy may depend on can be derived from the invariances of the conditional distribution — sufficiency and adequacy in the classical sense, reached by the same route as Jaynes’ clique features in Section 5.2. What the data are left to learn is the function of those statistics.
5. **The free energy.** The ELBO that drives these methods is $-\mathcal{F}(\psi)$ at $T = 1$ (Chapter 3), so variational inference on the learned energy is the minimization of $U - TS$ that runs beneath the whole book; and inference in the trained model — iterate toward the fixed point, then round — is energy-based reasoning on a discrete landscape in the sense of Chapter 8.

What is open is at least as interesting. A discrete EBM trainer under OS supervision whose objective *bounds* the likelihood does not yet exist; EquiVSet says so in as many words, and the candidates are gradient-informed discrete samplers for the pull-up (Grathwohl et al. 2021) and contrastive objectives against structured negatives (Section 4.6). Beyond mean field, structured variational families — the copula is a patch — would let the multi-step trajectory of Section 5.3 be used rather than avoided; that more steps *hurt* EquiVSet is a symptom worth diagnosing. Priors on the set function — diminishing returns (Bilmes and Bai 2017), bounded submodularity ratio (Bian et al. 2017) — would let the guarantees of continuous DR-submodular maximization apply to the learned F_θ rather than be assumed of it. And the setting generalizes: multisets

and integer lattices already have provable mean-field inference (Sahin et al. 2020); sequences, permutations, and graphs are the natural next domains for the same recipe — a discrete energy learned from its own optimizers. That recipe is what we mean by discrete maximum-entropy learning, and compound selection is only its first scientific case (Chapter 10).

The most immediate bottleneck is the Monte Carlo estimate in Equation 5.10, repeated at every unrolled step. A preprint of ours (Shi et al. 2026) removes it by learning the continuous relaxation directly — a neural set-function extension (Karalias et al. 2022) climbed by gradient ascent in place of the sampled fixed point — with a constant-factor guarantee under weak DR-submodularity and a reading of the ELBO as a free energy whose temperature is learned rather than fixed at one. It is not yet published, and we record it here as the direction the strand is taking rather than as a result.

5.5 Maximum-entropy reinforcement learning: energies over trajectories

Reinforcement learning (RL) supplies the largest instance of the family outside our own work, and the one that most closely resembles Section 5.4. We do not survey the field here. The aim of this section is to place maximum-entropy RL on the three levels of Section 5.1, and to show that inverse RL is the optimal subset oracle with trajectories in place of subsets.

5.5.1 A Gibbs distribution over trajectories

Consider a Markov decision process with states s , actions a , transition kernel $p(s' | s, a)$, and reward $r(s, a)$, and let $\tau = (s_1, a_1, \dots, s_H, a_H)$ denote a trajectory. The representation level is a Gibbs distribution over trajectories,

$$\begin{aligned} p(\tau) &= \frac{1}{Z} h(\tau) \exp\left(\frac{1}{\alpha} \sum_{t=1}^H r(s_t, a_t)\right), \\ h(\tau) &= p(s_1) \prod_{t=1}^{H-1} p(s_{t+1} | s_t, a_t) \prod_{t=1}^H u(a_t), \end{aligned} \tag{5.13}$$

with energy $E(\tau) = -\sum_t r(s_t, a_t)$, temperature α , and a base measure that is not uniform: h is the law of the trajectories generated by the dynamics under a uniform action distribution u , i.e., the *passive dynamics* of the system in the terminology of KL control (Todorov 2006, 2009; Kappen 2005). The base measure is what keeps the model from placing mass on physically impossible transitions. For deterministic dynamics h is uniform over the feasible trajectories, and Equation 5.13 is exactly the trajectory distribution of maximum-entropy inverse RL (Ziebart et al. 2008); for stochastic dynamics the same paper carries an explicit transition factor together with an approximation, and the principled repair is the *maximum causal entropy* discussed next (Ziebart et al. 2010).

5.5.2 Soft Bellman equations as free energies

The inference level restricts $\mathcal{Q}$ to the distributions an agent can actually realize, namely those induced by a policy under the true dynamics, $q_\pi(\tau) = p(s_1) \prod_t p(s_{t+1} | s_t, a_t) \pi(a_t | s_t)$. This restriction has a pleasant consequence. In $\text{KL}(q_\pi \| h)$ the transition factors cancel, and what remains is $\mathbb{E}_{q_\pi} [\sum_t \log \pi(a_t | s_t) - \log u(a_t)]$, i.e., minus the *causal entropy* of the policy (Ziebart et al. 2010) up to a constant. Minimizing the free energy Equation 5.2 over q_π is therefore

$$\max_{\pi} \mathbb{E}_{q_\pi} \left[\sum_t r(s_t, a_t) + \alpha H(\pi(\cdot | s_t)) \right], \quad (5.14)$$

which is the objective of maximum-entropy RL (Ziebart et al. 2010; Fox et al. 2016; Haarnoja et al. 2017). Its optimality conditions are the *soft* Bellman equations,

$$\begin{aligned} Q(s, a) &= r(s, a) + \mathbb{E}_{s' \sim p(\cdot | s, a)} [V(s')], \\ V(s) &= \alpha \log \sum_a \exp(Q(s, a)/\alpha), \\ \pi^*(a | s) &= \exp((Q(s, a) - V(s))/\alpha), \end{aligned} \quad (5.15)$$

and the soft value $V(s)$ is a log-partition function, i.e., a free energy at temperature α , one per state. As $\alpha \rightarrow 0$ one recovers the hard Bellman equations and the greedy policy, exactly as $T \rightarrow 0$ recovers energy minimization in Section 5.1.2. The reading of RL as inference in a graphical model, in which the policy-induced family plays the role of $\mathcal{Q}$, goes back to Todorov, Toussaint, and Kappen (Todorov 2009; Toussaint 2009; Kappen et al. 2012) and is reviewed by Levine (2018); the theory of entropy-regularized Markov decision processes states the same facts in the language of convex regularization (Neu et al. 2017; Geist et al. 2019). When the reward is given, as in soft Q-learning and the soft actor-critic (Haarnoja et al. 2017, 2018), this is the whole story: a maximum-entropy model with variational inference and no learner, i.e., the same class as the cooperative games of Section 5.3, with α a genuine dial. The KL-regularized policy optimization of Equation 6.1 is the same construction on a single decision, with a reference policy as the base measure and the KL coefficient as the temperature; imposing the KL as a constraint instead, as trust-region methods do (Schulman et al. 2015), is the constrained form of the same problem. Chapter 9 develops this reading for the post-training of pretrained models.

5.5.3 Inverse reinforcement learning: the optimal subset oracle on trajectories

The learning level becomes active when the reward is unknown and one observes expert trajectories τ^* instead, which is the problem of inverse RL (Ng and Russell 2000). Maximum-entropy inverse RL fits a parametric reward r_θ by maximum likelihood under Equation 5.13. For a linear reward $r_\theta(s, a) = \theta^\top \phi(s, a)$, the likelihood equations Equation 5.5 read

$$\mathbb{E}_{\text{expert}} \left[\sum_t \phi(s_t, a_t) \right] = \mathbb{E}_{p_\theta} \left[\sum_t \phi(s_t, a_t) \right], \quad (5.16)$$

i.e., the learned reward makes the model reproduce the expert’s expected feature counts. This is the *feature-expectation matching* that apprenticeship learning had imposed directly (Abbeel and Ng 2004), and the model expectation is computed by the soft Bellman recursion of Equation 5.15, i.e., the inference of the second level runs inside the training loop. At zero temperature the same objective becomes a max-margin structured prediction problem over policies, termed maximum margin planning (Ratliff et al. 2006).

The parallel with Section 5.4 is now line by line, and Table 5.2 records it. In both cases the supervision consists of *observed optimal solutions* of an unknown objective: a chosen subset in one case, a demonstrated trajectory in the other. In both cases the finite temperature turns the optimality constraints into a likelihood, the zero-temperature limit is a perceptron or max-margin loss, the inner problem is a free-energy minimization over a tractable family (mean field on the multilinear extension; soft value iteration over policies), and the learning signal is the difference between an expert statistic and a model statistic. The difference lies in the base measure and in the family: the power set has no dynamics and h is uniform, whereas trajectories are constrained by the transition kernel, which is why the causal entropy replaces the Bernoulli entropies of Equation 5.9. We regard this parallel as the strongest external evidence that “learning an energy from its own optimizers” is a definition and not the description of one paper.

Table 5.2: The optimal subset oracle and maximum-entropy inverse RL, read through the same three levels.

	Optimal subset oracle (Section 5.4)	Maximum-entropy inverse RL
Observed optimal solution	subset $S^* \subseteq V$	trajectory τ^*
Energy to be learned	$-F_\theta(S; V)$	$-\sum_t r_\theta(s_t, a_t)$
Base measure h	uniform on 2^V	passive dynamics
Constrained reading	optimality constraints (inverse optimization)	feature-expectation matching (Abbeel and Ng 2004)
Zero-temperature limit	structured perceptron; max-margin submodular learning (Lin and Bilmes 2012; Sipos et al. 2012)	maximum margin planning (Ratliff et al. 2006)
Inner problem	mean field on the multilinear extension	soft value iteration
Entropy that does the work	Bernoulli entropies of q_ψ	causal entropy of π
Learning signal	marginal-based loss through unrolled steps	expert minus model feature counts

5.6 What the family adds, and where it stops

Having placed the strands, we close the chapter with three general facts that the definition of Section 5.1 makes available, and with a statement of what the definition excludes.

5.6.1 Maximum likelihood as a saddle-point problem

For the Gibbs model $p_\theta = h e^{-E_\theta/T} / Z(\theta)$, the variational representation $-T \log Z(\theta) = \min_q \mathcal{F}_T(q; E_\theta)$ of Equation 5.2 turns the negative log-likelihood into

$$\begin{aligned} T \cdot \text{NLL}(\theta) &= \mathbb{E}_{\hat{p}}[E_\theta] + T \log Z(\theta) \\ &= \max_q \left\{ \mathbb{E}_{\hat{p}}[E_\theta] - \mathbb{E}_q[E_\theta] - T \text{KL}(q \| h) \right\} + \text{const}, \end{aligned} \quad (5.17)$$

so maximum likelihood is $\min_\theta \max_q$: a saddle-point, or primal–dual, problem in which the inner player is the inference of Equation 5.3 and the outer player shapes the energy (Kim and Bengio 2016; Dai et al. 2019). This is the form in which the pull-up of Equation 4.5 appears as a variational player, and it is also the right place to locate the estimators of Table 4.1. Maximum likelihood with MCMC samples from the inner maximizer; contrastive divergence truncates the chain that would reach it; the fourth road of Section 5.4 solves the inner problem over a tractable family; score matching and noise-contrastive estimation do something different in kind — they replace the Kullback–Leibler divergence behind Equation 5.17 by another divergence — and we return to them in Section 5.6.3. We call Equation 5.17 the *saddle-point form* and we do not call it minimax, since that word is reserved for the different nesting of Chapter 7.

5.6.2 Maximum entropy inside, minimum entropy outside: the bridge to minimax entropy

Let the energy be linear in its last layer, $E_\theta(x) = -w^\top \phi_{\theta'}(x)$, as is the case for every network whose output is a linear read-out of its penultimate features, e.g., the last layer of ρ in a DeepSets set function. For fixed θ' , the maximum-likelihood w solves the Jaynes problem Equation 5.1 with $\mathbf{T} = \phi_{\theta'}$ and μ the empirical moments, and the likelihood equations in w state that $\mathbb{E}_{\hat{p}}[\phi_{\theta'}] = \mathbb{E}_{p_\theta}[\phi_{\theta'}]$. At such a point, and for uniform h ,

$$\begin{aligned} \mathbb{E}_{\hat{p}}[-\log p_\theta] &= -w^\top \mathbb{E}_{\hat{p}}[\phi_{\theta'}] + \log Z \\ &= -w^\top \mathbb{E}_{p_\theta}[\phi_{\theta'}] + \log Z = H(p_\theta), \end{aligned} \quad (5.18)$$

i.e., the cross-entropy of the data under the model equals the entropy of the model itself. Consequently, minimizing the negative log-likelihood over the feature parameters θ' is minimizing the entropy of the maximum-entropy model built on the features $\phi_{\theta'}$:

$$\min_{\theta'} \text{NLL}(\theta', w^*(\theta')) = \min_{\theta'} H(p_{\theta', w^*(\theta')}).$$

This is the minimax entropy principle of Chapter 7, with a continuous family of features and with gradient descent in place of greedy selection. The incremental version is classical: in feature induction for random fields, the gain of adding a feature is the reduction of the model’s entropy, which equals the reduction of the KL divergence to the data (Della Pietra et al. 1997), and in the FRAME model of texture the filters are selected by the largest entropy drop (Zhu et al. 1998). Replacing “select a filter from a bank” by “take a gradient step on the filter parameters”

is what a deep energy model does (Ngiam et al. 2011). The vision lineage of energy-based models describes itself in exactly these terms: the generative ConvNet is an exponential tilting of a Gaussian reference distribution by a ConvNet, and its authors present it as a hierarchical FRAME model, i.e., a maximum-entropy model whose features are learned filters (Xie et al. 2016). In this precise sense, every energy-based model trained by maximum likelihood is a minimax-entropy learner. The hypotheses should be kept in mind — a linear last layer (or a family closed under rescaling), an exact fit of w , and a uniform base measure — and Chapter 7 studies the explicit, budgeted, combinatorial version of the outer problem, which is where the open questions lie.

5.6.3 Where the family stops

The definition also says where the family stops, and which of its neighbours sit on which side of the line. We go through them in turn, starting with the estimators of Chapter 4, since the answer there is not the same for every estimator.

- Contrastive divergence (Hinton 2002), its persistent variant (Tieleman 2008), and short-run MCMC (Nijkamp et al. 2019) are maximum-entropy learners in the strict sense. They approximate the inner maximization of Equation 5.17 by sampling from, or truncating, a Markov chain whose target is the maximum-entropy distribution, so the entropy term is carried by that target. The truncated versions share the structure and the caveat of Section 5.4.2: what is fit is the output of a truncated inference procedure, the update is biased and in general not the gradient of any objective (Carreira-Perpiñán and Hinton 2005; Bengio and Delalleau 2009; Sutskever and Tieleman 2010), and the learned energy need not be a valid density (Nijkamp et al. 2020). In this respect the optimal subset oracle and CD- k stand or fall together, and we keep both inside the family.
- Score matching (Hyvärinen 2005) and noise-contrastive estimation (Gutmann and Hyvärinen 2010) fit the same Gibbs model by minimizing a different divergence, and we call them maximum-entropy learners *in the generalized sense*. The construction behind the word is that of proper scoring rules. Every proper scoring rule S defines a generalized entropy $H_S(p) = \mathbb{E}_p[S(x, p)]$ and a Bregman divergence $d_S(p, q) = \mathbb{E}_p[S(x, q)] - H_S(p)$, and maximizing H_S under constraints is dual to minimizing the worst-case expected score, exactly as for the Shannon entropy and the log score (Grünwald and Dawid 2004; Dawid 2007). Score matching minimizes the Fisher divergence, which is d_S for the Hyvärinen score — the basic second-order local proper scoring rule, computable without the normalizing constant (Parry et al. 2012) — and the corresponding generalized entropy is $-\frac{1}{2} \mathbb{E}_p \|\nabla_x \log p\|^2$, i.e., minus one half of the Fisher information. Score matching is therefore a *minimum-Fisher-information* learner, a principle with its own history in physics (Frieden 1990). Its bridges to the strict case are quantitative: it is the infinitesimal-step limit of Langevin contrastive divergence (Hyvärinen 2007), the derivative of the KL divergence under Gaussian smoothing (Lyu 2009), and, with likelihood weighting in diffusion models, an upper bound on the negative log-likelihood (Song, Durkan, et al. 2021). Noise-contrastive estimation is the maximum-entropy classifier of Section 5.1.4 with the energy as its single learned feature; it is consistent, and for normalized models

it becomes asymptotically as efficient as maximum likelihood as the noise-to-data ratio grows (Gutmann and Hyvärinen 2012). The constant it estimates in place of $\log Z$ plays the role of a free-energy difference between the model and the noise distribution, and the estimator is, up to the roles of data and noise samples, Bennett’s acceptance-ratio method of statistical physics, which is itself a logistic-regression maximum-likelihood estimator (Bennett 1976; Shirts et al. 2003). Both estimators, together with ratio matching, belong to one Bregman-divergence framework for unnormalized models (Gutmann and Hirayama 2011), and the Kullback–Leibler divergence of maximum likelihood is itself a Bregman divergence. Their derivations remain in Chapter 4.

- The zero-temperature losses — perceptron, hinge, maximum margin planning — are the $T \rightarrow 0$ limits of the family and lie on its boundary: the entropy term has been switched off.
- Directed graphical models fitted by maximum likelihood or EM are learners at levels ii)–iii): the E-step is the I-projection of Equation 5.1, and the free-energy view of EM is that of Section 4.8. But their model class is not a linear exponential family (Section 5.2), so they are maximum-entropy learners without being maximum-entropy models.
- Minimum-entropy learning (Chapter 6) uses the same inner Gibbs tilt (Equation 6.2) but reverses the outer criterion; it is excluded by
 - ii) by design, and the two chapters are the two directions of the dial of Chapter 3.
- Entropy bonuses added to an arbitrary objective as a heuristic, without a Gibbs model behind them, are in the penumbra: they become maximum-entropy learning exactly when the objective can be written as Equation 5.2.

Two remarks connect the definition to finite samples and to decision theory. When the moments μ are estimated from data, relaxing the constraints of Equation 5.1 to a box $|\mathbb{E}_q[\mathbf{T}] - \hat{\mu}| \leq \beta$ is dual to ℓ_1 -regularized maximum likelihood, so constraint slack and regularization strength are one quantity (Dudík et al. 2007; Altun and Smola 2006). And the maximum-entropy distribution is the minimax strategy of a zero-sum game against nature under logarithmic loss (Topsøe 1979; Grünwald and Dawid 2004), which gives the principle a decision-theoretic rationale that is independent of Jaynes’ informational one.

5.7 Takeaway

Maximizing entropy reframes choosing a distribution over coalitions as the process of specifying a constraint; mean-field inference on this distribution transforms the question from “which valuation should I adopt?” to “how far along the mean-field path should I iterate?”—placing Shapley, Banzhaf, and related values on a continuous spectrum rather than as disconnected axioms. This echoes the main probabilistic graphical models insight from earlier in the chapter: a strictly positive MRF corresponds to a maximum-entropy model with structured constraints, and it is the mean-field relaxation that unifies different valuation principles as points along a single trajectory.

If we no longer assume the game is given, the same framework becomes a learning algorithm: we model a Gibbs distribution over subsets, where the energy is a neural set function, and

learn this energy directly from observations of optimal subsets—by differentiating through mean-field inference on the multilinear extension. This is maximum-entropy learning as defined in Section 5.1, but now on a discrete space with an energy function discovered (not specified) from data. Here, the ELBO becomes the free energy to optimize, and the resulting energy landscape is itself shaped by the training data—a direct example of the chapter’s central idea.

Read through the three levels of Section 5.1, the chapter has one shape. Every strand shares the representation and inference levels—a Gibbs model and a free-energy minimization—and the strands differ in whether the learning level is active and in what supervises it. The cooperative games and maximum-entropy RL with a given reward stop at inference; the optimal subset oracle and inverse RL learn the energy from observed optimal solutions, and the two are parallel line by line (Table 5.2). Finally, by the identity Equation 5.18, maximum likelihood over the features of any such model is already a minimax-entropy problem, which is related to Chapter 7.

6 Minimum-Entropy Learning

A model that answers the same question five different ways is telling us something. This chapter is about turning that signal — a model’s disagreement with itself, measurable without any ground truth answer — into a training objective, and about what such an objective can and cannot deliver.

The true result is more limited than people often say, and it is important to say this clearly from the beginning. Minimising the entropy of a model’s own answers is a strikingly effective way to *elicit* capability the model already has: on mathematics it matches verifier-based reinforcement learning while using no labels at all. It cannot add capability, and that is not a defect of present implementations but a consequence of a training loop that never touches ground truth (Section 6.7). Its characteristic failure is equally structural. Left unchecked, the objective drives the model to a single confident answer, right or wrong; Section 6.6 shows that this is not an implementation nuisance to be tuned away but a first-order transition in a mean-field free energy — an abrupt jump rather than a gradual drift — with a computable critical point and, whenever three or more distinct answers compete, hysteresis: a memory effect that makes the jump one-way. That is why the methods that survive their own objective all hold the entropy above a floor — by reset, by fairness term, or by filter — and why no choice of regularisation strength can be made to serve instead. In the implementation this chapter examines most closely, the floor currently sits at zero — a configuration its own ablations arrived at, and one whose published runs completed stably. Section 6.6 argues there is no contradiction in that, once the floor’s role is named correctly: it is a stopping rule that decides how much of the model’s diversity sharpening spends, not a stability device, and in the published runs a cruder stopping rule — the finite training budget — was doing its work.

Three cautions run through what follows. The name is unfortunate: minimum-entropy learning is a family of *objectives* and has nothing to do with the Rényi min-entropy H_∞ of Chapter 2. The physics is routinely oversold: Schrödinger’s “order from order” turns out to be a genuine and clarifying reading of what these methods do, whereas Prigogine’s minimum-entropy-production principle does not survive contact with the details, and Section 6.9 says why in more detail. And the history is longer than the modern literature usually recounts — the objective and the term that balances it appear side by side as early as 1991, and the pairing has since been arrived at independently in several communities (Section 6.3).

6.1 Two regimes, and a warning about the name

The maximum-entropy principle of Section 2.3 answers a question about *honesty*: given that we have measured a few things and nothing else, which distribution commits us to the least? The

minimum-entropy principle answers a question about *commitment*, and it applies in a regime the first one never contemplates. Suppose the distribution in front of us is not a summary of our ignorance but the output of a model that has already absorbed an enormous amount of information. Its dispersion is then no longer a virtue to be preserved; it is a symptom. A model that answers the same question five different ways is telling us something, and what it is telling us can be turned into a training signal.

The two principles are therefore not rivals. They act on different objects at different moments: maximum entropy shapes the distribution we *assume*, minimum entropy sharpens the distribution a trained model *produces*. A system that only maximises entropy generates candidates forever and never decides; one that only minimises it becomes rigid and confidently wrong. This chapter mainly discusses the second failure, as it is the main problem for the methods covered here.

Before going further, a terminological issue has to be cleared. Chapter 2 introduced the **min-entropy** $H_\infty(p) = -\log \max_x p(x)$, the $\alpha \rightarrow \infty$ member of the Rényi family, used in cryptography to bound an adversary’s first-guess probability. That quantity has nothing to do with the subject of this chapter. Here, “minimum-entropy learning” names a class of *objectives* — procedures that reduce the entropy of a model’s predictive distribution — not a particular entropy functional. The collision is unfortunate and it is now standard in both literatures, so it is best met head on. As it happens, the connection is not quite empty: Section 6.5 shows that the objective actually implemented by the method at the centre of this chapter is the Rényi entropy at $\alpha = 2$, a different member of the same family.

6.2 From word-level to meaning-level uncertainty

Turning “the model contradicts itself” into a number requires deciding what counts as a contradiction, and the obvious choice fails immediately. Ask a language model for the capital of France five times and it may return *Paris*, *It is Paris*, *The capital of France is Paris*, *Paris, France*, and *Paris — it has been since 987*. These are five distinct token sequences. If we calculate entropy at the token level, based on the distribution over next words, it might suggest high uncertainty—even when, in reality, the model is certain. For example, if you ask which paper first documented an obscure phenomenon, the model might return five distinct and seemingly plausible citations, each actually incompatible with the others. In such cases, the token-level entropy could be the same as, or even lower than, situations where the correct answer is always given. This happens because the model generates convincing but fabricated responses with confidence on a word-by-word basis.

The solution is to group together outputs with the same meaning before making any measurement. Let $\sim$ be an equivalence relation on responses that identifies those with the same meaning, and let $\mathcal{C}$ be the resulting set of clusters. Sampling G responses $o_1, \dots, o_G$ from the model for a prompt q and writing $\hat{p}_c = |c|/G$ for the empirical frequency of cluster c , the **semantic entropy** is

$$H_{\text{sem}}(q) = - \sum_{c \in \mathcal{C}} \hat{p}_c \log \hat{p}_c.$$

This is Shannon’s formula applied to a coarser σ -algebra, which is the whole of the idea. In the Paris case all five responses fall into one cluster and $H_{\text{sem}} = 0$. The measurement now tracks what we care about, and the repetition exploit disappears, because repeating oneself does not create a new cluster.

Semantic entropy was introduced for uncertainty quantification (Kuhn et al. 2023) and reached wide attention as a hallucination detector (Farquhar et al. 2024): resample an answer ten times, cluster by meaning, and flag the high-entropy cases before a user sees them. Its appeal is that the signal is available with no answer key, no reference database, and no change to the model — which makes it deployable exactly where verification is hardest. But it is only a detector. It says when not to trust an answer; it does not produce a better one. The rest of this chapter concerns the obvious next question: if this uncertainty can be measured, can it be optimised?

6.3 Entropy minimisation as an objective: a short history

The idea of adding an entropy term to a loss and descending it is not new, and its history is worth tracing, because the same pair of ideas has surfaced independently in several communities, under different names and with different emphases — a pattern that is common wherever one piece of mathematics serves problems with different constraints. Everything the next sections struggle with has been met before.

Both terms are already present in 1991. Bridle et al. (1992) asked how to train a classifier on unlabelled data alone, and proposed maximising the mutual information between the input and the predicted class. Decomposed, that objective reads

$$I(c; x) = \underbrace{H(\bar{y})}_{\text{fair}} - \underbrace{H(y)}_{\text{firm}},$$

where y is the predictive distribution for one input and $\bar{y}$ its average over the dataset. The second term is entropy minimisation exactly as it is practised today: make each individual prediction decisive. The first is the thing that stops the second from being satisfied by fraud, since a classifier that answers “class 3” to everything is perfectly decisive and carries no information at all. Bridle and his co-authors state the point in so many words, observing that it is trivial to achieve either desideratum alone, and their summary of what a working objective must be — *firm but fair* — remains the clearest available description of what is at stake.

The two terms then travelled separately. Grandvalet and Bengio (2004) brought entropy minimisation to semi-supervised classification: augment the likelihood on labelled data with $-\lambda \sum_{x \in \mathcal{U}} H(p_\theta(y | x))$ over the unlabelled set, pushing the classifier towards confident predictions on data it has never been told the answer for. The justification is the cluster assumption — that decision boundaries should fall in low-density regions — and under it, confident predictions on unlabelled points are evidence of a boundary in the right place. It is the firm term on its own, and the setting explains why that is a reasonable formulation there: the labelled likelihood anchors the classifier to all K classes, so the firm term can stand alone in a way it cannot in the fully unsupervised case, and a fairness term has no obvious work to do. This anchored formulation became the standard reference point for the modern literature, and

pseudo-labelling (Lee 2013) — whose author explicitly identifies it with entropy regularisation — operates in the same anchored regime. The fairness term returned to wide use in 2020, when Berthelot et al. (2020) introduced it under the name *distribution alignment* and traced the idea back to Bridle’s mutual-information objective, observing that it had been proposed more than twenty-five years earlier and had, to their knowledge, not been in use in contemporary semi-supervised learning. Read as a whole, the record is less a story of anything being lost than of the two terms mattering in different regimes: as long as a supervised anchor was present, the firm term alone was enough; the fair term becomes essential exactly when the anchor is removed — which is the situation of this chapter.

What the field eventually converged on is not one fix but a three-term structure. Drawing on what has proven effective, any objective for unlabelled data must include all of the following components, with each omission leading to its own characteristic failure:

term	what it buys	what its absence costs
conditional entropy down — <i>firmness</i>	decisive predictions	predictions stay diffuse and carry no usable signal
marginal entropy up, or an equipartition constraint — <i>fairness</i>	use of the whole label space	everything collapses onto one class
a capacity, smoothness or invariance constraint	classes that mean something	the partition fragments, or becomes balanced and arbitrary

That two of the three do not suffice is not a conjecture. Gomes et al. (2010) demonstrate both failures side by side on data constructed to expose them: dropping fairness lets decision boundaries vanish and clusters disappear, dropping the capacity term shatters the data into a large number of categories. The equipartition methods make the third failure vivid — a constant encoder paired with a uniform code satisfies a balance constraint *exactly*, which is why Asano et al. (2020) and Caron et al. (2020) must couple it to an augmentation-invariance objective. And Caron et al. (2021) separate the two collapse attractors quantitatively: without centring the output entropy falls to zero, without sharpening it rises to $\log K$, and only the two together hold a representation in between.

One mechanism, six notations. The most striking feature of this literature is how many communities arrived at the fairness term independently — as a bonus in the loss, as a reweighting of the pseudo-label, as an edit to the logits, and as a constraint on the assignment matrix. These are objects of four different kinds, and it is worth saying precisely in what sense they are the same thing, because the ways in which they are *not* turn out to matter.

Fix the notation for the rest of this section. A network maps an input x to logits $z_c(x)$ over K classes, giving predictions $p_c(x) = \text{softmax}(z(x))_c$, and over a batch or dataset of N inputs the **class marginal** is the average prediction

$$\bar{p}_c = \frac{1}{N} \sum_{n=1}^N p_c(x_n), \quad \sum_c \bar{p}_c = 1.$$

π_c denotes the **target class distribution** for class c —that is, the desired proportion of examples

assigned to each class. In most typical cases, we want all classes to be equally represented, so π_c is set to the uniform distribution: $\pi_c = 1/K$, where K is the total number of classes. In some scenarios, prior knowledge may justify setting a non-uniform target distribution.

Collapse is the statement that $\bar{p}$ concentrates on one class; fairness is any mechanism that resists it. The unifying observation is that all six act on the predictions in the same way:

i Each replaces the logits $z_c(x)$ by $z_c(x) - \gamma_c$ for a **per-class offset** γ_c increasing in how over-represented class c currently is. What distinguishes them is how γ_c is estimated, and how far the resulting marginal is driven towards its target.

mechanism	offset γ_c	enforcement	acts on
marginal-entropy bonus $+\lambda H(\bar{p})$	$\propto p_c(x)(\log \bar{p}_c - \mathbb{E}_{p(x)} \log \bar{p})$	soft penalty, one gradient step	the loss
$-\lambda D_{\text{KL}}(\bar{p} \parallel \text{Unif})$	as above	soft penalty, one gradient step	the loss
distribution alignment	$\log \bar{p}_c - \log \pi_c$	one Sinkhorn step	the pseudo-label
logit debiasing	$\lambda \log \bar{p}_c$	one Sinkhorn step ($\lambda = 1$)	the logits
centring	$\bar{z}_c$, a moving average	one step, on a lagged estimate	the teacher's logits
Sinkhorn equipartition	the γ making $\bar{p}$ exactly uniform	hard constraint, at convergence	the assignments

Reading in order: the marginal-entropy bonus of Bridle et al. (1992), Hu et al. (2017) and Liang et al. (2020); the same quantity written as a divergence from uniform; the distribution alignment of Berthelot et al. (2020); logit debiasing (Menon et al. 2021; Wang et al. 2022); the centring operation of Caron et al. (2021); and the Sinkhorn equipartition constraint of Asano et al. (2020) and Caron et al. (2020).

Rows three, four and six are the same operation at different stopping times. With a uniform target $\pi_c = 1/K$, distribution alignment reweights the prediction to $\tilde{q}_c \propto p_c(x)/\bar{p}_c$, which in logits is $z_c(x) - \log \bar{p}_c$ up to a per-example constant — logit debiasing at $\lambda = 1$. And if the predictions are arranged into the matrix $Q_{cn} = p_c(x_n)/N$, whose columns already sum to $1/N$, its row sums are exactly $\bar{p}_c$, so the first row-normalisation of the Sinkhorn iteration towards $U(\frac{1}{K}\mathbf{1}, \frac{1}{N}\mathbf{1})$ rescales row c by $1/(K\bar{p}_c)$: the same offset once more. **Logit debiasing is one Sinkhorn step; Sinkhorn is logit debiasing run to convergence.**

This distinction is important, and it is a common source of confusion. One debiasing step does not make the marginal uniform. It makes the *row* sums uniform, but the per-example softmax that follows is a column renormalisation, and that undoes part of the correction — which is precisely why Sinkhorn alternates rather than stopping. The residual shrinks by a roughly constant factor of about three per pass: with $K = 10$ classes and $N = 256$ examples drawn from a moderately confident network, the total-variation distance from uniform runs $0.060 \rightarrow 0.020 \rightarrow 0.0064 \rightarrow 0.0021$, falling to about 10^{-7} after a dozen passes and to machine

precision after about thirty. A single step therefore buys most of the correction and none of the guarantee, which is the operative difference between a method that *discourages* collapse and one that *forbids* it. Row two, by contrast, is identical to row one: $D_{\text{KL}}(\bar{p} \parallel \text{Unif}) = \log K - H(\bar{p})$, so the two objectives differ by a constant and have the same gradient everywhere.

The remaining two relations are approximations, and both gaps have a definite sign. The gradient of $H(\bar{p})$ is *not* the flat offset $\lambda \log \bar{p}_c$: it carries a factor $p_c(x)$, so it acts on each example only through the classes that example is already predicted to belong to, and it is centred, so within an example it merely redistributes. The hand-designed offset applies the full correction to every example unconditionally. They agree on sign and on monotonicity in $\bar{p}_c$, and on little else — which is also why the two are calibrated differently: $\lambda = 1$ is a canonical, tuning-free setting for the logit correction, being exactly the Bayes adjustment that maps a model with implicit prior $\bar{p}$ onto a uniform one (Menon et al. 2021), whereas the entropy bonus has no distinguished coefficient and must be tuned.

Centring differs along another axis entirely. Subtracting the mean *logit* $\bar{z}_c$ is, because $\log p_c(x) = z_c(x) - \log Z(x)$ and the log-partition is common to all classes, the same as subtracting $\overline{\log p_c}$ — the logarithm of the **geometric** mean of the predictions, where debiasing subtracts the logarithm of the **arithmetic** mean:

$$\begin{aligned} \gamma_c^{\text{debias}} &= \log \left(\frac{1}{N} \sum_n p_c(x_n) \right) \\ &\geq \frac{1}{N} \sum_n \log p_c(x_n) = \gamma_c^{\text{centring}} - \frac{1}{N} \sum_n \log Z(x_n), \end{aligned}$$

by Jensen, with equality only if p_c is constant across the dataset; the trailing term is the same for every class and so invisible to the softmax. The gap widens with the variance of p_c over examples, so centring discounts classes that dominate a few inputs strongly and penalises only classes predicted uniformly often — a weaker and smoother intervention, which is consistent with Caron et al. (2021) needing to pair it with sharpening rather than with an explicit capacity term.

i Gradient of the marginal-entropy bonus

Differentiating $H(\bar{p}) = -\sum_c \bar{p}_c \log \bar{p}_c$ through $\bar{p}_c = \frac{1}{N} \sum_n p_c(x_n)$ and the softmax Jacobian $\partial p_c / \partial z_j = p_c(\delta_{cj} - p_j)$ gives, for one example x ,

$$\begin{aligned} \frac{\partial H(\bar{p})}{\partial z_j(x)} &= \frac{1}{N} \sum_c (-\log \bar{p}_c - 1) p_c(\delta_{cj} - p_j) \\ &= -\frac{p_j(x)}{N} \left(\log \bar{p}_j - \sum_c p_c(x) \log \bar{p}_c \right), \end{aligned}$$

the -1 terms cancelling because $\sum_c p_c = 1$. Ascending H therefore *subtracts* this quantity from the logits, which is the offset in the first row of the table. Two features are worth noting. The bracket has p -weighted mean zero, so the update is a pure redistribution within each example; and the prefactor $p_j(x)$ vanishes for classes the example is confidently not in, so an example already committed elsewhere contributes almost nothing to correcting

an over-represented class. Neither property is shared by $\gamma_j = \lambda \log \bar{p}_j$.

Six communities, six notations, one mechanism — with the caveat that the two notations most often treated as interchangeable, the entropy bonus and the logit correction, are the two that are genuinely distinct.

A second and orthogonal question — what the penalty does *at* the boundary of the simplex — matters for what follows, and appears not to be remarked on anywhere. Tanaka et al. (2018) penalise $D_{\text{KL}}(\pi \parallel \bar{p})$, a fixed prior π against the marginal prediction $\bar{p}$ — with the prior in the *first* slot, so the term *diverges* as any class’s marginal mass approaches zero: an infinitely high barrier that no finite reward can buy through. The marginal entropy $H(\bar{p})$ is bounded on the simplex, so collapsing costs at most $\log K$, and a sufficiently strong firmness term will simply pay it. One is a hard floor and the other a soft one, and the distinction decides whether collapse is impossible or merely expensive.

Test-time adaptation is the closest analogue to the unsupervised setting. When a deployed network adapts to a shifted input distribution by minimising the entropy of its own predictions on the incoming stream (Wang et al. 2021), the labelled anchor is gone and collapse stops being hypothetical. Niu et al. (2023) devote a subsection to it, under a heading that says all that needs saying — online entropy minimisation tends to predict all samples to the same class. Their measurements are the strongest empirical statement available that this objective can be actively harmful when nothing holds it — accuracy on ImageNet-C at the highest corruption severity, averaged over all fifteen corruptions, with a batch-normalised ResNet-50:

stream	no adaptation	Tent	EATA
imbalanced label order, batch 64	18.0	2.1	0.9
balanced labels, batch size 1	18.0	0.1	0.1

Adaptation by entropy minimisation takes a network from 18% to a fraction of one per cent — far worse than leaving it alone. Both of these are deliberately adverse streams rather than the standard benchmark, which is the point: the objective is fine until the conditions that were implicitly propping it up are withdrawn. Note also that the weight regularisation of Niu et al. (2022) does not rescue it here, though it does help on backbones whose normalisation is not batch-dependent, which is itself evidence that anchoring is a weaker remedy than a structural one.

Two details of the fix Niu et al. (2023) propose are worth carrying forward. The first is that they monitor a moving average of the entropy loss and *reset the model* when it falls below a threshold, set to 0.2 — an entropy floor placed strictly inside the simplex, implemented as a discrete intervention rather than a filter. The same object returns in Section 6.4 with its role transformed: here it is a stability device rescuing a network from destruction; there, where the degenerate endpoint turns out to be quiet rather than catastrophic, a floor would serve as a stopping rule — and Section 6.6 argues that distinction is the key to reading its current setting of zero correctly. The second is that in their diagnostic experiment, on a single corruption with a group-normalised backbone, the model collapses at severity level five while at level three

the gradients stay in a stable range throughout: the behaviour changes sharply as a control parameter is turned. That is the closest thing in the empirical literature to the critical value derived in Section 6.6, though it should not be read as a general safe threshold — under mixed corruptions the same backbone collapses at level three as well. Running the other way, Lee et al. (2024) argue that entropy is not a trustworthy reliability signal in the first place, since under spurious correlation confidently wrong samples are exactly the harmful ones — a result that weakens the entire filtering family and, by elimination, strengthens the case for structural remedies.

The language-model literature has adopted two of the three families and not the third. Reinforcement learning from internal feedback has taken up filtering — the two-sided entropy band described in Section 6.4, the substitution of self-certainty for entropy in Zhao et al. (2025) — and anchoring, in the form of covariance-targeted clipping and KL penalties (Cui et al. 2025). A fairness term is absent: no work in this line maximises a dataset-level marginal entropy. Part of the reason is a genuine type mismatch. The answer space of a language model is prompt-dependent, so “cluster 1” for one question has nothing to do with “cluster 1” for another and the marginal $\bar{p}$ has no well-defined referent. Candidate substitutes exist — a prompt-independent output attribute such as length, format or refusal rate; a population-level coverage or pass@ k target; or a floor on cluster mass within each prompt, which is what δ_{low} would approximate if it were set anywhere above zero — but none has been tried. That the gap is real is confirmed by what happens without them: in the plain entropy-minimisation baseline that Zhao et al. (2025) run against their own method, the model “converges to producing the same character regardless of the prompt” after a few updates. That is the vision literature’s collapse, arriving in the language literature under a different name and from the same cause.

What is new in the method described next is precisely the removal of the supervised anchor. There is no labelled term at all. This is what makes the approach interesting, and it is also what returns the degeneracy in full force — a point developed in Section 6.6, where it turns out to be sharper than a mere risk of a bad optimum.

6.4 EMPO: reward agreement, not correctness

EMPO — entropy-minimised policy optimisation — applies the objective to the elicitation of reasoning in large language models, in a fully unsupervised setting (Zhang et al. 2025). The context is the reinforcement-learning recipe behind recent reasoning models (Shao et al. 2024; DeepSeek-AI 2025): sample several attempts at a question, score each against a ground-truth answer with an automatic verifier, and reinforce the winners,

$$r_i = \mathbb{1}[\text{verify}(o_i, a) = \text{true}].$$

The recipe works remarkably well, but the bottleneck sits entirely in the symbol a . Every training question needs a gold answer and a cheap, reliable checker. Mathematics has one — compare the final expression — and code has a test suite. A differential diagnosis does not; neither does a proposed mechanism, a study design, or a molecule nobody has synthesised. The algorithm is not the scarce resource. The verifier is.

EMPO removes a from the loop. Given an unlabelled question q , it samples G responses, partitions them by meaning as in Section 6.2, and pays each response the share of the group that agrees with it:

$$r_i = \hat{p}_{c(i)} = \frac{|c(i)|}{G},$$

where $c(i)$ is the cluster containing o_i . Responses in the majority cluster are reinforced and outliers are suppressed, with no reference to whether the majority is right. In the released implementation, the equivalence relation is a pipeline of answer extraction and symbolic equality for mathematics, and a small dedicated verifier model for open-ended reasoning — cheap in both cases, and cheap in a way that does not require knowing the answer, though for the learned verifier that last clause turns out to need a careful audit, taken up in Section 6.7.

Two guards accompany the reward. The first is a band on the entropy: training proceeds only on questions with $\delta_{\text{low}} < H_{\text{sem}}(q) < \delta_{\text{high}}$. Too much dispersion and the majority cluster is noise; too little and there is nothing left to sharpen but overconfidence. The second is a set of anti-gaming penalties — an empty answer is penalised outright, and an answer that merely echoes the question earns nothing — which prevent the model from manufacturing agreement without saying anything.

It is worth examining how the first guard is realized in practice, because the implementation ended up stricter than the design intended—and understanding why is revealing. The results above were obtained with the band active. When the experiments were later rebuilt on a different training framework, an ablation found the thresholds made limited difference, and they were relaxed to keep any group whose entropy is neither zero nor close to its ceiling: $0 < H_{\text{sem}} < 0.6 \log G$, over a group of G samples. **The two endpoints (0 and $\log G$) are redundant.** A group in which all G responses agree has $H_{\text{sem}} = 0$ and pays every response the same reward, 1; a group in which no two agree has $H_{\text{sem}} = \log G$ and again pays every response the same reward, $1/G$. Since the advantage subtracts the group mean, both cases contribute an identically zero update whether they are filtered or not. A filter acting at the endpoints removes only groups the optimiser was already ignoring — consistent with the ablation’s verdict that the original thresholds could be relaxed at little cost.

What remains is the upper bound alone, and it is a real constraint: at $G = 16$ it admits a group only if its meanings are no more dispersed than roughly five equally-sized clusters. But it guards against one failure only — a majority that is really noise. Nothing in the current configuration bounds the entropy from *below* at any value strictly inside the simplex. That asymmetry is not a detail, but reading it correctly takes some care, and Section 6.6 supplies the reading: the floor turns out to be a *stopping rule* — the mechanism that decides where sharpening halts, per prompt — rather than a stability device, which is why its absence coexists peacefully with stable published runs governed by a short horizon, and why the ablation that retired it, run at exactly such a horizon, was not positioned to see what it does.

Empirically the method closes most of the gap to supervision without using any. On five mathematics benchmarks with a 7B base model, averaging over the suite:

Method	What it needs	Score
Base model	—	33.7
Supervised fine-tuning	$\{q, r, a\}$	39.0
Verifier-based RL (GRPO)	$\{q, a\}$	51.7
EMPO	$\{q\}$	51.6

Here q is the question, a the gold answer, and r a reasoning trace. Two things in this table deserve to be said plainly. The first is that supervised fine-tuning, which sees the most, does worst of the three — imitation of traces is a weaker signal than selection among the model’s own attempts. The second is that the tie between EMPO and verifier-based RL holds on mathematics; on open-ended reasoning a real gap of several points remains. The honest summary is that removing labels costs little in a domain where the model is already strong, and more elsewhere.

6.5 What “reward agreement” actually optimises

It is worth computing what the reward above does in expectation, because the answer is both cleaner and different from the one usually stated.

Averaging r_i over the group, each response contributes the frequency of its own cluster, so a cluster of size $|c|$ contributes $|c|$ times $\hat{p}_c$:

$$\bar{r} = \frac{1}{G} \sum_{i=1}^G \hat{p}_{c(i)} = \sum_{c \in \mathcal{C}} \hat{p}_c^2 = \exp(-H_2(\hat{p})),$$

where $H_2(p) = -\log \sum_c p_c^2$ is the **Rényi entropy of order two**, also called the collision entropy (Rényi 1961). The quantity being maximised is the *collision probability*: the chance that two independent samples from the model land on the same meaning.

This deserves a frank note, and then a correction to the way such paper-versus-code mismatches are usually described. The objective as written in the paper is the Shannon semantic entropy of Section 6.2; the objective the code implements is H_2 . These are genuinely different functions: their gradients differ, and H_2 weights the dominant cluster more heavily than Shannon does. But they are not different *kinds* of object, and naming the family they belong to turns the discrepancy from an inconsistency into a choice that can be argued for.

A unified family, parametrized by order. Consider the Rényi entropy H_α from Section 2.1.4. Instead of rewarding responses based on their cluster’s simple share, generalize by awarding a power of that share:

$$r_\alpha(c) = \frac{p_c^{\alpha-1} - 1}{\alpha - 1}, \quad \mathbb{E}_{c \sim p}[r_\alpha] = \frac{\sum_c p_c^\alpha - 1}{\alpha - 1},$$

an expectation monotone in H_α for every $\alpha > 1$, so that maximising it minimises the entropy of that order. Two members are already in front of us. As $\alpha \rightarrow 1$ the reward tends to $\log p_c$ and its expectation to $-H_{\text{sem}}$, which is the paper’s objective. At $\alpha = 2$ the reward is $p_c - 1$, which

differs from the implemented reward p_c by a constant — and constants are invisible here, since the advantage subtracts the group mean before anything is done with it. **The stated objective and the implemented one are the $\alpha \rightarrow 1$ and $\alpha = 2$ members of a single one-parameter family.**

A third member turns up in the filters. The surviving band is written in terms of H_{sem} , but the majority-share cutoffs that have at various points stood in for it are thresholds on $\max_c p_c$, which is e^{-H_∞} . Between the derivation, the reward and the gate, the algorithm therefore touches three orders at once: $\alpha \rightarrow 1$, $\alpha = 2$, $\alpha = \infty$. The min-entropy that Section 2.1.4 was at pains to separate from minimum-entropy learning turns out to appear here after all — not as the objective, but as the gatekeeper.

Why order two is the defensible choice. Three reasons, of which the first is decisive. A group holds only G samples — seven to sixteen in the released configurations — so p_c is never known, only estimated by $\hat{p}_c = |c|/G$. Shannon entropy admits *no unbiased estimator at any finite G* : the expectation of any estimator is a polynomial in p , and entropy is not a polynomial (Paninski 2003, Prop. 8). Collision probability is a polynomial, of degree two, and is therefore estimable without bias by $\sum_c |c|(|c| - 1)/G(G - 1)$. The gap is not academic: at $G = 8$ over five clusters the plug-in Shannon estimator is biased downward by 0.28 nats, a fifth of the quantity being measured, while the degree-two estimator is exact. Second, the reward stays bounded: a singleton cluster earns $1/G$, where the order-one reward $\log \hat{p}_c$ hands it $-\log G$, a spike that grows with the group size. Third —

$$\log K - H_\alpha = \frac{\alpha}{2K} \sum_c \varepsilon_c^2 + O(\varepsilon^3), \quad p_c = \frac{1 + \varepsilon_c}{K}, \quad \sum_c \varepsilon_c = 0,$$

so near the uniform distribution every order is the *same* function of the distribution, differing only by the factor α . A Shannon derivation and a collision-entropy implementation give parallel gradients wherever the meanings are near-balanced, which — since the band admits only high-entropy questions — is where training begins. They part company as p concentrates, which is precisely the regime Section 6.6 is about.

They are not, for all that, the same function. Both are Schur-concave, so they agree on any pair ordered by majorization, and both are maximised by the uniform distribution and minimised by a point mass. But majorization is a partial order, and on incomparable pairs the two can disagree outright. A group placing half its mass on one meaning and scattering the rest over nine others, $p = (0.5, 0.056^{\times 9})$, has Shannon entropy 1.79 against 1.39 for a group split evenly four ways — while its collision entropy is 1.28 against 1.39. One says the first group is the more diverse; the other says it is the less. Which order is used is a modelling decision, not a formality.

A second and independent axis: from a partition to a kernel. Everything above holds the hard clustering fixed. Relaxing it is a separate generalisation, and the two compose. Replace the equivalence relation with a positive semi-definite similarity kernel over the sampled answers, normalise it to unit trace as $\rho = K/\text{tr } K$, and read the entropy off the spectrum, $H_\alpha(\rho) = \frac{1}{1-\alpha} \log \text{Tr } \rho^\alpha$, with $\alpha \rightarrow 1$ giving the von Neumann entropy of Chapter 2. The classical cases are recovered exactly rather than approximately: when K is block-diagonal with all-ones

blocks, the non-zero eigenvalues of ρ are the cluster masses p_c (Pasarkar and Dieng 2024), so $\alpha \rightarrow 1$ returns H_{sem} and $\alpha = 2$ returns $-\log \sum_c p_c^2$. The resulting two-parameter family — an order, and a choice of hard or soft — is the order- q Vendi score (Friedman and Dieng 2023; Pasarkar and Dieng 2024), which coincides with the matrix-based Rényi entropy introduced independently by Sánchez Giraldo et al. (2015), and which is, under the name Hill numbers, the standard family of diversity indices in ecology (Hill 1973; Leinster and Cobbold 2012).

This is worth spelling out because of one claim it would be easy to make and wrong. The kernel relaxation is *not* available only for collision probability. Nikitin et al. (2024) build exactly the $\alpha \rightarrow 1$ version for language models, under the name kernel language entropy, and prove that for any semantic clustering there exists a kernel whose von Neumann entropy equals the semantic entropy — by the same block-diagonal construction. What distinguishes $\alpha = 2$ is cost, not existence: $\text{Tr } \rho^2 = \|\rho\|_F^2$ is a sum of squares needing no eigendecomposition, while every other order needs the spectrum. Over a group of sixteen that is free either way; over a corpus it is not.

The gap worth recording is the empty cell. The kernel literature has developed the $\alpha \rightarrow 1$ corner and the clustering literature the $\alpha = 2$ one, and a soft, order-two semantic objective — the natural home for a reward that is already a collision probability — appears not to have been tried.

There is also a pleasing coincidence in the expression $\sum_c p_c^2$. It is exactly the chance-agreement term p_e in Cohen’s κ , the standard correction applied when two human raters are scored for agreement (Cohen 1960). In the inter-rater setting one computes the agreement two independent raters would reach by luck alone, and subtracts it. Here the same expression is computed over one model’s repeated answers and *maximised*. The statistical object that measures how much of an agreement is worthless is, read in the other direction, the training signal.

6.6 The free-energy view, and why entropy collapses

Here is the conclusion of this section, before the argument for it. Rewarding agreement is a rich-get-richer process, and whether it runs away is decided by a single number crossing one. On one side of that threshold, an accidental imbalance between meanings dies out and the model stays honest about what it does not know; on the other side, the imbalance feeds on itself and the model commits to one meaning — and nothing in the objective checks whether that meaning is correct. Two properties of the crossing make it worse than an ordinary tuning problem. It is abrupt: the entropy barely moves for most of the approach, then falls off a cliff. And when more than two meanings compete, it has memory: the collapsed state is already stable on the *safe* side of the threshold, so a single large fluctuation can tip the system over, and moving the knob back afterwards does not undo it. Setting the knob near its critical value is therefore not a defence. A floor on the state is — though what exactly it defends, and what stood in for it in the published runs, takes the rest of the section to state precisely.

Everything below is the derivation, and the machinery is that of Chapter 3.

The control parameter is a temperature, and it is a familiar one. Anyone who has sampled from a language model has already met the object doing the work here. Sampling at temperature τ

replaces the model's distribution by $p(o)^{1/\tau}$ and renormalises: large τ flattens the distribution, small τ sharpens it, and $\tau \rightarrow 0$ is arg-max decoding. What follows is that same operation with two modifications. The exponent is a reward rather than the model's own log-probability, and — this is the entire source of the difficulty — the reward is itself the probability being adjusted. It is a softmax whose logits are its own outputs.

Start from the canonical form of the policy-optimisation problem, in which reward is traded against a Kullback–Leibler penalty that keeps the updated policy near a reference π_0 :

$$\max_{\pi} \mathbb{E}_{\pi}[r] - \beta D_{\text{KL}}(\pi \| \pi_0). \quad (6.1)$$

This is the minimisation of a free energy, with $-r$ in the role of energy and β in the role of temperature, and it can be solved in closed form. Add a Lagrange multiplier λ to enforce that the probabilities sum to one, and differentiate with respect to $\pi(o)$:

$$\begin{aligned} \frac{\partial}{\partial \pi(o)} \left[\sum_{o'} \pi(o') r(o') - \beta \sum_{o'} \pi(o') \log \frac{\pi(o')}{\pi_0(o')} + \lambda \left(1 - \sum_{o'} \pi(o') \right) \right] \\ = r(o) - \beta \log \frac{\pi(o)}{\pi_0(o)} - \beta - \lambda = 0. \end{aligned}$$

Solving for $\pi(o)$ and absorbing everything independent of o into a normaliser gives the **Gibbs tilt of the reference**,

$$\pi^*(o) = \frac{\pi_0(o) e^{r(o)/\beta}}{Z}, \quad Z = \sum_o \pi_0(o) e^{r(o)/\beta}, \quad (6.2)$$

which is the unique maximiser, since the KL term is strictly convex in π and the reward term is linear. Substituting back, the optimal value is $\beta \log Z$ — a log-partition function, exactly the free energy of Chapter 3. This is the same exponential family that Section 2.3 produces from the other direction: there, maximising entropy under a moment constraint; here, maximising reward under a divergence constraint.

The temperature reading is worth making concrete. Take three semantic clusters, a uniform reference $\pi_0 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, and rewards $r = (1.0, 0.5, 0.0)$. Then Equation 6.2 gives

β	π^*	$\mathbb{E}[r]$
∞	(0.333, 0.333, 0.333)	0.500
2	(0.419, 0.326, 0.254)	0.582
1	(0.506, 0.307, 0.186)	0.660
0.25	(0.867, 0.117, 0.016)	0.925
$\rightarrow 0$	(1, 0, 0)	1.000

At high temperature the update does nothing and the policy stays at its reference; at zero temperature it becomes greedy and places all mass on the best cluster. The KL coefficient is precisely the knob that decides how far towards the arg-max a single update is allowed to

travel — the role played by τ in temperature sampling, now attached to a reward instead of a log-probability.

Nothing so far is specific to entropy minimisation. What *is* specific is that under EMPO the reward is not an external function of the response: by Section 6.4, r is the cluster frequency of the very policy being optimised. Writing p_c for the probability the policy assigns to cluster c and p_c^0 for the reference, substituting $r(o) = p_{c(o)}$ into Equation 6.2 turns an explicit formula into a **self-consistency condition**:

$$p_c = \frac{p_c^0 e^{p_c/\beta}}{\sum_{c'} p_{c'}^0 e^{p_{c'}/\beta}}. \quad (6.3)$$

This is a mean-field equation, of exactly the kind that describes a magnet whose spins feel the average field they themselves generate. The consequences are worth spelling out, because they turn a vague worry about instability into a quantitative statement. Take the KL anchor seriously for the moment and suppose π_0 is genuinely fixed; the paragraphs after next remove that supposition, which is what real implementations do.

A critical value. Take a reference that is uniform over K clusters, $p_c^0 = 1/K$, and nudge the balanced solution slightly: $p_c = 1/K + \varepsilon_c$ with $\sum_c \varepsilon_c = 0$. Substituting into Equation 6.3, the factor $e^{1/(K\beta)}$ is common to every cluster and cancels, leaving $p_c \propto e^{\varepsilon_c/\beta} \approx 1 + \varepsilon_c/\beta$ to first order, so that after normalisation

$$\varepsilon_c \mapsto \frac{\varepsilon_c}{K\beta}.$$

An imbalance between meanings is thus multiplied by $1/(K\beta)$ at every step. This is the number promised at the opening, and the rest of the section is a matter of where it sits relative to one. The balanced state survives a perturbation if and only if $K\beta > 1$, so there is a critical value

$$\beta_c = \frac{1}{K},$$

below which the policy commits to a single cluster. The appearance of K is not an artefact: the more distinct meanings a model is entertaining, the smaller each one's share of the group, the weaker the positive feedback any one of them can exert, and so the less braking is needed to contain it. For $K = 2$ the self-consistency condition can be solved exactly — writing $m = p_1 - p_2$, Equation 6.3 becomes

$$m = \tanh\left(\frac{m}{2\beta}\right),$$

the Curie–Weiss equation — the textbook description of a magnet that spontaneously magnetises when cooled below a critical temperature — with β in the place of temperature and $\beta_c = 1/2$ in the place of that critical temperature (the Curie point). One warning for readers arriving from physics, where β conventionally denotes *inverse* temperature: here it is the KL coefficient and it plays the role of T itself. Large β means strong regularisation, a policy held near its reference and high entropy — the hot, disordered phase; small β is the cold, ordered, collapsed one.

i Derivation of the $K = 2$ case

With two clusters and a uniform reference, the factor $\frac{1}{2}$ cancels from numerator and denominator of Equation 6.3, leaving a softmax whose logits are the cluster masses themselves. Since $p_1 + p_2 = 1$ there is only one degree of freedom; write $p_{1,2} = \bar{p} \pm \delta$ with $\bar{p} = \frac{1}{2}$ and $\delta = m/2$. Then

$$m = \frac{e^{p_1/\beta} - e^{p_2/\beta}}{e^{p_1/\beta} + e^{p_2/\beta}} = \frac{e^{\bar{p}/\beta}(e^{\delta/\beta} - e^{-\delta/\beta})}{e^{\bar{p}/\beta}(e^{\delta/\beta} + e^{-\delta/\beta})} = \tanh\left(\frac{\delta}{\beta}\right) = \tanh\left(\frac{m}{2\beta}\right),$$

by the definition of the hyperbolic tangent. The cancellation of $e^{\bar{p}/\beta}$ is worth pausing on, because it is the same one that occurs at general K in the perturbation argument above: a softmax is blind to a common shift of its logits, so only *differences* between clusters drive the dynamics. It is also why subtracting the group baseline $\bar{r}$ leaves the update direction untouched.

Whether a non-zero solution exists is then settled entirely by the slope at the origin,

$$\left. \frac{d}{dm} \tanh\left(\frac{m}{2\beta}\right) \right|_{m=0} = \frac{1}{2\beta},$$

which is the amplification factor $1/(K\beta)$ evaluated at $K = 2$, as it must be — linear stability analysis is exactly this derivative. Because $\tanh$ is concave for $m > 0$ and bounded by one, a slope below one lets the curve fall away from the diagonal immediately and never return, so $m = 0$ is the only solution; a slope above one carries the curve above the diagonal, and boundedness forces it back across, creating a pair of solutions $\pm m^*$. The two regimes are separated by $1/(2\beta) = 1$.

Expanding $\tanh x = x - \frac{1}{3}x^3 + O(x^5)$ and dividing through by $m \neq 0$ gives $1 = 1/(2\beta) - m^2/(24\beta^3)$, hence

$$m^* = 2\beta\sqrt{3(1 - 2\beta)} \simeq \sqrt{3\left(1 - \frac{\beta}{\beta_c}\right)}$$

near the critical point. The imbalance m — the *order parameter*, in physics language — grows continuously out of zero with exponent $1/2$, the standard mean-field value. That continuity is what makes the $K = 2$ transition *second order*: the state changes smoothly, with no jump. It is precisely this smoothness that fails for $K \geq 3$.

The two regimes, step by step. The algebra is easier to trust once one has watched it happen. Both runs start from the same barely-perturbed state $p = (0.36, 0.32, 0.32)$ with $K = 3$; only β changes.

Above the threshold ($\beta = 1.5\beta_c$), the perturbation is erased and the entropy returns to its ceiling $\log 3 = 1.0986$:

step	p	H
0	0.360, 0.320, 0.320	1.0970
2	0.345, 0.327, 0.327	1.0983

step	p	H
4	0.339, 0.331, 0.331	1.0985
6	0.336, 0.332, 0.332	1.0986
8	0.334, 0.333, 0.333	1.0986

Below the threshold ($\beta = 0.75 \beta_c$), the same perturbation compounds and the distribution is most of the way to a point mass:

step	p	H
0	0.360, 0.320, 0.320	1.0970
2	0.384, 0.308, 0.308	1.0931
4	0.432, 0.284, 0.284	1.0775
6	0.539, 0.231, 0.231	1.0097
8	0.750, 0.125, 0.125	0.7358

The column that deserves attention is the entropy in the second table. Over the first four steps it falls by less than two per cent while the gap between the leading cluster and the others nearly quadruples, and only then does it drop. Entropy is a *lagging* indicator of a process that is already well advanced, which is exactly why collapse is reported as sudden: from the outside it looks like training proceeding normally and then failing without warning. The warning was in the gap between clusters, not in the entropy.

For more than two clusters the transition is discontinuous. At $K = 2$ the dominant cluster's mass grows continuously from $1/2$ as β falls through β_c . For $K \geq 3$ it does not, and the cleanest evidence is a decisive experiment: hold β fixed, start the iteration from two different places, and see whether they agree. The balanced start carries a 10^{-3} perturbation, because starting from perfect symmetry would test nothing: a perfectly balanced state stays balanced by symmetry at every β . For $K = 3$, reporting the converged $\max_c p_c$:

β	from balanced start	from concentrated start	
$1.200 \beta_c$	0.333	0.333	balanced only
$1.080 \beta_c$	0.333	0.674	both stable
$1.035 \beta_c$	0.333	0.768	both stable
$1.020 \beta_c$	0.333	0.788	both stable
$1.000 \beta_c$	0.811	0.811	collapsed only

Throughout the window $\beta \in [1.000, 1.093] \beta_c$ the two states are *both* stable, and which one the system occupies depends on where it came from. That path dependence is called *hysteresis* — the system remembers its history — and it is the defining signature of a *first-order* transition, one where the state jumps rather than drifts. The window is empty at $K = 2$, widens to $[1.000, 1.403] \beta_c$ at $K = 5$ and $[1.000, 1.882] \beta_c$ at $K = 8$ — the more meanings in play, the larger the region of ambiguity. This matches a classical model of statistical physics, the

mean-field Potts model, whose transition is likewise smooth at $q = 2$ states and discontinuous for $q \geq 3$ (Wu 1982) — so the behaviour is a structural property of self-referential dynamics of this type rather than a peculiarity of the present problem.

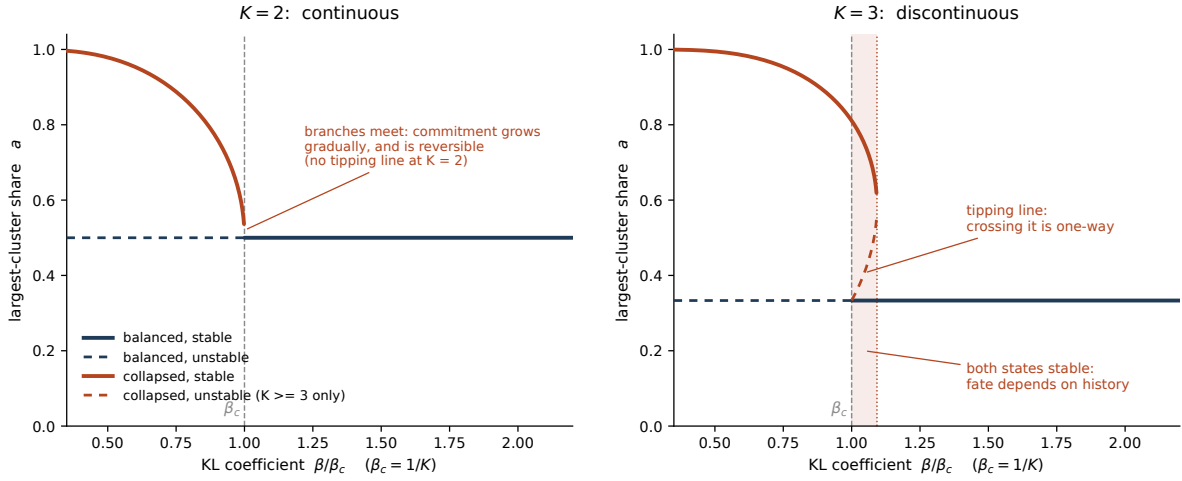


Figure 6.1: Where training can come to rest, and why the number of competing meanings changes everything. Each panel shows the resting states of the self-consistency condition Equation 6.3 as the anchor strength β varies: strong anchor on the right, weak on the left. Height is the share of the largest meaning cluster — $1/K$ means the model is keeping its options open (navy), 1 means it has committed to a single answer (rust). Solid curves are states the system can rest in; dashed curves are unstable and cannot be occupied. **Left ($K = 2$):** weaken the anchor past β_c and commitment grows *gradually* out of the balanced state; strengthen it again and the system walks back. No red dashed line appears because none exists at $K = 2$ — there is no tipping line to cross, so the transition is continuous and reversible. **Right ($K = 3$):** the committed state is already available and stable *before* the threshold is reached. In the shaded window both states are stable, separated by an unstable ridge (red dashed): which state the system occupies depends on where it came from, and a fluctuation across the ridge is one-way. This is the figure’s practical message — with three or more meanings in play, setting β “just above the critical value” parks the system inside the window where a single large fluctuation commits it irreversibly. Generated by `code/min-entropy-phase-diagram.py`.

Figure 6.1 makes the practical consequence visible. The threshold $\beta_c = 1/K$ is not where the danger begins; it is only the last point of the safe regime — what physicists call a *spinodal*, the place where the balanced state finally stops surviving even small knocks. The collapsed state has already been available and stable for some distance above it. Tuning β to sit just above β_c is therefore not a safety margin — it places the system exactly in the region where both states coexist, and where a single fluctuation large enough to cross the dividing line is never recovered from. What is needed is not a carefully chosen distance from the critical point but a mechanism that acts on the state itself.

In practice there is no such reference, and collapse becomes unconditional. The stabilising mechanism above depends entirely on the KL term in Equation 6.1 anchoring the policy to a *fixed* π_0 . It is worth checking whether real implementations have one, and the answer is that they do not. In the released EMPO code both available training stacks disable the penalty outright: the reference-model implementation sets its KL coefficient `beta: 0.0`, and the large-scale implementation switches the KL penalty off everywhere it can appear — both the penalty added to the reward and the auxiliary KL loss, along with both of their coefficients — in every one of its published run scripts. The entropy bonus is likewise set to zero. There is no term

anywhere in the optimiser that opposes entropy reduction.

What remains is only the implicit restraint of taking small steps. There is a standard way to read each gradient step in the language of this section (formally, as *mirror descent*): every update applies Equation 6.2 with the *current* policy in place of π_0 and the step size in place of β — instead of tilting a fixed reference, each step tilts wherever the policy already stands, so the cluster probabilities evolve as $p_c \propto p_c e^{p_c/\beta}$. The same perturbation analysis now gives

$$\varepsilon_c \mapsto \varepsilon_c \left(1 + \frac{1}{K\beta}\right),$$

an amplification factor strictly greater than one for *every* $\beta > 0$. No step size stabilises the balanced state. A trust region — whether imposed by a small learning rate or by the unusually tight clipping range the released configuration uses — bounds how fast the policy moves, not where it ends up. Within this idealised dynamics, collapse is not a risk that a step size can tune away; it is the asymptotic behaviour, on a clock the trust region can stretch but not stop.

Two features of the group-relative advantage sharpen this rather than soften it. Advantages are computed as $A_i = (r_i - \bar{r})/s_r$ within the group of G samples. The baseline $\bar{r}$ cancels exactly in Equation 6.2, since a constant shift in the exponent is absorbed by the normaliser, so subtracting the mean has no effect on the direction of the update. The division by the group standard deviation s_r does have an effect: it acts as an adaptive inverse temperature — rescaling the pull without redirecting it — and because $s_r \rightarrow 0$ as the group becomes homogeneous, the effective temperature *shrinks* precisely as collapse proceeds. The pull towards the dominant cluster does not weaken near the degenerate state; it is held back only by clipping.

This is the sharpest justification available for the entropy floor of Section 6.4, and stating it correctly also dissolves what looks, at first sight, like a contradiction. The chapter has argued that a floor strictly inside the simplex is structurally necessary; the released configuration sets $\delta_{\text{low}} = 0$; and yet the published runs completed stably and an ablation found the thresholds made little difference. If the floor were a *stability device* — a thing that keeps the optimiser from blowing up — those three facts could not sit together. They sit together because that is not what the floor is, and seeing what it is instead requires separating three questions that the word *collapse* runs together.

Does training diverge without a floor? No — and not because anything opposes the dynamics, but because the endpoint silences itself. A prompt whose group has fully agreed pays every response the same reward, the advantage is identically zero, and the prompt stops contributing gradient. The fully-agreed state is what dynamicists call *absorbing* — once a prompt reaches it, it never leaves — but it is also quiet: nothing visibly breaks, and reaching it looks like convergence, not failure. The catastrophic collapses of Section 6.3 — a classifier sending every input to one class, a policy emitting one character for every prompt — are *cross-input* degeneracies, and EMPO’s structure blocks the direct route to them: per-prompt clusters leave no shared label space for a marginal to collapse onto, and the anti-gaming penalties tax the cheapest constant policies.

Where does each prompt end without a floor? With all of its probability on a single meaning — at a corner of the simplex. That is what the mean-field analysis establishes: once the anchor is gone

the balanced state is unstable at every β , the transition is first order, and past the point of no return the commitment cannot be undone by a luckier sample later. Nothing in the optimiser selects a stopping point short of full commitment, and no setting of β can be made to select one — with an anchor it merely positions the system inside the window where both states coexist, and without one there is nothing to position.

Is full commitment what the method wants? Half of it is. Sharpening all the way to that corner is exactly the elicitation the method is for, on prompts where the majority is right. The other half is the cost, and it is not hypothetical, because the paper’s own measurements price it: on prompts where the majority is wrong, the commitment is equally final — and the $\text{pass}@k$ curves of Section 6.7, converging to the base model and crossing below it on hard benchmarks, are the coverage bill for sharpening run to the end.

On this reading the floor is a **stopping rule**, not a safety rail: it decides how much of the distribution’s diversity sharpening is allowed to spend, per prompt, and it is the only instrument in the design that can make that decision at the level of the state. The ceiling cannot substitute — it rejects dispersion, and full commitment approaches from the other side. What operated in its place in the published runs was a second, cruder stopping rule: the finite training budget, a global and undirected version of what the floor would do per prompt. A fixed horizon stops the average prompt in time, the already-committed prompt too late, and the still-diffuse prompt too early; but under a horizon short enough, and clipping as tight as the released configuration’s, most prompts never enter the region where a per-prompt floor would bind. The ablation’s null verdict is then the expected signature rather than a refutation — it compared two configurations both governed, in effect, by the horizon — and the same reasoning says where the floor’s value should become measurable: longer runs, sparser prompts, weaker base models, and any deployment in which locking in the current majority early is expensive. The degeneracy is the one Section 6.3 traced through three decades of this idea, now out in the open because the supervised anchor that used to contain it has been removed. Read against that history, even the band as designed is a *filtering* remedy — the weakest of the three families Section 6.3 describes, since it acts on which examples are used rather than on the objective, and it has no counterpart to the fairness term that the semi-supervised and self-supervised literatures found necessary. The analysis also predicts the failure mode reported in follow-up work on this family of methods, in which performance improves for a while and then falls below the untrained model: that is what running past a first-order transition looks like from the outside — a horizon that outlasted its usefulness as a stopping rule.

Three caveats on the scope of this analysis. It is an idealised treatment of the *reward dynamics* in the space of cluster probabilities, not of the neural network; it treats each prompt independently, ignoring the fact that all prompts share one set of weights, so an update for one prompt shifts the policy for all others. It treats p_c as known rather than estimated from G samples, which for the group sizes actually used adds sampling noise of the same order as the effects being described. And it reads the optimiser as taking the idealised update of Equation 6.2 exactly, whereas the clipped surrogate actually used departs from that whenever the clip binds. What survives these caveats is the qualitative structure — self-consistency, a critical temperature, a discontinuous transition, and the absence of any stabilising term — not the precise numerical threshold.

A note on what here is new, since the ingredients are not. Reading a learning objective as a free energy and locating its critical temperature is the method of deterministic annealing, where Rose et al. (1990) computed the temperature at which a cluster splits in two — the same construction, run in the opposite direction, since there cooling *creates* structure out of a single-cluster symmetric state whereas here it *destroys* the balanced one. The information bottleneck has its own sharp transition, below which the trivial representation becomes the global optimum (Wu et al. 2019; Wu and Fischer 2020), and entropy-regularised learning dynamics are known to switch behaviour abruptly at a critical exploration rate (Kianercy and Galstyan 2012). Closest of all, GX-Chen et al. (2025) analyse precisely the object of Equation 6.2 and conclude, as this section does, that collapse is a property of the objective rather than of the optimiser, and that β is the parameter controlling it. What that work characterises is the optimum; what is added here is the self-consistency that arises when the reward is the policy’s own cluster frequency, and with it a critical value, an order of transition, and a stability analysis. The claim worth defending is not that this is a phase transition — any physicist would frame it so — but the chain that follows from the transition being first order: collapse has memory, the threshold at which the balanced state loses stability is consequently the wrong safety criterion, and no choice of β can substitute for a floor on the state.

6.7 The information ceiling

There is a limit on what any of this can achieve, and it is worth stating formally because it is easy to lose sight of amid the empirical results.

Let π_0 be the base policy, π_T the policy after training, and a^* the true answer to a question q . Every update EMPO performs is computed from samples drawn from the current policy and from the equivalence relation $\sim$; the ground truth enters nowhere. Two facts about that loop do all the work: the randomness of sampling — seeds, batch order; call it R — carries no information about the answer, and $\sim$ is frozen before training begins — whatever it is, it does not change in response to anything the run produces. Everything the run ends up with is therefore determined by four ingredients: the base model, the question, the metric, and the dice. The trained policy is a deterministic function $\pi_T = f(\pi_0, q, \sim, R)$, and the data-processing inequality — the rule that no amount of computation can create information about a quantity its inputs do not already carry — gives

$$I(\pi_T; a^* \mid \pi_0, q, \sim) = 0.$$

All the content sits in the words after the bar. The statement is not that the trained policy knows nothing about the truth — it plainly does — but that *given what the base model already was and what the equivalence relation already knew*, the training run added nothing. The same argument, run once more (the callout below has the steps), yields the inequality of this section’s title, and it has two terms:

$$I(\pi_T; a^* \mid q) \leq \underbrace{I(\pi_0; a^* \mid q)}_{\text{pre-training}} + \underbrace{I(\sim; a^* \mid \pi_0, q)}_{\text{the metric}}.$$

Information about the truth can be rearranged and expressed more reliably, but the budget is fixed before the first update — by pre-training, plus whatever the equivalence relation itself carries about the answer. On mathematics the second term is not merely small but identically zero: the clustering there is answer extraction followed by symbolic equality, a hand-written rule with no trained parameters and no access to the question, so the ceiling is pre-training alone, exactly as the one-term reading suggests. On open-ended reasoning the second term cannot simply be declared zero, because the equivalence relation is itself a trained model — and that turns out to deserve a section of its own, taken up below. Either way, whatever the trained policy gained, it gained by redistributing probability mass over a support that the base model already had.

i Derivation, and a world small enough to check it

The assumption doing the quiet work is that the training randomness R — sampling seeds, batch order — is independent of a^* given $(\pi_0, q, \sim)$, and that $\sim$ is frozen before the run. Both hold in the released code, where rewards and filters are computed from cluster frequencies alone. Then, in three steps. *Determinism*: every update uses only the G samples and their cluster frequencies, so $\pi_T = f(\pi_0, q, \sim, R)$ with f deterministic. *Markov chain*: given $(\pi_0, q, \sim)$, π_T is a function of R alone and R is conditionally independent of a^* , so $P(\pi_T | \pi_0, q, \sim, a^*) = P(\pi_T | \pi_0, q, \sim)$ — the chain $a^* \rightarrow (\pi_0, q, \sim) \rightarrow \pi_T$. *Data processing*: $0 \leq I(\pi_T; a^* | \pi_0, q, \sim) \leq I(R; a^* | \pi_0, q, \sim) = 0$. The two-term ceiling then takes two short steps: since π_T is a function of $(\pi_0, \sim, R)$ once q is fixed, data processing gives $I(\pi_T; a^* | q) \leq I(\pi_0, \sim, R; a^* | q)$; and expanding the right side with the chain rule, the R term vanishes and the other two are the budget and the metric term. Supervision breaks the chain at the first step: the verifier reward $\mathbb{1}[\text{verify}(o_i, a)]$ makes π_T a function of a^* as well, and each verification deposits bits about the answer into the weights.

The theorem is easiest to trust in a world small enough to enumerate. Let a^* be uniform on $\{0, 1\}$; let pre-training set a parameter θ_0 equal to a^* with probability 0.8; let the base policy answer θ_0 with probability 0.7; and let training sample $G = 5$ answers and sharpen towards their majority, so that θ_T is a function of θ_0 and the sampling noise alone. Exact enumeration of the joint distribution gives $I(\theta_T; a^* | \theta_0) = 0$ — machine-exact, not approximately — while $I(\theta_T; a^*) = 0.121$ bits against a pre-training budget of $I(\theta_0; a^*) = 0.278$ bits, and replacing the majority vote by a verifier-guided update lifts the conditional information to 0.358 bits. Two details repay attention. The inherited information is *less* than the budget: majority voting is a noisy compression of θ_0 , and processing loses information more readily than it preserves it. And one loophole sits outside the theorem entirely — selecting a checkpoint on a labelled validation set is itself an update that touches ground truth, worth at most $\log_2$ of the number of candidates in bits; small, but not zero. The numbers are reproduced by `code/min-entropy-information-ceiling.py`.

The metric is not free. The second term of the ceiling deserves to be followed into the implementation, because what sits there is more interesting than a technicality. For open-ended reasoning, the released code clusters answers with a small generative verifier (Ma et al. 2025)

— a mathematics-specialised 1.5B model fine-tuned on roughly 230,000 questions *with ground-truth answers*, whose equivalence verdicts were annotated by a much larger commercial model. The weights of $\sim$ are, in other words, a deterministic function of a labelled dataset. Three details of how it is called sharpen the point. The prompt contains the full question, not just the two answers being compared. One of the two policy outputs is placed in a slot literally named *Ground Truth Answer* and the other in a slot named *Student Answer* — the model’s native supervised-verification format, with a sampled answer standing in for the gold one. And the prompt instructs the model not to solve the question itself, which is an instruction, not an architectural constraint: nothing prevents the forward pass from forming its own belief about the answer on the way to a verdict.

So can the metric smuggle supervision into a “fully unsupervised” run? The answer has a clean structure: leakage requires two conditions *jointly*, and either one alone transmits nothing. First, **knowledge** — the verifier must actually carry information about this question’s answer, $I(\sim; a^* \mid \pi_0, q) > 0$, whether from partially solving q , from having seen related questions in its training set, or from the annotation model whose judgments it distils. Second, a **behavioural channel** — its same-or-not-the-same verdicts must *vary* with that knowledge: merging a pair more readily when both members agree with its own belief, or fragmenting clusters of answers it considers wrong. A verifier that knows the answer but judges equivalence without ever consulting that knowledge leaks nothing: if the verdicts do not vary with what it knows, nothing of what it knows can reach the training run. The toy world above extends to check this exactly: add a verifier belief V that equals a^* with probability 0.9, and a merging rule that consolidates clusters agreeing with V while fragmenting the rest.

clustering metric	$I(\theta_T; a^* \mid \theta_0)$	$I(\theta_T; a^*)$ vs. one-term ceiling 0.278
uninformed rule (regex-like)	0	0.044 — holds
informed, correctness-blind verdicts	0	0.121 — holds
informed, verdicts follow the belief	0.114	0.340 — exceeded

In the third row the training run genuinely adds information about the truth — the quantity the one-term reading says is zero is 0.114 bits, and the trained model ends up knowing more than pre-training supplied. No theorem is violated: conditioning on the verifier still gives exactly zero in every row, and the two-term ceiling (0.636 bits in this world) holds throughout. The bits came from the verifier’s training labels, carried into the run through pairwise equivalence verdicts. The numbers are reproduced by `code/min-entropy-verifier-leak.py`.

Two things follow. The defence offered for such metrics — they answer only “are these two the same”, never “which is better”, and provide no gradient — is right in spirit, because it narrows the behavioural channel; the middle row of the table is exactly that defence working. But it does not close the channel while the question sits in the prompt, and how far the verdicts of this particular verifier follow its beliefs has not yet been measured — a natural piece of follow-up work, and a cheap one. The measurements are easy to specify: cluster with the question replaced by an empty string and compare training curves; measure how often the

verdict flips when the two answers swap slots; on a labelled held-out set, compare the merge rate for pairs of equivalent *wrong* answers against pairs of equivalent correct ones. Until then, a careful accounting prices the two settings differently. The mathematics results — including the $\text{pass}@k$ curves below — rest on a parameter-free symbolic rule and inherit the one-term ceiling exactly. The open-ended results are unsupervised *at the level of per-question labels*, with a metric distilled from labelled verification data; their ceiling is the two-term one, and the second term — plausibly small, since the verifier is a 1.5B model and the benchmarks in question sit well above its own solving ability — awaits measurement.

A last distinction keeps the accounting honest in the other direction. Even a metric carrying zero bits about a^* imports something real: a definition of *what counts as the same meaning*, learned from labelled entailment or equivalence data (Kuhn et al. 2023). That is supervision about how the space of answers is carved into meanings, not about which meaning is the true one — and it is load-bearing, since an earlier attempt with a weaker entailment model produced clusters poor enough that training collapsed into reward hacking. The theorem is silent about this kind of supervision, and should be: it is the same status as the tokeniser, fixed machinery that defines the problem rather than labels that answer it. The information ceiling concerns the other kind, and for the metric actually deployed on open-ended tasks, the other kind is not ruled out — only unmeasured.

The ceiling is not a technicality, and it has a testable consequence: a method that adds no information cannot enlarge the set of problems the model is able to solve given enough attempts. Measuring $\text{pass}@k$ — the probability that at least one of k sampled attempts is correct — the trained and untrained models should converge as k grows, and on some benchmarks the untrained model should win, since sharpening a distribution costs coverage. That cost is not a side effect — the objective works at it directly: a correct answer rare enough to appear once among G samples earns the group’s lowest reward and a negative advantage, so the objective actively pushes down exactly the answers that only $\text{pass}@k$ can see. Both predictions are observed — on mathematics benchmarks, where the metric is the parameter-free rule and the ceiling argument applies in its exact form. At $k = 1$ the trained model is far ahead; by $k = 256$ the curves meet, and on harder benchmarks the base model overtakes. Minimum-entropy learning changes which answer comes out first. It does not change which answers are reachable.

6.8 Schrödinger, read carefully

The preceding result has an old precedent, and getting the precedent right requires more care than the usual invocation supplies.

In *What Is Life?* (Schrödinger 1944), Schrödinger asks how an organism avoids the decay to thermodynamic equilibrium that the second law seems to demand. His answer, in the sections titled “It feeds on ‘negative entropy’ ” and “The statistical meaning of entropy,” is that it does not evade the law but pays for its exemption:

“Thus a living organism continually increases its entropy — or, as you may say, produces positive entropy — and thus tends to approach the dangerous state of

maximum entropy, which is death. It can only keep aloof from it, i.e. alive, by continually drawing from its environment negative entropy.”

and, a few pages later, that the organism “attract[s], as it were, a stream of negative entropy upon itself, to compensate the entropy increase it produces by living and thus to maintain itself on a **stationary and fairly low entropy level**.”

Two details in that sentence are usually dropped and both matter here. First, the claim is a *balance*, not a minimisation: a level is held constant because what flows out offsets what is produced inside. Nothing is being pushed to its lowest possible value. Second, the level is “fairly low,” not zero. Schrödinger is describing a floor, and Section 6.6 gives the machine-learning analogue of why a floor rather than a minimum is the right target.

The part of the book that bears most directly on Section 6.7 is a different argument, the one Schrödinger frames as **order from order** as against order from disorder. Ordinary physical law, he observes, is statistical: the regularity of a gas law is order extracted from the disorder of enormous numbers. Heredity cannot work that way, because the structure carrying it is far too small for statistics to smooth anything. It must instead be order copied from pre-existing order — his “aperiodic crystal,” a structure whose information resides in not repeating.

The mapping onto the $\text{pass}@k$ result is exact enough to be worth stating as a point-by-point correspondence rather than a loose analogy. Pre-training is the aperiodic crystal: the reservoir of structure, acquired at enormous cost from an external source. Entropy minimisation is the copying mechanism: it makes that structure come out reliably, and it makes it come out first. It creates nothing. An organism does not synthesise its own order either; it imports it, and Schrödinger’s whole point is that the import is what has to be accounted for.

Where the analogy must stop is also clear. Schrödinger’s entropy is the physical entropy of an open system exchanging real heat with a real environment, carrying units of k_B and constrained by the second law. The entropy of Section 6.2 is the Shannon entropy of a predictive distribution over meanings. Chapter 2 argued that these are the same functional applied to different objects, which licenses the shared vocabulary and the shared mathematics — but not the transfer of a thermodynamic theorem. Nothing in this chapter is derived from the second law, and nothing in it needs to be.

One further detail is worth recording, because it turns out to favour the treatment given here. In a note appended to the chapter in later editions, Schrödinger conceded to his critics that he would have framed the discussion in terms of **free energy** had he been writing for physicists alone, choosing entropy only because it connects more transparently to Boltzmann’s order–disorder principle for a general reader. The concession is narrower than it is often reported to be — he does not retract the argument — but it points in a useful direction. Section 6.6 shows that the object doing the real work in minimum-entropy learning *is* a free energy, with the KL coefficient as its temperature. On this point Schrödinger’s second thoughts and the mathematics agree.

6.9 Prigogine, and an analogy that does not hold

There is a second thermodynamic principle with “minimum entropy” in its name, formulated a year after Schrödinger’s book by a scientist who worked on open systems and later won a Nobel Prize for the thermodynamics of self-organisation. The temptation to enlist it here is considerable. It should be resisted, and setting out why is more instructive than the endorsement would have been.

Prigogine’s theorem (Prigogine 1945, 1947) concerns the **entropy production rate**. For an open system the second law is written as a balance, $dS/dt = d_e S/dt + d_i S/dt$, separating entropy exchanged across the boundary from entropy generated internally, with the internal term $\sigma \equiv d_i S/dt \geq 0$. Near equilibrium, each flow responds in proportion to the push that drives it — heat flow to a temperature difference, diffusion to a concentration difference: $J_i = \sum_j L_{ij} X_j$, and

$$\sigma = \sum_i J_i X_i = \sum_{i,j} L_{ij} X_i X_j.$$

The theorem states that if some pushes are held fixed by external constraints and the rest are free, minimising σ over the free ones is equivalent to the vanishing of the flows they drive — that is, to the steady state. In that regime σ only ever decreases as the system evolves (it is what dynamicists call a Lyapunov function), which is what makes the steady state stable.

Three things about this result rule it out as a foundation for anything in this chapter.

It is about a rate, not a level. The quantity minimised is $d_i S/dt$, with units of entropy per unit time, and it is not the system’s entropy. Schrödinger’s “fairly low entropy level” and Prigogine’s minimum entropy production are different claims about different quantities, and they are logically independent: a system could hold Schrödinger’s balance while producing entropy at any rate whatever (Gnaiger 2009). What Section 6.2 minimises is a level — the entropy of a distribution — with no rate anywhere in sight.

Its hypotheses have no counterpart here, and they are more restrictive than usually advertised. The theorem requires the flow–push relations to be linear, symmetric ($L_{ij} = L_{ji}$, Onsager reciprocity), with coefficients L_{ij} that are *constant* rather than state-dependent — and it requires the system’s variables to look the same when the film is run backwards (Verschaffelt 1954; Maes and Netočný 2013). That last condition is the one most often omitted, and it is precisely the one Landauer (1975) exploited: a resistance in series with an inductance settles at the current $I = E/R$, which manifestly does not minimise $\sigma = RI^2/T$ — because a current reverses sign when the film runs backwards, and the theorem’s hypotheses exclude such variables. Jaynes noted that the principle already fails for a network of resistors held at different temperatures (Jaynes 1980). Extending the statement from a point to a whole region, by minimising $\int_V \sigma dV$ to determine spatial profiles, produces field equations that contradict the conservation laws (Barbera 1999; Martyushev et al. 2007). Modern work has settled the status of the whole construction: the theorem is a small-driving approximation to an exact variational principle, accurate only when the system is barely pushed (Maes and Netočný 2007). It is a narrow theorem, not a general law.

It does not apply to living systems, and Prigogine said so. In his Nobel lecture he was explicit that the result holds only in “a ‘strictly’ linear theory in which the deviations from equilibrium are so small that the phenomenological coefficients may be treated as constants,” and that “far from equilibrium the thermodynamic behavior could be quite different, in fact, even opposite to that indicated by the theorem of minimum entropy production” (Prigogine 1978). Elsewhere he noted that the linear relations, adequate for heat conduction and diffusion, “are generally not valid for the conditions of chemical kinetics” — which is to say, for biochemistry. The modern verdict is blunter. Martyushev and Seleznev (2013) point out that if minimum entropy production governed development, the goal of an organism’s life would be the soonest possible death, since σ attains its minimum value of zero at equilibrium.

The history here contains its own warning. Prigogine did apply the principle to biology, in a 1946 paper with Wiame proposing that living systems evolve towards states of least entropy production and offering, in the authors’ words, “a physicochemical interpretation of Lamarckism” (Prigogine and Wiame 1946). That paper is a substantial part of why the principle took hold among biologists, and its destination is a fair indication of what happens when a theorem with narrow hypotheses is carried into a domain that satisfies none of them.

Two further cautions belong here, since they generalise beyond this case. The first is that minimum and maximum entropy production are not the contradiction they appear to be: one minimises and the other maximises the same function, under different constraints and with respect to different variables, and which applies depends on the choice of thermodynamic variables and their behaviour under time reversal (Martyushev and Seleznev 2006). The second is a piece of history that ought to make anyone cautious about arguing from the word “entropy” in a title. The founding paper of the maximum-entropy-production literature in climate science was published as a theory of *minimum* entropy exchange; the two descriptions are the same physics, differing only in whether one counts the flux at the boundary as leaving or entering (Paltridge 1975).

The conclusion for this chapter is that the shared phrase is a coincidence. Minimum-entropy learning is not an instance of the minimum entropy production principle, does not inherit its guarantees, and would gain nothing if it did, since those guarantees hold only in a regime with no analogue here. There is one genuine structural inversion worth noticing on the way out. In Prigogine’s setting, low entropy production is where the dynamics *ends up on its own*; the principle is descriptive. In the setting of Section 6.6, low entropy is what we *pay* for, and having paid, we must then pay again to stop the system going lower. The two are not the same shape of statement at all.

6.10 So why call it minimum-entropy learning?

Given the preceding two sections, the name deserves a defence. It is defensible on two of three possible grounds, and it is worth being explicit about which.

As a literal description, it holds. The method minimises a named entropy functional over a well-defined space — the Shannon entropy of a distribution over semantic clusters as stated, the Rényi entropy at $\alpha = 2$ as implemented. Nothing metaphorical is happening; a specific

entropy of a specific distribution goes down.

As a structural claim, it holds, and this is the more interesting case. The objects that appear when the method is analysed are not borrowed imagery. The KL-regularised optimum is a Gibbs distribution; the objective is a free energy; the coefficient β enters exactly where a temperature enters; the self-consistency condition is a mean-field equation with a genuine critical point and a genuine first-order transition. These are the same mathematics as statistical physics, not a picture of it — which is the same standard that justifies calling a Boltzmann machine a Boltzmann machine.

As a claim to physical law, it does not hold and is not needed. No thermodynamic theorem is being invoked, no second law is doing any work, and no result transfers from Prigogine’s setting or from Schrödinger’s. The biological reading of Section 6.8 is an analogy about the *provenance of order* — a claim about where structure comes from, which the pass@k result independently establishes on its own terms — and not a derivation.

Keeping these three levels apart is the discipline this whole area most needs. The entropy vocabulary is powerful enough to describe learning objectives precisely, and seductive enough to make unearned claims sound rigorous. The first two grounds are earned. The third would not be.

6.11 When agreement means truth, and when it does not

The method rests on treating consensus as a proxy for correctness, and it is worth being precise about when that substitution is legitimate.

It works when the knowledge is already in the model but unstably expressed. The model can reach the answer; it simply does not do so reliably, and sharpening the distribution converts an occasional success into a dependable one. This is the regime Section 6.7 describes: the answer is already among the things the model can say, and training moves probability onto it. It also requires the question to have a stable meaning, since grouping by meaning is the only thing preventing “be consistent” from degenerating into “repeat yourself.”

It fails in two ways. The model may never have seen the material, in which case there is nothing to sharpen — no amount of self-agreement conjures knowledge the model never had. More dangerously, the model may be confidently and consistently wrong, in which case low entropy certifies a falsehood and training reinforces it. The paper’s own appendix contains a clean instance: on a probability problem the model silently substitutes a different goal, then twice announces that something looks wrong and reworks the solution — each rework landing on the arithmetic and never on the substituted goal. It returns 325 with high confidence; the answer is 265. Nothing in the reward can see the substitution: a model that re-derives the same misreading on resampling lands in the same cluster, the semantic entropy is low, and agreement is paid for being wrong. Fluent self-correction is not verification.

Three design rules follow, and they summarise most of what this chapter has established.

Threshold, do not minimise. Stop reducing entropy at a floor, leaving the model room to change its mind. Section 6.6 shows this is not a matter of taste: without a floor the fixed point is degenerate

and the approach to it is discontinuous. Schrödinger’s “fairly low,” not zero.

Group meanings, not strings. Otherwise consistency collapses into repetition, which is free to produce and worth nothing.

Never evaluate on the training signal. Consistency drives the optimisation; held-out ground truth is what gets reported. The moment those become the same number, the method is measuring its own success by the criterion it was trained to satisfy, and the failure case above becomes invisible.

6.12 Takeaway

Minimum-entropy learning turns a model’s disagreement with itself into a training signal, which buys reliable elicitation of existing capability without any labels at all — and buys nothing beyond that, by an information-theoretic argument that the $\text{pass}@k$ curves on mathematics confirm in its exact form. Its central pathology, degeneration to a single confident answer, is not an implementation nuisance but a phase transition in a mean-field free energy, which is why the durable remedies are floors on the entropy of the state — stopping rules for how much diversity sharpening may spend — rather than settings of a knob.

Set against Chapter 5, the pair is now complete: maximum entropy supplies possibility and minimum entropy supplies commitment, and neither is a general recipe for intelligence on its own. The next chapter shows that the two are not merely complementary but composable — that choosing *which features to model* turns out to require maximising entropy on the inside and minimising it on the outside, simultaneously.

7 Minimax-Entropy Learning

7.1 Which features should a model include?

Maximum entropy (Chapter 5) answers a precise question: given a set of measured features, which model is least biased? It is silent on a prior question — *which features should we measure at all?*

Every model is a lossy compression that trades complexity against uncertainty. Include too few features and the model stays vague; include too many and it ceases to be a compression. For a fixed budget of features, we want the description that leaves the least uncertainty.

7.2 The minimax entropy principle

Starting from the **minimum description length** (MDL) principle (Grünwald 2007), two results follow in sequence. First, among all models consistent with a feature set F , the shortest description is the maximum-entropy model (Feder 1986). Second, for such models the description length equals the model's own entropy, $L(P_F) = S(P_F)$. Choosing features to minimize description length therefore means choosing them to minimize that entropy.

The optimal features are the ones producing the maximum-entropy model with *minimum* entropy (Carcamo et al. 2025):

$$F^* = \arg \min_F S(P_F), \quad P_F = \arg \max_P \{ S(P) : \mathbb{E}_P[f_\mu] = \langle f_\mu \rangle_{\text{exp}}, f_\mu \in F \}.$$

The nesting gives the principle its name: maximize entropy on the inside, minimize it on the outside. The best features are those that maximally constrain our expectations while leaving the model maximally random about everything unobserved. The principle was first proposed for texture modeling in computer vision (Zhu et al. 1997).

The principle is closer to everyday practice than its combinatorial statement suggests. Let an energy-based model (Chapter 4) be linear in its last layer, $E_\theta(x) = -\sum_\mu w_\mu \phi_\mu(x; \theta')$, so that for fixed θ' the functions $\phi_\mu(\cdot; \theta')$ play the role of the feature set F . For fixed θ' the maximum-likelihood w is the maximum-entropy model P_F of these features, and at that point the cross-entropy of the data equals $S(P_F)$; this is the identity Equation 5.18 of Section 5.6.2, whose hypotheses are a linear last layer, an exact fit of w , and a uniform base measure. Maximum likelihood over θ' is therefore $\min_F S(P_F)$ over a continuous family of feature sets: minimax entropy, with gradient descent in place of greedy selection. In this sense every neural energy-based model trained by maximum likelihood is a minimax-entropy learner, whether or not anyone says so. What this chapter adds, and what the implicit version lacks, is the *explicit*

selection of a feature set under a description-length budget — a combinatorial outer problem whose solution can be inspected, bounded, and compared across budgets.

7.3 Three equivalent views

Minimizing $S(P_F)$ optimizes three quantities at once. In what follows, P_{true} denotes the distribution that actually generated the data, and P_{ind} the featureless baseline model in which the variables are treated as independent.

- **Shortest description.** Since $L(P_F) = S(P_F)$, the optimum is the most compressed account of the data.
- **Closest to the truth.** The entropy gap to the true distribution is exactly a divergence, $S(P_F) - S(P_{\text{true}}) = D_{\text{KL}}(P_{\text{true}} \| P_F)$, so the optimum is also the most accurate model.
- **Most informative features.** The information carried by the selected features, $I_F = S(P_{\text{ind}}) - S(P_F)$, is maximized at the same point.

Compression, accuracy, and informativeness coincide — a reassuring sign that the criterion is the right one.

7.4 Solving it

The difficulty is combinatorial. Each candidate feature set requires solving a maximum-entropy problem for its parameters, and the number of candidate sets grows super-exponentially, so brute-force search is infeasible beyond small systems (Carcamo et al. 2025). The practical workhorse is a **greedy** algorithm that repeatedly adds whichever remaining feature reduces the entropy most.

Structure sometimes restores exactness. For trees of pairwise correlations the entropy reduction decomposes into a sum of mutual informations, the problem becomes a maximum-spanning-tree problem, and greedy is optimal. More generally, if entropy reduction is **submodular** — diminishing returns as features accumulate — greedy comes within $1 - 1/e$ of the optimum (Nemhauser et al. 1978). Whether the minimax entropy objective is submodular in general remains open; where it is not, guarantees can still be recovered in terms of how far the objective departs from submodularity (Bian et al. 2017).

7.5 Takeaway

Minimax entropy couples the two preceding principles into a single criterion: **maximize entropy to model the world, minimize entropy to commit to what matters**. Maximum entropy supplies possibility, minimum entropy supplies certainty, and the minimax formulation decides which details deserve to be modelled at all.

The principle has recently yielded exactly solvable models of large neuronal populations (Lynn et al. 2025). Its implicit form runs inside every energy-based model trained by maximum

likelihood (Section 5.6.2), but its explicit form — the selection of a feature set under a description-length budget, with guarantees on sample complexity — remains largely untapped in machine learning. Open directions include scalability and sample complexity in high dimensions, realizations for LLMs, agents, and multimodal models, and the relationship to the free-energy principle (Friston 2010).

8 Energy-Based Guidance and Refinement

The previous chapters treated the energy as something to be learned from data. This chapter and the next treat a different situation, which is by now the most common one in practice: a large generative model is already trained, and we want to change what it generates without training it from scratch. We call the trained model the *reference*, written p_{ref} , and we call the change a *tilt*. Both chapters are organized around one object,

$$p^*(x) = \frac{p_{\text{ref}}(x) \exp(-E(x)/T)}{Z}, \quad Z = \sum_x p_{\text{ref}}(x) \exp(-E(x)/T), \quad (8.1)$$

which is the free-energy form of Section 5.1.1 with the reference as base measure, $h = p_{\text{ref}}$: p^* is the distribution closest to the reference, in Kullback–Leibler divergence, among those with a prescribed expected energy, and equivalently the minimizer of $\mathbb{E}_q[E] + T \text{KL}(q \| p_{\text{ref}})$. In the vocabulary of Section 5.1, Equation 8.1 is the representation level. The two chapters differ in *where the tilt is executed*. In this chapter it is executed at inference time: the reference is frozen, and a sampler does the work — gradient guidance, reranking, importance sampling, sequential Monte Carlo, or iterative refinement. This is the inference level of Section 5.1.2 carried out at test time. In Chapter 9 the tilt is absorbed into the weights of the reference, or the energy is learned through the reference; that is the learning level. These are not rival approaches, but rather two ways to obtain the same target distribution: one pays the cost in computation at inference (test) time, while the other pays it upfront through additional training.

What is fixed and what is fitted. It is worth being precise about the phrase “the energy is given”, since several methods in this chapter *train* an energy: a residual energy by noise-contrastive estimation, a verifier by pairwise ranking, a guidance network by regression. The distinction we emphasize is not whether the energy itself is trained or untrained, but rather which part in Equation 8.1 is being held fixed and which is being adjusted. In this chapter the *reference* is fixed; its parameters are never updated, and the energy, however it was obtained, enters only through the sampler. Training the energy is a learning problem in its own right — a maximum-entropy learner in the sense of Section 5.1.4 whose data are labels, preferences, or contrastive noise — but it is a separate learner, and its output is then handed to the sampler. So “energy given” should be read as “energy given to the sampler, reference frozen”. Two methods at the end of the chapter blur even this line, because in them the energy *is* the generator: Energy Matching and energy-based transformers have no separate reference, and sampling is refinement on the energy itself. We treat them as the limit in which the tilt has nothing left to tilt.

8.1 The tilted reference

Three remarks on Equation 8.1 organize the chapter.

First, the temperature has the status of a genuine dial (Section 5.1.4): the energy is given, no constraint level is prescribed, and T is chosen. It appears in the literature under several names. In classifier guidance for diffusion models it is the reciprocal of the guidance scale w ; in KL-regularized fine-tuning it is the KL coefficient β ; in best-of- N selection, as we show in Section 8.4, the number of candidates N plays the role of an inverse temperature. Small T concentrates p^* on the low-energy region and moves it far from the reference; large T leaves the reference almost untouched. The zero-temperature limit is energy minimization on the support of the reference, and it is where guidance stops being a probabilistic method (Section 5.6.3).

Second, the energy is additive. If two energies E_1 and E_2 encode two requirements, the tilt by $E_1 + E_2$ is the product of the two tilts normalized once, i.e., a product of experts (Hinton 2002). This is why guidance composes: a classifier, a constraint, and a reward can be added without retraining anything. The price is a partition function that depends on all of them together, and most of the technical content of the chapter is about how each sampler avoids or estimates that partition function.

Third, the constrained reading is available. By the duality of Section 5.1.1, p^* is the I-projection of the reference onto the set $\{q : \mathbb{E}_q[E] = U\}$ for the level U implied by T . The reading matters for one practical question: how far from the reference a tilt is allowed to travel. The Kullback–Leibler divergence $\text{KL}(p^* \parallel p_{\text{ref}})$ is a budget, and several results below are bounds on that budget.

8.2 Continuous guidance

For continuous data the reference is usually a diffusion or flow model, and the sampler has a gradient available. Classifier guidance (Dhariwal and Nichol 2021) is the original instance of Equation 8.1 in this setting. To generate from $p(x \mid y) \propto p(x)p(y \mid x)$ with a pretrained unconditional model $p(x)$ and a classifier $p(y \mid x)$, one adds the classifier gradient to the score at every noise level,

$$\nabla_{x_t} \log p_t(x_t \mid y) \approx \nabla_{x_t} \log p_t(x_t) + w \nabla_{x_t} \log p(y \mid x_t), \quad (8.2)$$

with a guidance scale $w \geq 1$ that sharpens the conditional. In our notation the energy is $E = -\log p(y \mid x)$ and $T = 1/w$; the target is a product of experts, and the classifier must be trained on noisy inputs x_t . Classifier-free guidance (Ho and Salimans 2022) removes the separate classifier by training the generator with and without the condition and extrapolating, $\hat{\epsilon} = (1 + w) \epsilon_\theta(x_t, y) - w \epsilon_\theta(x_t)$, which amounts to using the implicit classifier $p(y \mid x_t) \propto p(x_t \mid y)/p(x_t)$ as the energy.

For a general energy J on clean data, i.e., a target $p'(x_0) \propto p(x_0) e^{-J(x_0)}$, the difficulty is that the sampler runs on noisy states x_t and needs the energy *there*. The exact intermediate energy is

$$E_t(x_t) = -\log \mathbb{E}_{p(x_0|x_t)}[e^{-J(x_0)}], \quad (8.3)$$

a free energy of the posterior over clean samples given the noisy one, and its gradient is the exact guidance. Lu et al. (Lu et al. 2023) show that Equation 8.3 is the only guidance consistent with the marginals of the reverse process and propose contrastive energy prediction to learn it; the common shortcut evaluates J at the posterior mean $\hat{x}_0(x_t)$ instead, which is a first-order approximation. Feng et al. (Feng et al. 2025) carry the analysis to general flow matching, where the reference is a vector field v_t generating a probability path from noise to data. They ask for a guidance field g_t such that $v_t + g_t$ generates the path to $p'(x) \propto p(x) e^{-J(x)}$, and derive three families: an asymptotically exact Monte Carlo guidance that reweights samples of the coupling by e^{-J} , a local guidance obtained by Taylor expansion that is proportional to ∇J and recovers the classical gradient guidance of diffusion models under affine Gaussian paths, and training-based exact guidance with its own regression losses. The value of this framework for us is the hierarchy it makes explicit: exact guidance is an expectation under the posterior of the reference, and gradient guidance is its cheapest approximation.

When several energies are composed, the approximation error of gradient guidance compounds, since the reverse process only sees the sum of approximate scores. Du et al. (Du et al. 2023) show that running a few steps of Markov chain Monte Carlo (Langevin or Hamiltonian) on the composed energy inside each reverse step corrects the composition; the sampler then targets the product of experts rather than the sum of its approximations. This is refinement in the sense of Section 8.5, applied locally in noise level.

Energy Matching (Balcerak et al. 2025) takes the opposite route to the same end. Instead of tilting a flow with an energy, it learns a single time-independent scalar potential $V_\theta(x)$ whose gradient plays two roles: far from the data manifold it is a transport field along optimal transport paths, and near the manifold it is the drift of Langevin dynamics, so that the sampler equilibrates in a Boltzmann well $\propto e^{-V_\theta}$ around the data. Additional requirements are added as energy terms — the authors use an interaction energy to spread samples over modes in protein design — which is the additivity of Section 8.1 with the potential itself as the reference. Here the energy and the generator coincide; there is no frozen reference to tilt, and we return to this case in Section 8.5.

8.3 Discrete guidance: residual energies over a proposal

For sequences the sampler has no gradient in the data space, and the tilt is implemented by *proposing* from the reference and *reweighting* by the energy. The construction is the residual energy-based model of Deng et al. (Deng et al. 2020),

$$p(x) = \frac{p_{\text{AR}}(x) \exp(-E_\phi(x))}{Z_\phi}, \quad (8.4)$$

in which a pretrained autoregressive language model p_{AR} is the proposal, E_ϕ is a bidirectional scorer of the whole sequence, and generation draws n candidates from p_{AR} and resamples

them with weights $\propto e^{-E_\phi}$. The energy is trained by noise-contrastive estimation with the proposal as the noise distribution (Gutmann and Hyvärinen 2010), so that, in the classification of Section 5.6.3, the residual EBM is a maximum-entropy learner in the generalized sense whose inference is importance sampling from the tilt. The same construction at the token level is Electric (Clark et al. 2020b), an energy-based cloze model trained by conditional noise-contrastive estimation; its authors show that ELECTRA’s replaced-token detection (Clark et al. 2020a) is a closely related negative-sampling objective, which gives a principled reading of what that discriminative pre-training learns: an energy over tokens in context.

Discrete diffusion language models sharpen the need for a residual energy. A masked diffusion model decodes several tokens in parallel at each step, but its denoiser factorizes over positions, $p_\theta(x_{t-1} \mid x_t) = \prod_i p_\theta(x_{t-1}^i \mid x_t)$, so the tokens decoded together are conditionally independent given the current state; with few steps the independence assumption is badly violated and quality degrades. The energy-based diffusion language model (EDLM) of Xu et al. (Xu et al. 2025) puts a residual energy on each denoising step,

$$p_{\text{EDLM}}(x_{t-1} \mid x_t) \propto p_{\text{diffusion}}(x_{t-1} \mid x_t) \exp(-E(x_{t-1}, x_t)), \quad (8.5)$$

and shows that the energy can be read off a pretrained autoregressive model or fitted by noise-contrastive estimation from a bidirectional transformer; the sampler is parallel importance sampling over candidates from the diffusion proposal. On language-modeling benchmarks EDLM approaches autoregressive perplexity and reaches the same generation quality as the diffusion baseline in fewer steps (a $1.3\times$ sampling speedup is reported).

Our own contribution to this line is the *unified energy* (Uni-E) of Yan et al. (Yan et al. 2026), which starts from a decomposition of the gap between a diffusion language model and the data distribution. Writing the Kullback–Leibler divergence along a decoding order and expanding it, the gap splits into three factors: a *capacity* factor, i.e., how well the denoiser fits each position; a *dependency* factor, i.e., whether the tokens decoded in one step are independent given the context; and an *invariance* factor, i.e., whether a token decoded now would be decoded the same way once more context is revealed. The second factor is handled by the independent energy of EDLM, which compares the denoiser with a proxy autoregressive model on the fully decoded sequence,

$$E_{\text{ind}}(x_0, x_t) = \log \frac{p_\theta(x_0 \mid x_t)}{p_{\text{AR}}(x_0 \mid x_t)} - \log F_{\text{ind}}(x_t). \quad (8.6)$$

For the third factor the paper introduces an *invariant energy*. Let x_s be a partially revealed sequence between x_t and x_0 , obtained by re-masking; the decoding of x_s from x_t is a “future step”, and a token is invariant if revealing it does not change what the proxy model believes about the rest. The invariant energy is

$$E_{\text{inv}}(x_0, x_s, x_t) = \log \frac{p_\theta(x_s \mid x_t)}{p_{\text{AR}}(x_s \mid x_t)} + \log \frac{p_{\text{AR}}(x_0 \mid x_t)}{p_{\text{AR}}(x_0 \mid x_s)} - \log F_{\text{inv}}(x_t), \quad (8.7)$$

whose second term is small exactly when the intermediate reveal x_s leaves the proxy’s posterior

over x_0 unchanged. By the additivity of energies the two are combined into the unified energy $E_{\text{uni}} = E_{\text{inv}} + E_{\text{ind}}$, parameterized without its normalizer as

$$E_{\phi}(x_0, x_s, x_t) = \log \frac{p_{\theta}(x_s | x_t)}{p_{\text{AR}}(x_s | x_t)} + \log \frac{p_{\theta}(x_0 | x_t)}{p_{\text{AR}}(x_0 | x_s)}. \quad (8.8)$$

The property that distinguishes Uni-E from other residual energies is that its partition function is available in closed form. Summing $p_{\theta}(x_0 | x_t) e^{-E_{\phi}}$ over x_0 , the denoiser cancels against the second log-ratio and what remains is $p_{\text{AR}}(x_0 | x_s)$, which sums to one; hence

$$\log F(x_t) = \log \frac{p_{\text{AR}}(x_s | x_t)}{p_{\theta}(x_s | x_t)}, \quad (8.9)$$

and the tilted denoiser is the proxy model’s posterior given the revealed sequence x_s . No Monte Carlo estimate of the normalizer is needed, which is the usual bottleneck of residual EBMs. The paper proves that the resulting decoder has a strictly smaller gap to the data distribution than the diffusion model whenever the dependency or invariance factor is non-zero, and reports that on diffusion language models Uni-E reaches quality comparable to autoregressive decoding with about a 22% speedup, while on diffusion large language models it mitigates the degradation that otherwise grows with the degree of parallelism. Two caveats should be stated. The invariant energy relies on a proxy autoregressive model and on a sampling estimate of the invariance term whose accuracy is characterized rather than exact (the paper’s Lemma 4.1 gives the approximation factor). And the whole construction is a tilt of a *frozen* denoiser: the diffusion model is not retrained, and the energy, whether read off the proxy or fitted by noise-contrastive estimation, enters only through the reweighting of candidates. It is discrete guidance in the sense of this chapter, and its closed-form normalizer is what makes it exact rather than approximate.

8.4 Verifiers as energies, best-of- N , and sequential Monte Carlo

The simplest energy for a language model is a *verifier*: a scalar score of a complete output, low for good outputs and high for bad ones. Outcome reward models trained to predict correctness (Cobbe et al. 2021) and process reward models trained on step-level labels (Lightman et al. 2024) are energies of this kind. The learning signal is usually pairwise: given a preferred output y_w and a rejected one y_l , the Bradley–Terry model (Bradley and Terry 1952)

$$P(y_w \succ y_l) = \sigma(E(y_l) - E(y_w)) \quad (8.10)$$

is fitted by maximum likelihood, which is logistic regression on the difference of energies and hence the maximum-entropy classifier of Section 5.1.4 applied to pairs. The energy outcome reward model (EORM) of Jiang et al. (Jiang et al. 2025) shows how little is needed: a 55M-parameter encoder with a scalar energy head, trained with the pairwise loss over all correct–incorrect pairs of sampled chains of thought, using outcome labels only. At inference the verifier reranks a pool of N candidates from the language model and returns $\arg \min_y E_{\theta}(y)$;

the partition function is never needed, since the candidates share it. In the terms of this chapter, reranking is the zero-temperature limit of the tilt on the finite support of the candidate pool, and the reported gains (Llama 3 8B rises to 90.7% on GSM8K and 63.7% on MATH) are those of best-of- N with a learned energy in place of a large reward model.

Best-of- N itself deserves a precise statement, because it is the most widely used inference-time tilt and it is *not* the exponential tilt of Equation 8.1. Drawing N samples from the reference and keeping the one with the highest reward induces a distribution π_{BoN} that is a monotone reweighting of p_{ref} by the rank of the reward. Beirami et al. (Beirami et al. 2024) show that the expression $\log N - (N - 1)/N$, often quoted as the KL divergence between π_{BoN} and the reference, is an *upper bound*,

$$\text{KL}(\pi_{\text{BoN}} \| p_{\text{ref}}) \leq \log N - \frac{N - 1}{N}, \quad (8.11)$$

and that the win rate of π_{BoN} against the reference is bounded by $N/(N + 1)$. Gui, Gârbasea, and Veitch (Gui et al. 2024) embed best-of- N and the distributions learned by alignment methods in a common class of tiltings of the reference and show that, within this class, best-of- N is essentially optimal for the trade-off between win rate and KL divergence. Read through Equation 8.1, N is an inverse temperature: the KL budget grows like $\log N$, exactly as the budget of an exponential tilt grows as T falls. The cost is N forward passes per output; distilling the best-of- N distribution into the weights is a post-training method, and we return to it in Section 9.2.

Between reranking whole outputs and gradient guidance sits sequential Monte Carlo. Zhao et al. (S. Zhao et al. 2024) observe that a large part of language model alignment, red-teaming, and constrained generation can be cast as sampling from a target $\sigma(x_{1:T}) \propto p_{\text{ref}}(x_{1:T}) \phi(x_{1:T})$ with a potential ϕ on complete sequences, i.e., Equation 8.1 with $\phi = e^{-E/T}$, and that the right inference-time tool is sequential Monte Carlo with *twist functions* $\psi_t(x_{1:t}) \approx \mathbb{E}[\phi(x_{1:T}) \mid x_{1:t}]$, the expected future potential given a prefix. The twists are soft value functions: their logarithms satisfy the soft Bellman recursion of Section 5.5.2, and the paper’s contrastive method for learning them is a soft reinforcement learning algorithm. Resampling partial sequences by their twists focuses computation on promising prefixes, and the authors derive bidirectional bounds on $\log Z$ from the same machinery, which let one measure how far an inference method is from the target in KL divergence in both directions. Controlled decoding (Mudgal et al. 2024) is the same idea with a learned prefix scorer used either token by token or blockwise; in both cases the reference is frozen and the energy acts through the prefix values.

8.5 Refinement and reasoning as energy minimization

Rather than accepting a single forward pass, one can iteratively lower the energy of a candidate — an optimization view of “thinking longer”. Pure descent,

$$x \leftarrow x - \eta \nabla_x E(x), \quad (8.12)$$

is the zero-temperature limit of Chapter 3: it consults the energy and ignores the entropy, and inherits the classical weakness of local minima. Adding noise and lowering it over the course of the process, i.e., annealing (Kirkpatrick et al. 1983), is the finite-temperature version, and it is the schedule that diffusion samplers implement under another name. For text, where the data space is discrete, COLD decoding (Qin et al. 2022) runs Langevin dynamics on a continuous relaxation of the sequence under a composed energy and rounds at the end; for images, the MCMC corrections of Section 8.2 are refinement inside each reverse step. In every case the number of steps is a compute knob that buys quality, and the target of the refinement is a tilt of a frozen reference.

Energy-based transformers (EBTs) (Gladstone et al. 2026) carry refinement to the point where it becomes the model. An EBT assigns an energy to a pair of input and candidate prediction and produces its prediction by gradient descent on the candidate until convergence; the number of descent steps is chosen at inference time, which the authors frame as System-2 thinking learned from unsupervised pre-training alone, without verifiers or verifiable rewards. They report that EBTs scale faster than a strong autoregressive Transformer baseline during pre-training and improve with inference-time computation, with larger gains on data farther from the training distribution. Two points matter for this book. Learning here is the shaping of the landscape of Section 4.3 so that the correct continuation is the minimizer, and inference is the zero-temperature inference of Section 5.1.2 with a variable budget; and, as in Energy Matching, there is no separate reference: the energy is the generator, so the “tilt” of Equation 8.1 degenerates to the energy itself. This is the boundary case announced at the beginning of the chapter, and it is instructive because it shows the two roles of the energy — verifier and generator — collapsing into one function.

The unifying perspective, then, is the following. A reasoning problem defines an energy landscape over candidate solutions; reasoning is the search for low-energy candidates, by guidance, reranking, sampling, or refinement; and learning shapes the landscape so that correct or self-consistent solutions sit at its minima. Maximum-entropy learning (Chapter 5) defines the landscapes, minimum-entropy learning (Chapter 6) sharpens them from the model’s own agreement, and the methods of this chapter navigate them at test time.

8.6 Guidance on the three levels

Table 8.1 reads the methods of this chapter through the three levels of Section 5.1. In every row the reference is fixed and the inference level is where the method lives; the energy column records where the energy came from, which is a separate learning problem when it was trained.

Table 8.1: Inference-time methods read through the three levels. The base measure is never updated; where an energy was trained, that training is a separate maximum-entropy learner whose output is handed to the sampler.

Method	Base measure (frozen)	Energy and its source	Sampler	Temperature
Classifier guidance (Dhariwal and Nichol 2021); classifier-free guidance (Ho and Salimans 2022)	diffusion prior	$-\log p(y x_t)$: a classifier trained on noisy inputs, or the implicit classifier $p(x_t y)/p(x_t)$	score plus gradient of the energy	guidance scale: $T = 1/w$ (classifier), $1/(1 + w)$ (classifier-free)
Exact energy guidance, CEP (Lu et al. 2023); flow-matching guidance (Feng et al. 2025)	diffusion / flow	given J on clean data; intermediate energy Equation 8.3, exact (Monte Carlo, learned) or local (gradient)	guided score or vector field	T of the tilt
Compositional generation with MCMC (Du et al. 2023)	diffusion prior	sum of given energies (product of experts)	Langevin / HMC steps inside each reverse step	T ; number of MCMC steps
Residual EBMs (Deng et al. 2020); Electric (Clark et al. 2020b)	autoregressive LM (Deng et al.); a cloze-model noise distribution (Electric)	residual energy, NCE-trained with the base measure as noise	sample-then-resample by e^{-E_ϕ} ; reranking	absorbed in the scale of E_ϕ
EDLM (Xu et al. 2025); Uni-E (Yan et al. 2026)	diffusion LM denoiser	log-ratio to a proxy AR model, or NCE-trained	parallel importance sampling over candidates; closed-form Z for Uni-E	absorbed
COLD decoding (Qin et al. 2022)	LM, entering as a fluency energy	composed energies: fluency, constraints	Langevin on soft tokens, then rounding	T ; step count
Verifier reranking: ORM (Cobbe et al. 2021), PRM (Lightman et al. 2024), EORM (Jiang et al. 2025)	LM candidate pool	verifier trained from labels (ORM, PRM) or Bradley–Terry pairs (EORM)	arg min of the energy over the pool	$T \rightarrow 0$
Best-of- N (Beirami et al. 2024; Gui et al. 2024)	LM	reward, given	keep the best of N	not a Gibbs tilt; KL budget $\leq \log N - (N-1)/N$

Method	Base measure (frozen)	Energy and its source	Sampler	Temperature
Twisted SMC (S. Zhao et al. 2024); controlled decoding (Mudgal et al. 2024)	LM	potential $\phi = e^{-E/T}$, given; twists / prefix scorer learned	resampling of prefixes by twists; token-wise or blockwise reweighting by the prefix scorer	T of the potential
Energy Matching (Balcerak et al. 2025); EBTs (Gladstone et al. 2026)	— (energy is the generator)	learned potential / energy	Langevin in the Boltzmann well; gradient descent on the candidate	T ; number of steps

The table also makes the trade-off with the next chapter visible. Every row pays for the tilt at test time, in candidates, steps, or particles, and the payment is repeated for every query. Post-training pays once, by fitting the reference so that a single forward pass approximates the tilt, and gives up the ability to change the energy afterwards without retraining. The choice between them is a choice between computation and storage, or, in the terms of Gladstone et al. (2026), between System 2 and System 1.

8.7 Takeaway

Guidance, reranking, sequential Monte Carlo, and refinement are one operation performed by different samplers: the tilt of a frozen reference by an energy, Equation 8.1, which is the maximum-entropy model of Section 5.1 with the pretrained generator as base measure. The energy may be a classifier, a composed constraint, a residual log-ratio to a proxy model, a learned verifier, or a reward; when it was trained, that training is a separate learner, and the sampler treats its output as given. The temperature is a genuine dial that appears as a guidance scale, a KL coefficient, or the size N of a candidate pool, and the zero-temperature limit — descent and reranking — is where the method leaves the family. The next chapter asks what changes when the reference itself is allowed to move.

9 Maximum-Entropy Post-Training

Post-training is the name that has become standard for everything done to a pretrained model after pre-training: supervised fine-tuning on demonstrations, reinforcement learning from human or verifier feedback, preference optimization, and their many variants. The methods look different and are usually presented as a list. In this chapter we argue that most of them are one thing, and that the one thing is maximum-entropy learning in the sense of Section 5.1 with the pretrained model as base measure. Specifically, the object every method approximates is the tilt of the reference policy,

$$\begin{aligned}\pi^*(y \mid x) &= \frac{\pi_{\text{ref}}(y \mid x) \exp(r(x, y)/\beta)}{Z_\beta(x)} \\ &= \arg \min_{\pi} \mathbb{E}_{\pi}[-r(x, y)] + \beta \text{KL}(\pi(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x)),\end{aligned}\tag{9.1}$$

which is Equation 8.1 with the pretrained model as reference, the negative reward as energy, and the KL coefficient β as temperature; it was derived in Section 6.6 as Equation 6.2. Chapter 8 sampled from this tilt at inference time with the reference frozen. Post-training moves to the learning level: either the reference is fitted so that a single forward pass approximates π^* — we call this *amortizing* the tilt — or the energy r is itself learned from preferences, outcomes, or the model’s own behaviour. We treat the two in turn, then the continuous case of diffusion and flow models, and finally the boundary of the family.

9.1 From the reference to its tilt

The KL-regularized objective on the right of Equation 9.1 has a longer history than its current use. In control it is the KL-control problem of Section 5.5.1, in which the base measure is the passive dynamics (Todorov 2006, 2009). For sequence models it was introduced by Jaques et al. (Jaques et al. 2017), who fine-tuned a recurrent network with a reward while penalizing the divergence from the maximum-likelihood model it started from, precisely to keep what the model had learned from data. Ziegler et al. (Ziegler et al. 2019) used the same penalty to fine-tune language models from human preferences, and the recipe of reward model plus KL-regularized policy optimization became reinforcement learning from human feedback (RLHF) (Christiano et al. 2017; Ouyang et al. 2022). It is worth noting why the penalty is not optional. Without it, maximizing $\mathbb{E}_{\pi}[r]$ over all policies places the entire mass on the arg-max of the reward, a degenerate distribution; Korbak, Perez, and Buckley (Korbak et al. 2022) make the point sharply and show that the KL-regularized problem is equivalent to variational inference: minimizing $\text{KL}(\pi \parallel \pi^*)$ for the target Equation 9.1, i.e., approximating a Bayesian posterior in which the reference is the prior and $e^{r/\beta}$ the likelihood. In the terms of this book, the entropy

term is load-bearing (Section 5.1.4), and removing it is the zero-temperature boundary of Section 5.6.3.

The constrained form of Section 5.1.1 has an instance of its own. Generation with distributional control (GDC) (Khalifa et al. 2021) asks for the distribution p closest to a pretrained language model a that satisfies moment constraints on chosen features,

$$\begin{aligned} \min_p \text{KL}(p \parallel a) \quad \text{s.t.} \quad \mathbb{E}_p[\phi_i(x)] &= \bar{\mu}_i, \quad i = 1, \dots, K, \\ \implies p(x) &= \frac{a(x) \exp(\lambda^\top \phi(x))}{Z}. \end{aligned} \quad (9.2)$$

The constraints can be pointwise (every sample mentions sports) or distributional (half of the samples mention female characters), the solution is an energy-based model over the reference, and the authors describe the framework as a generalization of the maximum-entropy principle — which it is: Equation 9.2 is Equation 5.1 with $h = a$. The two forms sit side by side in Table 9.1. RLHF declares an energy and a temperature; GDC declares features and moments; the multipliers of one are the temperature of the other, and both are tilts of the same reference. Go et al. (Go et al. 2023) close the circle by showing that RLHF minimizes the reverse KL divergence to its implicit target while GDC minimizes the forward KL divergence to its explicit one, and by allowing any f -divergence in between; in their experiments the Jensen–Shannon divergence often gives the best balance between alignment and diversity. In the vocabulary of Section 5.6.3, the choice of divergence at the amortization step is the generalized sense of maximum-entropy learning, now applied to the fitting of the policy rather than of the energy.

Table 9.1: The two forms of the maximum-entropy model, instantiated in post-training.

	Free-energy form (RLHF, KL-control)	Constrained form (GDC)
Declared	reward r , temperature β	features ϕ , moments $\bar{\mu}$
Target	$\pi_{\text{ref}} e^{r/\beta} / Z$	$a e^{\lambda^\top \phi} / Z$
Temperature	a genuine dial	determined by the moments
Amortization	reverse KL, policy gradient	forward KL, distributional policy gradient
Generalization	any f -divergence (Go et al. 2023)	

The temperature of Equation 9.1 is a genuine dial in the sense of Section 5.1.4: the reward is given, no level is prescribed, and β is chosen. Small β moves the policy far from the reference and toward the arg-max of the reward; large β leaves it nearly unchanged. The same trade-off appears as a KL constraint rather than a penalty in trust-region methods (Schulman et al. 2015), which is the constrained form once more.

9.2 Amortizing the tilt

Amortization fits a parametric policy π_θ to $\pi^\star$ so that generation is one forward pass. The methods differ in the divergence they minimize and in how they estimate its gradient; none of them changes the target.

Reverse KL and policy gradients. Minimizing $\text{KL}(\pi_\theta \parallel \pi^\star)$ is, up to a constant, maximizing the right-hand side of Equation 9.1 in θ , and its gradient is a policy gradient with a shaped reward,

$$\nabla_\theta \mathcal{J}(\theta) = \mathbb{E}_{y \sim \pi_\theta(\cdot | x)} \left[\left(r(x, y) - \beta \log \frac{\pi_\theta(y | x)}{\pi_{\text{ref}}(y | x)} - b(x) \right) \nabla_\theta \log \pi_\theta(y | x) \right], \quad (9.3)$$

with any baseline $b(x)$ (Williams 1992). Proximal policy optimization (Schulman et al. 2017) estimates this with a learned value baseline and a clipped surrogate, and is the algorithm behind the RLHF of Ouyang et al. (2022). Two recent variants remove the value network. Group relative policy optimization (GRPO) (Shao et al. 2024) samples a group of G outputs $\{o_i\}$ for each prompt and uses the group as its own baseline,

$$\begin{aligned} \hat{A}_i &= \frac{r_i - \text{mean}(r_1, \dots, r_G)}{\text{std}(r_1, \dots, r_G)}, \\ \mathcal{L}_{\text{GRPO}} &= -\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left[\min(\rho_{i,t} \hat{A}_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_i) - \beta \widehat{\text{KL}}_{i,t} \right], \end{aligned} \quad (9.4)$$

where $\rho_{i,t}$ is the token-level probability ratio to the sampling policy and the KL term is estimated per token by the unbiased, non-negative estimator

$$\widehat{\text{KL}}_{i,t} = \frac{\pi_{\text{ref}}(o_{i,t} | \cdot)}{\pi_\theta(o_{i,t} | \cdot)} - \log \frac{\pi_{\text{ref}}(o_{i,t} | \cdot)}{\pi_\theta(o_{i,t} | \cdot)} - 1. \quad (9.5)$$

REINFORCE leave-one-out (RLOO) (Ahmadian et al. 2024) is the sequence-level version: the baseline for each sample is the mean reward of the other $G - 1$ samples, there is no clipping, and the authors show that this simpler estimator matches or outperforms PPO in the RLHF setting. What we want the reader to see is that Equation 9.4 and RLOO are estimators of the same gradient Equation 9.3. The group normalization and the clipping are variance reduction; the target is unchanged, and the KL term is the entropy term of the free energy, estimated rather than computed. Every variant in this family is therefore an approximate maximum-entropy learner in the strict sense of Section 5.1.4, in which the inner problem is solved by amortization rather than by sampling.

Forward KL and distribution matching. The distributional policy gradient (DPG) of Parshakova, Andreoli, and Dymetman (Parshakova et al. 2019), which GDC (Khalifa et al. 2021) adopts, minimizes the forward divergence $\text{KL}(p \parallel \pi_\theta)$ to the explicit target Equation 9.2 by importance sampling,

$$\nabla_{\theta} \text{KL}(p \parallel \pi_{\theta}) = -\mathbb{E}_{x \sim q} \left[\frac{p(x)}{q(x)} \nabla_{\theta} \log \pi_{\theta}(x) \right], \quad (9.6)$$

with a proposal q that is periodically updated to the current policy and an estimate of Z . GFlowNet fine-tuning (Hu et al. 2024) trains the policy so that complete sequences are sampled with probability proportional to a reward, $\pi_{\theta}(x) \propto R(x)$, which is the tilt at $\beta = 1$ with $r = \log R$ and the reference absorbed into R ; the training objective is a subtrajectory-balance loss rather than a divergence, and the authors call the result amortized inference, which is the exact name. Rejection-sampling fine-tuning — sample from the current model, keep the outputs the reward accepts, and fit by maximum likelihood on the survivors (Dong et al. 2023; Gulcehre et al. 2023) — is forward-KL amortization of a truncated tilt, and distilling the best-of- N distribution of Section 8.4 into the weights (Gui et al. 2024) is the same move with the best-of- N tilt as target, implemented as maximum likelihood on the best sample together with a pairwise term on the best and the worst of the N . The divergence matters: reverse KL is mode-seeking and drives the policy toward the reward’s arg-max as β falls, whereas forward KL is mass-covering and needs samples from, or importance weights for, the target. Neither is right in general, which is the empirical finding of Go et al. (2023).

9.3 Learning the energy

The reward of Equation 9.1 is rarely given. It is learned, and the way it is learned is where post-training becomes a maximum-entropy learner in the sense of iii) in Section 5.1.4.

Reward models from pairs. The standard supervision is a comparison: for a prompt x , a preferred response y_w and a rejected one y_l . The Bradley–Terry model (Bradley and Terry 1952)

$$P(y_w \succ y_l \mid x) = \sigma(r_{\theta}(x, y_w) - r_{\theta}(x, y_l)) \quad (9.7)$$

is fitted by maximum likelihood, which is logistic regression on the difference of rewards and hence the maximum-entropy classifier of Section 5.1.4 applied to pairs; its likelihood equations match the empirical and predicted win frequencies. Outcome and process reward models (Cobbe et al. 2021; Lightman et al. 2024) and the small verifier of Section 8.4 are trained in this way, and the trained reward is then either handed to a sampler (Chapter 8) or amortized (Section 9.2).

Direct preference optimization: the energy through the closed form. Direct preference optimization (DPO) (Rafailov et al. 2023) removes the separate reward model by substituting the representation level into the learning level. Inverting Equation 9.1 gives

$$r(x, y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z_{\beta}(x), \quad (9.8)$$

and the partition function cancels in the difference that enters Equation 9.7. Replacing π^* by the policy being trained yields the loss

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right], \quad (9.9)$$

whose gradient weights each pair by how wrongly the implicit reward $\hat{r}_\theta = \beta \log(\pi_\theta/\pi_{\text{ref}})$ currently orders it. Nothing is sampled during training, which is the practical appeal; the conceptual point is that the log-ratio of the policy to the reference *is* the energy, so that learning the policy and learning the energy are one step. Rafailov et al. (Rafailov et al. 2024) extend the reading to the token level: in the token Markov decision process, the token-wise log-ratio $\beta \log \pi^*(a_t | s_t)/\pi_{\text{ref}}(a_t | s_t)$ is the soft advantage $Q^* - V^*$ of maximum-entropy reinforcement learning, so DPO fits soft Q -values of Section 5.5.2 with the reference as base measure. The limitations are those of any learned energy: the implicit reward is defined only relative to π_{ref} , the data are off-policy, and a small β lets the policy travel far from the reference on the strength of a few comparisons.

Energies from contrastive noise. The residual energies of Section 8.3 are the third route: noise-contrastive estimation with the reference as noise distribution (Deng et al. 2020; Xu et al. 2025; Yan et al. 2026). This is a maximum-entropy learner in the generalized sense of Section 5.6.3, and its output is used at inference time, so it belongs to both chapters — learned here, applied there.

9.4 Self-referential energies

A last class of methods takes the reward from the model itself. EMPO (Zhang et al. 2025) rewards a response by the frequency of its semantic cluster among the model’s own samples; test-time reinforcement learning (TTRL) (Zuo et al. 2025) does the same with an exact-match majority vote as the pseudo-label; and in test-time adaptation the analogous move is to minimize the entropy of the model’s predictions on the test stream, which Park et al. (Park et al. 2025) show should be paired with an explicit energy objective to be reliable. Formally these are instances of Equation 9.1 with r replaced by a functional of π itself, and the closed form becomes the self-consistency equation Equation 6.3. The consequence, worked out in Section 6.6, is a rich-get-richer dynamics with a collapse threshold, and the reason is that the outer criterion has been reversed: an external reward tilts the reference toward what the world values, whereas an internal one tilts it toward what the model already believes, and the entropy term now fights the objective rather than regularizing it. This is why we place these methods in Chapter 6 and not here. They share the inner Gibbs tilt with maximum-entropy post-training and differ in the sign of the outer problem; the two chapters are the two directions of the dial of Chapter 3.

9.5 The continuous case

The same structure recurs, almost verbatim, for diffusion and flow models, with the pretrained generator p_{base} as reference and a reward on clean samples as energy. Black et al. (Black et al. 2024) cast denoising as a multi-step decision process and optimize a reward by policy gradient (DDPO). It is worth noting that DDPO has no KL term: the authors control over-optimization

by stopping early, so that on its own the method sits at the boundary of the family described in Section 9.6, and it is the concurrent DPOK of Fan et al. (Fan et al. 2023), which adds a per-step KL penalty to the pretrained model, that brings the construction back inside. Wallace et al. (Wallace et al. 2024) carry DPO over to diffusion models by writing the preference likelihood on denoising trajectories. Uehara et al. (Uehara et al. 2024) make the target explicit: fine-tuning as entropy-regularized control against the pretrained model has the tilt $p_{\text{base}}(x) e^{r(x)/\alpha}$ as its solution, and the regularization is what prevents the “reward collapse” of over-optimizing an imperfect learned reward. Domingo-Enrich et al. (Domingo-Enrich et al. 2025) then show that the naive stochastic-optimal-control formulation carries an *initial value function bias*: the generated distribution picks up a term that depends on the initial noise and is not the tilt. The remedy is a particular memoryless noise schedule, under which the fine-tuned flow or diffusion model provably converges to $p_{\text{base}} e^r$, and their adjoint matching algorithm turns the control problem into a regression. Seen in light of Section 8.2, guidance and reward fine-tuning are two perspectives on the same underlying tilt: guidance injects the gradient correction from Equation 8.3 during sampling, while fine-tuning incorporates that same adjustment directly into the model’s drift.

9.6 Post-training on the three levels, and where it stops

Table 9.2 places the methods of this chapter on the three levels of Section 5.1. The representation level is the same in every row; the rows differ in what is computed and in whether the energy is given, learned, or self-referential.

Table 9.2: Post-training methods on the three levels. Amortization fits the reference to a fixed tilt; energy learning fits the tilt itself; self-referential rewards reverse the outer criterion and belong to minimum-entropy learning. DDPO is listed for contrast: without a KL term it sits at the boundary of the family.

Method	Base measure π_{ref}	Energy	What is fitted	Level	Divergence or objective
RLHF with PPO (Ouyang et al. 2022); GRPO (Shao et al. 2024); RLOO (Ahmadian et al. 2024)	pretrained LM	reward, given: a learned reward model or a rule-based verifier	policy $\pi_{\theta} \approx \pi^*$	amortized inference	reverse KL, by KL-penalized policy gradient
GDC, trained by DPG (Khalifa et al. 2021; Parshakova et al. 2019)	pretrained LM a	moments of declared features, i.e., $\lambda^{\top} \phi$	policy	amortized inference (constrained form)	forward KL, importance-weighted

Method	Base measure		What is fitted	Level	Divergence or objective
	π_{ref}	Energy			
f -DPG (Go et al. 2023)	pretrained LM	any unnormalized target P	policy	amortized inference	any f -divergence
GFlowNet fine-tuning (Hu et al. 2024)	pretrained LM (initialization; also evaluates R)	reward R , e.g., a posterior of the same LM	policy $\propto R$	amortized inference	subtrajectory-balance objective
RAFT (Dong et al. 2023), ReST (Gulcehre et al. 2023)	current / pretrained LM	reward, given	policy	amortized inference	forward KL to a truncated tilt: maximum likelihood on accepted samples
BoNBoN (Gui et al. 2024)	pretrained LM	reward, given	policy $\approx \pi_{\text{BoN}}$	amortized inference	maximum likelihood on the best of n plus an IPO-type term on best–worst pairs
Reward models (Christiano et al. 2017; Ouyang et al. 2022); ORM (Cobbe et al. 2021); PRM (Lightman et al. 2024); EORM (Jiang et al. 2025)	—	learned from pairs, or from outcome / step labels	energy	learning	log score on pairs (Bradley–Terry) or on labels
DPO (Rafailov et al. 2023, 2024)	pretrained LM	implicit: $\beta \log \pi_{\theta} / \pi_{\text{ref}}$	policy = energy	learning through the closed form	log score on pairs
Residual energies (Deng et al. 2020); EDLM (Xu et al. 2025); Uni-E (Yan et al. 2026)	AR LM / DLM denoiser	residual, NCE-trained or read off a proxy AR model	energy	learning (generalized sense), or none when read off	NCE, i.e., the logistic score

Method	Base measure		What is fitted	Level	Divergence or objective
	π_{ref}	Energy			
DDPO (Black et al. 2024)	diffusion prior	reward, given	denoising policy	reward maximization with no KL term (boundary case; early stopping)	—
DPOK (Fan et al. 2023); entropy-regularized control (Uehara et al. 2024); adjoint matching (Domingo-Enrich et al. 2025)	diffusion / flow prior	reward, given	denoising policy / generator drift	amortized inference	reverse KL: per-step KL penalty, or stochastic optimal control
Diffusion-DPO (Wallace et al. 2024)	diffusion prior	implicit, from pairs	generator = energy	learning through the closed form	log score on pairs
EMPO (Zhang et al. 2025); TTRL (Zuo et al. 2025); ReTTA (Park et al. 2025)	pretrained model	the model’s own agreement (EMPO, TTRL) or its own free energy (ReTTA)	policy	reversed outer criterion	— (Chapter 6)

The definition also says what is *not* maximum-entropy post-training. Reward maximization without a KL term is the zero-temperature boundary: its optimum is the degenerate distribution on the arg-max, which is the distribution collapse of Korbak et al. (2022). Best-of- N with $N \rightarrow \infty$ is the same limit reached from the sampling side. Supervised fine-tuning on demonstrations is maximum likelihood, i.e., forward KL from the data to the model, and involves no reference and no tilt; it is a maximum-entropy learner only in the trivial sense of Section 5.1.4, and it becomes part of the family here only as the fitting step of rejection-sampling fine-tuning, where the “data” are samples from a tilt. DPO with $\beta \rightarrow 0$ leaves the family in the same way as RLHF without a penalty.

What is open is, again, at least as interesting. Reward hacking is energy misspecification: the tilt faithfully concentrates on what the learned energy rewards, and the entropy term is the only thing that keeps an imperfect energy from being exploited, which is why the KL budget, the choice of β , and the choice of divergence (Go et al. 2023) are not implementation details. The trade-off between test-time computation (Chapter 8) and train-time amortization has no theory yet beyond the KL accounting of best-of- N ; a principled answer would say, for a given

energy and budget, how much of the tilt to execute at inference and how much to store. And the self-referential methods of Section 9.4 show that the boundary between maximum- and minimum-entropy learning runs through post-training itself: the same tilt, with an external or an internal energy, gives a method that stays honest or one that collapses.

9.7 Takeaway

Post-training is maximum-entropy learning with the pretrained model as base measure. Its object is the tilt Equation 9.1, which has both forms of Section 5.1.1 as instances — RLHF and KL-control declare a reward and a temperature, GDC declares features and moments. Amortization (PPO, GRPO, RLOO, DPG, GFlowNet fine-tuning, rejection sampling) fits the reference to the tilt and differs only in the divergence and its estimator; energy learning (reward models, DPO, contrastive residual energies) fits the tilt itself, and in DPO the two coincide because the policy’s log-ratio to the reference is the energy. The KL term is where the entropy is load-bearing, and removing it, or sending N or $1/\beta$ to infinity, is where the method leaves the family. The story so far is of a single distribution, accessible by paying in either of two currencies: computation expended at inference time, or training and storage invested in advance. The next part explores how and where these approaches enable progress in scientific discovery.

Part III

Part III — Applications

10 Entropy-Based Learning for Science

The ultimate aim of this research program is **scientific intelligence** — AI that is not only accurate and robust, but increasingly endowed with reasoning capabilities that enable new modes of scientific inquiry. Entropy- and energy-based learning offer principled tools for this goal.

10.1 Two synergistic directions

- **Advancing AI with Science.** Fundamental scientific and mathematical theories — statistical physics, information theory, the maximum-entropy principle — inspire principled AI methodologies. Energy-based games (Chapter 5), entropy-minimized reasoning (Chapter 6), and minimax-entropy model selection (Chapter 7) are three examples.
- **Advancing Science with AI.** The same methods accelerate discovery in the sciences, from molecular representation learning to chemical reasoning.

10.2 Molecular foundation models

A concrete thread runs from **self-supervised graph transformers** for molecular representation learning toward **molecule–language** models with reasoning capabilities. The trajectory mirrors the broader shift from *pretraining–finetuning* to *LLM-style reasoning*, and entropy-based objectives provide label-efficient signals where annotated data are scarce.

10.3 Outlook

A deeper unification may be within reach. The minimax view (Chapter 7) suggests that modelling the world and committing to a decision are two halves of one objective — an idea that echoes the **free-energy principle** proposed as a unified account of perception and action in the brain (Friston 2010). The quantity that principle asks the brain to minimize is the variational free energy of Chapter 3, which is what makes the correspondence more than a metaphor. Making it precise, and scaling it to LLMs, agents, and multimodal scientific models, is an open and appealing direction.

As the research matures, this book will grow to include worked examples, executable code, and reproducible experiments.

See the [Blue Whale Lab](#) and <https://yataobian.com/> for the latest developments.

11 Summary

This book introduced **entropy-based learning** through three complementary principles and their shared energy-based substrate:

- **Maximum entropy** yields the least-biased model consistent with given constraints — grounding exponential families, Boltzmann machines, maximum-entropy reinforcement learning, and a principled family of game-theoretic valuations (Bian et al. 2022). When the energy is learned rather than given, the same principle trains neural set functions from optimal subsets (Ou et al. 2022) and infers rewards from demonstrations (Ziebart et al. 2008). Applied to a pretrained model as base measure, it becomes energy-based guidance at inference time and maximum-entropy post-training — RLHF, GRPO, DPO — at training time, one Gibbs tilt paid for in two currencies.
- **Minimum entropy** sharpens a capable model’s own predictions, enabling *unsupervised* elicitation of reasoning (EMPO) in a latent semantic space (Zhang et al. 2025).
- **Minimax entropy** selects the features worth modelling at all: those whose maximum-entropy model carries the least entropy, simultaneously the shortest, most accurate, and most informative description (Carcamo et al. 2025).

Binding the three together is the **free energy** $F = U - TS$: temperature sets the exchange rate between energy and entropy, so the principles are three settings of one dial rather than three separate doctrines. It is also the equation that carried statistical physics into machine learning, from the Hopfield network and the Boltzmann machine through to the evidence lower bound — a lineage recognized by the 2024 Nobel Prize in Physics (The Royal Swedish Academy of Sciences 2024).

This is an early scaffold; future revisions will add worked examples, code, and experiments. For updates, see <https://yataobian.com/> and the [Blue Whale Lab](#).

References

- Abbeel, Pieter, and Andrew Y. Ng. 2004. "Apprenticeship Learning via Inverse Reinforcement Learning." *International Conference on Machine Learning (ICML)*.
- Abe, Sumiyoshi. 2000. "Axioms and Uniqueness Theorem for Tsallis Entropy." *Physics Letters A* 271 (1–2): 74–79. [https://doi.org/10.1016/S0375-9601\(00\)00337-6](https://doi.org/10.1016/S0375-9601(00)00337-6).
- Ackley, David H., Geoffrey E. Hinton, and Terrence J. Sejnowski. 1985. "A Learning Algorithm for Boltzmann Machines." *Cognitive Science* 9 (1): 147–69. https://doi.org/10.1207/s15516709cog0901_7.
- Aczél, János, and Zoltán Daróczy. 1963. "Über Verallgemeinerte Quasilineare Mittelwerte, Die Mit Gewichtsfunktionen Gebildet Sind." *Publicationes Mathematicae Debrecen* 10: 171–90. <https://doi.org/10.5486/PMD.1963.10.1-4.24>.
- Ahmadian, Arash, Chris Cremer, Matthias Gallé, et al. 2024. "Back to Basics: Revisiting REINFORCE-Style Optimization for Learning from Human Feedback in LLMs." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 12248–67.
- Ahuja, Ravindra K., and James B. Orlin. 2001. "Inverse Optimization." *Operations Research* 49 (5): 771–83.
- Altun, Yasemin, and Alexander J. Smola. 2006. "Unifying Divergence Minimization and Statistical Inference via Convex Duality." *Conference on Learning Theory (COLT)*.
- Asano, Yuki M., Christian Rupprecht, and Andrea Vedaldi. 2020. "Self-Labeling via Simultaneous Clustering and Representation Learning." *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1911.05371>.
- Baez, John C., and Tobias Fritz. 2014. "A Bayesian Characterization of Relative Entropy." *Theory and Applications of Categories* 29 (16): 422–56. <https://doi.org/10.70930/tac/myyolhqw>.
- Baez, John C., Tobias Fritz, and Tom Leinster. 2011. "A Characterization of Entropy in Terms of Information Loss." *Entropy* 13 (11): 1945–57. <https://doi.org/10.3390/e13111945>.
- Balcan, Maria-Florina, and Nicholas J. A. Harvey. 2018. "Submodular Functions: Learnability, Structure, and Optimization." *SIAM Journal on Computing* 47 (3): 703–54. <https://doi.org/>

[10.1137/120888909](https://doi.org/10.1137/120888909).

- Balcerak, Michal, Tamaz Amiranashvili, Antonio Terpin, et al. 2025. "Energy Matching: Unifying Flow Matching and Energy-Based Models for Generative Modeling." *Advances in Neural Information Processing Systems 38 (NeurIPS)*.
- Barbera, Elvira. 1999. "On the Principle of Minimal Entropy Production for Navier–Stokes–Fourier Fluids." *Continuum Mechanics and Thermodynamics* 11 (5): 327–30. <https://doi.org/10.1007/s001610050127>.
- Beirami, Ahmad, Alekh Agarwal, Jonathan Berant, et al. 2024. "Theoretical Guarantees on the Best-of- n Alignment Policy." *arXiv Preprint arXiv:2401.01879*.
- Belanger, David, and Andrew McCallum. 2016. "Structured Prediction Energy Networks." *International Conference on Machine Learning (ICML)*.
- Belanger, David, Bishan Yang, and Andrew McCallum. 2017. "End-to-End Learning for Structured Prediction Energy Networks." *International Conference on Machine Learning (ICML)*, 429–39.
- Bengio, Yoshua, and Olivier Delalleau. 2009. "Justifying and Generalizing Contrastive Divergence." *Neural Computation* 21 (6): 1601–21.
- Bennett, Charles H. 1976. "Efficient Estimation of Free Energy Differences from Monte Carlo Data." *Journal of Computational Physics* 22 (2): 245–68. [https://doi.org/10.1016/0021-9991\(76\)90078-4](https://doi.org/10.1016/0021-9991(76)90078-4).
- Berger, Adam L., Stephen A. Della Pietra, and Vincent J. Della Pietra. 1996. "A Maximum Entropy Approach to Natural Language Processing." *Computational Linguistics* 22 (1): 39–71.
- Berthelot, David, Nicholas Carlini, Ekin D. Cubuk, et al. 2020. "ReMixMatch: Semi-Supervised Learning with Distribution Alignment and Augmentation Anchoring." *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/1911.09785>.
- Besag, Julian. 1974. "Spatial Interaction and the Statistical Analysis of Lattice Systems." *Journal of the Royal Statistical Society: Series B (Methodological)* 36 (2): 192–236.
- Bian, Andrew An, Joachim M. Buhmann, Andreas Krause, and Sebastian Tschiatschek. 2017. "Guarantees for Greedy Maximization of Non-Submodular Functions with Applications." *International Conference on Machine Learning (ICML)*, 498–507.
- Bian, Yatao. 2020. *Awesome Energy-Based Models/Learning: A Comprehensive List of Energy-Based Learning Papers and Materials*. <https://github.com/yataobian/awesome-ebm>.

References

- Bian, Yatao, Joachim M. Buhmann, and Andreas Krause. 2019. "Optimal Continuous DR-Submodular Maximization and Applications to Provable Mean Field Inference." *International Conference on Machine Learning (ICML)*, 644–53.
- Bian, Yatao, Yu Rong, Tingyang Xu, Jiayang Wu, Andreas Krause, and Junzhou Huang. 2022. "Energy-Based Learning for Cooperative Games, with Applications to Valuation Problems in Machine Learning." *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=xLfAgCroImw>.
- Bilmes, Jeffrey A., and Wenruo Bai. 2017. "Deep Submodular Functions." *arXiv Preprint arXiv:1701.08939*.
- Black, Kevin, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. 2024. "Training Diffusion Models with Reinforcement Learning." *International Conference on Learning Representations (ICLR)*.
- Bloem-Reddy, Benjamin, and Yee Whye Teh. 2020. "Probabilistic Symmetries and Invariant Neural Networks." *Journal of Machine Learning Research* 21 (90): 1–61.
- Boltzmann, Ludwig. 1877. "Über Die Beziehung Zwischen Dem Zweiten Hauptsatze Der Mechanischen Wärmetheorie Und Der Wahrscheinlichkeitsrechnung Respektive Den Sätzen Über Das Wärmegleichgewicht." *Sitzungsberichte Der Kaiserlichen Akademie Der Wissenschaften in Wien* 76: 373–435.
- Bradley, Ralph Allan, and Milton E. Terry. 1952. "Rank Analysis of Incomplete Block Designs: I. The Method of Paired Comparisons." *Biometrika* 39 (3/4): 324–45.
- Bridle, John S., Anthony J. R. Heading, and David J. C. MacKay. 1992. "Unsupervised Classifiers, Mutual Information and 'Phantom Targets'." *Advances in Neural Information Processing Systems (NIPS)* 4, 1096–101.
- Burton, Didier, and Philippe L. Toint. 1992. "On an Instance of the Inverse Shortest Paths Problem." *Mathematical Programming* 53: 45–61.
- Calinescu, Gruia, Chandra Chekuri, Martin Pál, and Jan Vondrák. 2007. "Maximizing a Submodular Set Function Subject to a Matroid Constraint." *Integer Programming and Combinatorial Optimization (IPCO)*, 182–96.
- Callen, Herbert B. 1985. *Thermodynamics and an Introduction to Thermostatistics*. 2nd ed. Wiley.
- Carcamo, David P., Nicholas J. Weaver, Purushottam D. Dixit, and Christopher W. Lynn. 2025. "Minimax Entropy: The Statistical Physics of Optimal Models." *Physical Review E* 112 (6): 061001. <https://doi.org/10.1103/kr9x-q59y>.

References

- Caron, Mathilde, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. 2020. "Unsupervised Learning of Visual Features by Contrasting Cluster Assignments." *Advances in Neural Information Processing Systems (NeurIPS)*. <https://arxiv.org/abs/2006.09882>.
- Caron, Mathilde, Hugo Touvron, Ishan Misra, et al. 2021. "Emerging Properties in Self-Supervised Vision Transformers." *IEEE/CVF International Conference on Computer Vision (ICCV)*. <https://arxiv.org/abs/2104.14294>.
- Carreira-Perpiñán, Miguel Á., and Geoffrey E. Hinton. 2005. "On Contrastive Divergence Learning." *International Workshop on Artificial Intelligence and Statistics (AISTATS)*, Proceedings of machine learning research, vol. R5: 33–40.
- Christiano, Paul F., Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. "Deep Reinforcement Learning from Human Preferences." *Advances in Neural Information Processing Systems 30 (NeurIPS)*.
- Clark, Kevin, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2020a. "ELECTRA: Pre-Training Text Encoders as Discriminators Rather Than Generators." *International Conference on Learning Representations (ICLR)*.
- Clark, Kevin, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2020b. "Pre-Training Transformers as Energy-Based Cloze Models." *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 285–94.
- Clausius, Rudolf. 1865. "Über Verschiedene Für Die Anwendung Bequeme Formen Der Hauptgleichungen Der Mechanischen Wärmetheorie." *Annalen Der Physik Und Chemie* 125: 353–400.
- Cobbe, Karl, Vineet Kosaraju, Mohammad Bavarian, et al. 2021. "Training Verifiers to Solve Math Word Problems." *arXiv Preprint arXiv:2110.14168*.
- Cohen, Jacob. 1960. "A Coefficient of Agreement for Nominal Scales." *Educational and Psychological Measurement* 20 (1): 37–46. <https://doi.org/10.1177/001316446002000104>.
- Collins, Michael. 2002. "Discriminative Training Methods for Hidden Markov Models: Theory and Experiments with Perceptron Algorithms." *Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Csiszár, Imre. 1975. "I-Divergence Geometry of Probability Distributions and Minimization Problems." *Annals of Probability* 3 (1): 146–58.
- Csiszár, Imre. 2008. "Axiomatic Characterizations of Information Measures." *Entropy* 10 (3): 261–73. <https://doi.org/10.3390/e10030261>.

- Cui, Ganqu, Yuchen Zhang, Jiacheng Chen, et al. 2025. "The Entropy Mechanism of Reinforcement Learning for Reasoning Language Models." *arXiv Preprint arXiv:2505.22617*. <https://arxiv.org/abs/2505.22617>.
- Dai, Bo, Zhen Liu, Hanjun Dai, et al. 2019. "Exponential Family Estimation via Adversarial Dynamics Embedding." *Advances in Neural Information Processing Systems (NeurIPS)* 32.
- Daróczy, Zoltán. 1970. "Generalized Information Functions." *Information and Control* 16 (1): 36–51. [https://doi.org/10.1016/S0019-9958\(70\)80040-7](https://doi.org/10.1016/S0019-9958(70)80040-7).
- Dawid, A. Philip. 2007. "The Geometry of Proper Scoring Rules." *Annals of the Institute of Statistical Mathematics* 59 (1): 77–93. <https://doi.org/10.1007/s10463-006-0099-8>.
- Dawid, Anna, and Yann LeCun. 2024. "Introduction to Latent Variable Energy-Based Models: A Path Towards Autonomous Machine Intelligence." *Journal of Statistical Mechanics: Theory and Experiment* 2024 (10): 104011.
- Dayan, Peter, Geoffrey E. Hinton, Radford M. Neal, and Richard S. Zemel. 1995. "The Helmholtz Machine." *Neural Computation* 7 (5): 889–904. <https://doi.org/10.1162/neco.1995.7.5.889>.
- DeepSeek-AI. 2025. "DeepSeek-R1 Incentivizes Reasoning in LLMs Through Reinforcement Learning." *Nature* 645. <https://doi.org/10.1038/s41586-025-09422-z>.
- Della Pietra, Stephen, Vincent Della Pietra, and John Lafferty. 1997. "Inducing Features of Random Fields." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 19 (4): 380–93. <https://doi.org/10.1109/34.588021>.
- Deng, Yuntian, Anton Bakhtin, Myle Ott, Arthur Szlam, and Marc'Aurelio Ranzato. 2020. "Residual Energy-Based Models for Text Generation." *International Conference on Learning Representations (ICLR)*.
- Dhariwal, Prafulla, and Alexander Nichol. 2021. "Diffusion Models Beat GANs on Image Synthesis." *Advances in Neural Information Processing Systems (NeurIPS)*.
- Domingo-Enrich, Carles, Michal Drozdal, Brian Karrer, and Ricky T. Q. Chen. 2025. "Adjoint Matching: Fine-Tuning Flow and Diffusion Generative Models with Memoryless Stochastic Optimal Control." *International Conference on Learning Representations (ICLR)*.
- Domke, Justin. 2012. "Generic Methods for Optimization-Based Modeling." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 318–26.
- Domke, Justin. 2013. "Learning Graphical Model Parameters with Approximate Marginal Inference." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35 (10): 2454–67. <https://doi.org/10.1109/TPAMI.2013.31>.

- Dong, Hanze, Wei Xiong, Deepanshu Goyal, et al. 2023. "RAFT: Reward rAnked FineTuning for Generative Foundation Model Alignment." *Transactions on Machine Learning Research*.
- Du, Yilun, Conor Durkan, Robin Strudel, et al. 2023. "Reduce, Reuse, Recycle: Compositional Generation with Energy-Based Diffusion Models and MCMC." *Proceedings of the 40th International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 202.
- Du, Yilun, Shuang Li, and Igor Mordatch. 2020. "Compositional Visual Generation with Energy Based Models." *Advances in Neural Information Processing Systems (NeurIPS)*.
- Du, Yilun, and Igor Mordatch. 2019. "Implicit Generation and Modeling with Energy-Based Models." *Advances in Neural Information Processing Systems (NeurIPS)*.
- Dudík, Miroslav, Steven J. Phillips, and Robert E. Schapire. 2007. "Maximum Entropy Density Estimation with Generalized Regularization and an Application to Species Distribution Modeling." *Journal of Machine Learning Research* 8: 1217–60.
- Efron, Bradley. 1975. "Defining the Curvature of a Statistical Problem (with Applications to Second Order Efficiency)." *Annals of Statistics* 3 (6): 1189–242. <https://doi.org/10.1214/aos/1176343282>.
- Emmerich, Michael T. M., Ksenia Pereverdieva, and André H. Deutz. 2026a. "Maximum Solow–Polasky Diversity Subset Selection Is NP-Hard Even in the Euclidean Plane." *arXiv Preprint arXiv:2604.19484*. <https://arxiv.org/abs/2604.19484>.
- Emmerich, Michael T. M., Ksenia Pereverdieva, and André H. Deutz. 2026b. "Selecting a Maximum Solow–Polasky Diversity Subset in General Metric Spaces Is NP-Hard." *arXiv Preprint arXiv:2604.05495*. <https://arxiv.org/abs/2604.05495>.
- Faddeev, D. K. 1956. "On the Concept of Entropy of a Finite Probabilistic Scheme." *Uspekhi Matematicheskikh Nauk* 11 (1(67)): 227–31. <https://www.mathnet.ru/eng/rm7756>.
- Fan, Ying, Olivia Watkins, Yuqing Du, et al. 2023. "DPOK: Reinforcement Learning for Fine-Tuning Text-to-Image Diffusion Models." *Advances in Neural Information Processing Systems* 36 (NeurIPS).
- Farquhar, Sebastian, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. 2024. "Detecting Hallucinations in Large Language Models Using Semantic Entropy." *Nature* 630: 625–30. <https://doi.org/10.1038/s41586-024-07421-0>.
- Feder, M. 1986. "Maximum Entropy as a Special Case of the Minimum Description Length Criterion." *IEEE Transactions on Information Theory* 32: 847–49.

- Feng, Ruiqi, Chenglei Yu, Wenhao Deng, Peiyan Hu, and Tailin Wu. 2025. "On the Guidance of Flow Matching." *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 267: 16993–7029.
- Fox, Roy, Ari Pakman, and Naftali Tishby. 2016. "Taming the Noise in Reinforcement Learning via Soft Updates." *Conference on Uncertainty in Artificial Intelligence (UAI)*.
- Frieden, B. Roy. 1990. "Fisher Information, Disorder, and the Equilibrium Distributions of Physics." *Physical Review A* 41 (8): 4265–76. <https://doi.org/10.1103/PhysRevA.41.4265>.
- Friedman, Dan, and Adji Bousso Dieng. 2023. "The Vendi Score: A Diversity Evaluation Metric for Machine Learning." *Transactions on Machine Learning Research*. <https://arxiv.org/abs/2210.02410>.
- Friston, Karl. 2010. "The Free-Energy Principle: A Unified Brain Theory?" *Nature Reviews Neuroscience* 11 (2): 127–38.
- Geiger, Dan, David Heckerman, Henry King, and Christopher Meek. 2001. "Stratified Exponential Families: Graphical Models and Model Selection." *The Annals of Statistics* 29 (2): 505–29. <https://doi.org/10.1214/aos/1009210550>.
- Geist, Matthieu, Bruno Scherrer, and Olivier Pietquin. 2019. "A Theory of Regularized Markov Decision Processes." *International Conference on Machine Learning (ICML)*.
- Gibbs, Josiah Willard. 1902. *Elementary Principles in Statistical Mechanics*. Charles Scribner's Sons.
- Gladstone, Alexi, Ganesh Nanduru, Md Mofijul Islam, et al. 2026. "Energy-Based Transformers Are Scalable Learners and Thinkers." *International Conference on Learning Representations (ICLR)*.
- Gnaiger, Erich. 2009. "Open and Closed Systems: Styles of Thinking Explain Controversies on the 'Negative Entropy' Concept of Ludwig Boltzmann and Erwin Schrödinger." *Mitochondrial Physiology Network*. https://www.mitophysiology.org/images/2/23/Gnaiger_2009_OCESHS.pdf.
- Go, Dongyoung, Tomasz Korbak, Germán Kruszewski, Jos Rozen, Nahyeon Ryu, and Marc Dymetman. 2023. "Aligning Language Models with Preferences Through f -Divergence Minimization." *Proceedings of the 40th International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 202.
- Gomes, Ryan, Andreas Krause, and Pietro Perona. 2010. "Discriminative Clustering by Regularized Information Maximization." *Advances in Neural Information Processing Systems (NeurIPS)* 23.

References

- Grandvalet, Yves, and Yoshua Bengio. 2004. "Semi-Supervised Learning by Entropy Minimization." *Advances in Neural Information Processing Systems (NeurIPS)* 17.
- Grathwohl, Will, Kevin Swersky, Milad Hashemi, David Duvenaud, and Chris Maddison. 2021. "Oops I Took a Gradient: Scalable Sampling for Discrete Distributions." *International Conference on Machine Learning (ICML)*, 3831–41.
- Grathwohl, Will, Kuan-Chieh Wang, Jörn-Henrik Jacobsen, David Duvenaud, Mohammad Norouzi, and Kevin Swersky. 2020. "Your Classifier Is Secretly an Energy Based Model and You Should Treat It Like One." *International Conference on Learning Representations (ICLR)*.
- Gregor, Karol, and Yann LeCun. 2010. "Learning Fast Approximations of Sparse Coding." *International Conference on Machine Learning (ICML)*, 399–406.
- Grünwald, Peter D. 2007. *The Minimum Description Length Principle*. MIT Press.
- Grünwald, Peter D., and A. Philip Dawid. 2004. "Game Theory, Maximum Entropy, Minimum Discrepancy and Robust Bayesian Decision Theory." *Annals of Statistics* 32 (4): 1367–433.
- Gui, Lin, Cristina Gârbacea, and Victor Veitch. 2024. "BoNBoN Alignment for Large Language Models and the Sweetness of Best-of- n Sampling." *Advances in Neural Information Processing Systems* 37 (*NeurIPS*).
- Gulcehre, Caglar, Tom Le Paine, Srivatsan Srinivasan, et al. 2023. "Reinforced Self-Training (ReST) for Language Modeling." *arXiv Preprint arXiv:2308.08998*.
- Gutmann, Michael U., and Aapo Hyvärinen. 2012. "Noise-Contrastive Estimation of Unnormalized Statistical Models, with Applications to Natural Image Statistics." *Journal of Machine Learning Research* 13: 307–61.
- Gutmann, Michael, and Jun-ichiro Hirayama. 2011. "Bregman Divergence as General Framework to Estimate Unnormalized Statistical Models." *Conference on Uncertainty in Artificial Intelligence (UAI)*, 283–90.
- Gutmann, Michael, and Aapo Hyvärinen. 2010. "Noise-Contrastive Estimation: A New Estimation Principle for Unnormalized Statistical Models." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 297–304.
- GX-Chen, Anthony, Jatin Prakash, Jeff Guo, Rob Fergus, and Rajesh Ranganath. 2025. "KL-Regularized Reinforcement Learning Is Designed to Mode Collapse." *arXiv Preprint arXiv:2510.20817*. <https://arxiv.org/abs/2510.20817>.
- Haarnoja, Tuomas, Haoran Tang, Pieter Abbeel, and Sergey Levine. 2017. "Reinforcement Learning with Deep Energy-Based Policies." *International Conference on Machine Learning*

- (ICML), 1352–61.
- Haarnoja, Tuomas, Aurick Zhou, Pieter Abbeel, and Sergey Levine. 2018. “Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor.” *International Conference on Machine Learning (ICML)*, 1861–70.
- Hanel, Rudolf, and Stefan Thurner. 2011. “A Comprehensive Classification of Complex Statistical Systems and an Axiomatic Derivation of Their Entropy and Distribution Functions.” *EPL (Europhysics Letters)* 93 (2): 20006. <https://doi.org/10.1209/0295-5075/93/20006>.
- Hassani, Hamed, Mahdi Soltanolkotabi, and Amin Karbasi. 2017. “Gradient Methods for Submodular Maximization.” *Advances in Neural Information Processing Systems (NeurIPS)* 30.
- Havrda, Jan, and František Charvát. 1967. “Quantification Method of Classification Processes. Concept of Structural α -Entropy.” *Kybernetika* 3 (1): 30–35. <https://www.kybernetika.cz/content/1967/1/30>.
- Hershey, John R., Jonathan Le Roux, and Felix Weninger. 2014. “Deep Unfolding: Model-Based Inspiration of Novel Deep Architectures.” *arXiv Preprint arXiv:1409.2574*.
- Heuberger, Clemens. 2004. “Inverse Combinatorial Optimization: A Survey on Problems, Methods, and Results.” *Journal of Combinatorial Optimization* 8 (3): 329–61.
- Hill, M. O. 1973. “Diversity and Evenness: A Unifying Notation and Its Consequences.” *Ecology* 54 (2): 427–32. <https://doi.org/10.2307/1934352>.
- Hinton, Geoffrey E. 2002. “Training Products of Experts by Minimizing Contrastive Divergence.” *Neural Computation* 14 (8): 1771–800. <https://doi.org/10.1162/089976602760128018>.
- Hirsh, Jacob B., Raymond A. Mar, and Jordan B. Peterson. 2012. “Psychological Entropy: A Framework for Understanding Uncertainty-Related Anxiety.” *Psychological Review* 119 (2): 304–20. <https://doi.org/10.1037/a0026767>.
- Ho, Jonathan, Ajay Jain, and Pieter Abbeel. 2020. “Denoising Diffusion Probabilistic Models.” *Advances in Neural Information Processing Systems (NeurIPS)*.
- Ho, Jonathan, and Tim Salimans. 2022. “Classifier-Free Diffusion Guidance.” *arXiv Preprint arXiv:2207.12598*.
- Hopfield, John J. 1982. “Neural Networks and Physical Systems with Emergent Collective Computational Abilities.” *Proceedings of the National Academy of Sciences* 79 (8): 2554–58. <https://doi.org/10.1073/pnas.79.8.2554>.
- Hu, Edward J., Moksh Jain, Eric Elmoznino, et al. 2024. “Amortizing Intractable Inference in

- Large Language Models." *International Conference on Learning Representations (ICLR)*.
- Hu, Weihua, Takeru Miyato, Seiya Tokui, Eiichi Matsumoto, and Masashi Sugiyama. 2017. "Learning Discrete Representations via Information Maximizing Self-Augmented Training." *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/1702.08720>.
- Hyvärinen, Aapo. 2005. "Estimation of Non-Normalized Statistical Models by Score Matching." *Journal of Machine Learning Research* 6: 695–709.
- Hyvärinen, Aapo. 2007. "Connections Between Score Matching, Contrastive Divergence, and Pseudolikelihood for Continuous-Valued Variables." *IEEE Transactions on Neural Networks* 18 (5): 1529–31.
- Jalali, Mohammad, Cheuk Ting Li, and Farzan Farnia. 2023. "An Information-Theoretic Evaluation of Generative Models in Learning Multi-Modal Distributions." *Advances in Neural Information Processing Systems (NeurIPS)* 36: 9931–43. <https://doi.org/10.52202/075280-0434>.
- Jaques, Natasha, Shixiang Gu, Dzmitry Bahdanau, José Miguel Hernández-Lobato, Richard E. Turner, and Douglas Eck. 2017. "Sequence Tutor: Conservative Fine-Tuning of Sequence Generation Models with KL-Control." *Proceedings of the 34th International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 70: 1645–54.
- Jaynes, Edwin T. 1957. "Information Theory and Statistical Mechanics." *Physical Review* 106: 620.
- Jaynes, Edwin T. 1968. "Prior Probabilities." *IEEE Transactions on Systems Science and Cybernetics* 4 (3): 227–41.
- Jaynes, Edwin T. 1980. "The Minimum Entropy Production Principle." *Annual Review of Physical Chemistry* 31: 579–601. <https://doi.org/10.1146/annurev.pc.31.100180.003051>.
- Jiang, Eric H., Haozheng Luo, Shengyuan Pang, et al. 2025. "Learning to Rank Chain-of-Thought: Using a Small Model." *arXiv Preprint arXiv:2505.14999*.
- Jizba, Petr, and Toshihico Arimitsu. 2004. "The World According to Rényi: Thermodynamics of Multifractal Systems." *Annals of Physics* 312 (1): 17–59. <https://doi.org/10.1016/j.aop.2004.01.002>.
- Jordan, Michael I., Zoubin Ghahramani, Tommi S. Jaakkola, and Lawrence K. Saul. 1999. "An Introduction to Variational Methods for Graphical Models." *Machine Learning* 37 (2): 183–233.
- Jost, Lou. 2006. "Entropy and Diversity." *Oikos* 113 (2): 363–75. <https://doi.org/10.1111/j.2006.0030-1299.14714.x>.

References

- Jost, Lou. 2009. "Mismeasuring Biological Diversity: Response to Hoffmann and Hoffmann (2008)." *Ecological Economics* 68 (4): 925–28. <https://doi.org/10.1016/j.ecolecon.2008.10.015>.
- Jost, Lou, Philip DeVries, Thomas Walla, Harold Greeney, Anne Chao, and Carlo Ricotta. 2010. "Partitioning Diversity for Conservation Analyses." *Diversity and Distributions* 16 (1): 65–76. <https://doi.org/10.1111/j.1472-4642.2009.00626.x>.
- Kappen, Hilbert J. 2005. "Path Integrals and Symmetry Breaking for Optimal Control Theory." *Journal of Statistical Mechanics: Theory and Experiment* 2005 (11): P11011.
- Kappen, Hilbert J., Vicenç Gómez, and Manfred Oppen. 2012. "Optimal Control as a Graphical Model Inference Problem." *Machine Learning* 87 (2): 159–82.
- Karalias, Nikolaos, Joshua Robinson, Andreas Loukas, and Stefanie Jegelka. 2022. "Neural Set Function Extensions: Learning with Discrete Functions in High Dimensions." *Advances in Neural Information Processing Systems (NeurIPS)* 35: 15338–52.
- Khalifa, Muhammad, Hady Elsahar, and Marc Dymetman. 2021. "A Distributional Approach to Controlled Text Generation." *International Conference on Learning Representations (ICLR)*.
- Khinchin, A. I. 1957. *Mathematical Foundations of Information Theory*. Dover Publications.
- Kianercy, Ardeshtir, and Aram Galstyan. 2012. "Dynamics of Boltzmann q Learning in Two-Player Two-Action Games." *Physical Review E* 85 (4): 041145. <https://doi.org/10.1103/PhysRevE.85.041145>.
- Kim, Taesup, and Yoshua Bengio. 2016. "Deep Directed Generative Models with Energy-Based Probability Estimation." *arXiv Preprint arXiv:1606.03439*.
- Kirk, Robert, Ishita Mediratta, Christoforos Nalmpantis, et al. 2024. "Understanding the Effects of RLHF on LLM Generalisation and Diversity." *International Conference on Learning Representations (ICLR)*, 20620–53.
- Kirkpatrick, Scott, C. Daniel Gelatt, and Mario P. Vecchi. 1983. "Optimization by Simulated Annealing." *Science* 220 (4598): 671–80. <https://doi.org/10.1126/science.220.4598.671>.
- Koller, Daphne, and Nir Friedman. 2009. *Probabilistic Graphical Models: Principles and Techniques*. MIT Press.
- Korbak, Tomasz, Ethan Perez, and Christopher L. Buckley. 2022. "RL with KL Penalties Is Better Viewed as Bayesian Inference." *Findings of the Association for Computational Linguistics: EMNLP 2022*.
- Korbel, Jan. 2026. "Foundations of Entropy in Complex Systems." *arXiv Preprint*

- arXiv:2606.16312*. <https://arxiv.org/abs/2606.16312>.
- Krähenbühl, Philipp, and Vladlen Koltun. 2013. "Parameter Learning and Convergent Inference for Dense Random Fields." *International Conference on Machine Learning (ICML)*, 513–21.
- Kuhn, Lorenz, Yarin Gal, and Sebastian Farquhar. 2023. "Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation." *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2302.09664>.
- Kullback, Solomon. 1959. *Information Theory and Statistics*. Wiley.
- Kumar, Rithesh, Sherjil Ozair, Anirudh Goyal, Aaron Courville, and Yoshua Bengio. 2019. "Maximum Entropy Generators for Energy-Based Models." *arXiv Preprint arXiv:1901.08508*.
- Lafferty, John, Andrew McCallum, and Fernando Pereira. 2001. "Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data." *International Conference on Machine Learning (ICML)*, 282–89.
- Landauer, Rolf. 1975. "Inadequacy of Entropy and Entropy Derivatives in Characterizing the Steady State." *Physical Review A* 12 (2): 636–38. <https://doi.org/10.1103/PhysRevA.12.636>.
- Lazarev, Daniel. 2026. "A Structural Characterization of Entropy Functionals." *arXiv Preprint arXiv:2608.13917*. <https://arxiv.org/abs/2608.13917>.
- LeCun, Yann. 2022. "A Path Towards Autonomous Machine Intelligence." *OpenReview Preprint*.
- LeCun, Yann, Sumit Chopra, Raia Hadsell, Marc'Aurelio Ranzato, and Fu Jie Huang. 2006. "A Tutorial on Energy-Based Learning." In *Predicting Structured Data*, edited by Gökhan Bakir, Thomas Hofmann, Bernhard Schölkopf, Alexander J. Smola, and Ben Taskar. MIT Press.
- Lee, Dong-Hyun. 2013. "Pseudo-Label: The Simple and Efficient Semi-Supervised Learning Method for Deep Neural Networks." *ICML Workshop on Challenges in Representation Learning*.
- Lee, Jonghyun, Dahuin Jung, Saehyung Lee, et al. 2024. "Entropy Is Not Enough for Test-Time Adaptation: From the Perspective of Disentangled Factors." *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2403.07366>.
- Leinster, Tom. 2009. "A Maximum Entropy Theorem with Applications to the Measurement of Biodiversity." *arXiv Preprint arXiv:0910.0906*. <https://arxiv.org/abs/0910.0906>.
- Leinster, Tom. 2021. *Entropy and Diversity: The Axiomatic Approach*. Cambridge University Press. <https://doi.org/10.1017/9781108963558>.
- Leinster, Tom, and Christina A. Cobbold. 2012. "Measuring Diversity: The Importance of

- Species Similarity." *Ecology* 93 (3): 477–89. <https://doi.org/10.1890/10-2402.1>.
- Leinster, Tom, and Mark W. Meckes. 2016. "Maximizing Diversity in Biology and Beyond." *Entropy* 18 (3): 88. <https://doi.org/10.3390/e18030088>.
- Levine, Sergey. 2018. "Reinforcement Learning and Control as Probabilistic Inference: Tutorial and Review." *arXiv Preprint arXiv:1805.00909*.
- Liang, Jian, Dapeng Hu, and Jiashi Feng. 2020. "Do We Really Need to Access the Source Data? Source Hypothesis Transfer for Unsupervised Domain Adaptation." *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/2002.08546>.
- Lieb, Elliott H., and Jakob Yngvason. 1999. "The Physics and Mathematics of the Second Law of Thermodynamics." *Physics Reports* 310 (1): 1–96. [https://doi.org/10.1016/S0370-1573\(98\)00082-9](https://doi.org/10.1016/S0370-1573(98)00082-9).
- Lightman, Hunter, Vineet Kosaraju, Yuri Burda, et al. 2024. "Let's Verify Step by Step." *International Conference on Learning Representations (ICLR)*.
- Lin, Hui, and Jeff A. Bilmes. 2012. "Learning Mixtures of Submodular Shells with Application to Document Summarization." *Conference on Uncertainty in Artificial Intelligence (UAI)*, 479–90.
- Liu, Weitang, Xiaoyun Wang, John D. Owens, and Yixuan Li. 2020. "Energy-Based Out-of-Distribution Detection." *Advances in Neural Information Processing Systems (NeurIPS)*.
- Lu, Cheng, Huayu Chen, Jianfei Chen, Hang Su, Chongxuan Li, and Jun Zhu. 2023. "Contrastive Energy Prediction for Exact Energy-Guided Diffusion Sampling in Offline Reinforcement Learning." *Proceedings of the 40th International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 202.
- Lynn, Christopher W., Qiwei Yu, Rich Pang, Stephanie E. Palmer, and William Bialek. 2025. "Exact Minimax Entropy Models of Large-Scale Neuronal Activity." *Physical Review E* 111 (5): 054411. <https://doi.org/10.1103/PhysRevE.111.054411>.
- Lyu, Siwei. 2009. "Interpretation and Generalization of Score Matching." *Conference on Uncertainty in Artificial Intelligence (UAI)*, 359–66.
- Ma, Xueguang, Qian Liu, Dongfu Jiang, Ge Zhang, Zejun Ma, and Wenhui Chen. 2025. "General-Reasoner: Advancing LLM Reasoning Across All Domains." *arXiv Preprint arXiv:2505.14652*. <https://arxiv.org/abs/2505.14652>.
- MacArthur, Robert H. 1965. "Patterns of Species Diversity." *Biological Reviews* 40 (4): 510–33. <https://doi.org/10.1111/j.1469-185X.1965.tb00815.x>.

- Maes, Christian, and Karel Netočný. 2007. "Minimum Entropy Production Principle from a Dynamical Fluctuation Law." *Journal of Mathematical Physics* 48 (5): 053306. <https://doi.org/10.1063/1.2738753>.
- Maes, Christian, and Karel Netočný. 2013. "Minimum Entropy Production Principle." *Scholarpedia* 8 (7): 9664. <https://doi.org/10.4249/scholarpedia.9664>.
- Martyushev, Leonid M., A. S. Nazarova, and Vladimir D. Seleznev. 2007. "On the Problem of the Minimum Entropy Production in the Nonequilibrium Stationary State." *Journal of Physics A: Mathematical and Theoretical* 40 (3): 371–80. <https://doi.org/10.1088/1751-8113/40/3/002>.
- Martyushev, Leonid M., and Vladimir D. Seleznev. 2006. "Maximum Entropy Production Principle in Physics, Chemistry and Biology." *Physics Reports* 426 (1): 1–45. <https://doi.org/10.1016/j.physrep.2005.12.001>.
- Martyushev, Leonid M., and Vladimir D. Seleznev. 2013. "Entropy and Entropy Production: Old Misconceptions and New Breakthroughs." *Entropy* 15 (4): 1152–70. <https://doi.org/10.3390/e15041152>.
- Menon, Aditya Krishna, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. 2021. "Long-Tail Learning via Logit Adjustment." *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2007.07314>.
- Mironov, Mikhail, and Liudmila Prokhorenkova. 2025. "Measuring Diversity: Axioms and Challenges." *Proceedings of the 42nd International Conference on Machine Learning* 267: 44396–411. <https://proceedings.mlr.press/v267/mironov25a.html>.
- Mnih, Andriy, and Yee Whye Teh. 2012. "A Fast and Simple Algorithm for Training Neural Probabilistic Language Models." *International Conference on Machine Learning (ICML)*, 419–26.
- Moussouris, John. 1974. "Gibbs and Markov Random Systems with Constraints." *Journal of Statistical Physics* 10 (1): 11–33. <https://doi.org/10.1007/BF01011714>.
- Mudgal, Sidharth, Jong Lee, Harish Ganapathy, et al. 2024. "Controlled Decoding from Language Models." *Proceedings of the 41st International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 235.
- Neal, Radford M., and Geoffrey E. Hinton. 1998. "A View of the EM Algorithm That Justifies Incremental, Sparse, and Other Variants." In *Learning in Graphical Models*, edited by Michael I. Jordan. Kluwer Academic Publishers.
- Nehring, Klaus, and Clemens Puppe. 2002. "A Theory of Diversity." *Econometrica* 70 (3): 1155–98. <https://doi.org/10.1111/1468-0262.00321>.

References

- Nemhauser, George L., Laurence A. Wolsey, and Marshall L. Fisher. 1978. "An Analysis of Approximations for Maximizing Submodular Set Functions—i." *Mathematical Programming* 14: 265–94.
- Neu, Gergely, Anders Jonsson, and Vicenç Gómez. 2017. "A Unified View of Entropy-Regularized Markov Decision Processes." *arXiv Preprint arXiv:1705.07798*.
- Neumann, John von. 1927. "Thermodynamik Quantenmechanischer Gesamtheiten." *Nachrichten von Der Gesellschaft Der Wissenschaften Zu Göttingen, Mathematisch-Physikalische Klasse* 1927: 273–91.
- Ng, Andrew Y., and Stuart J. Russell. 2000. "Algorithms for Inverse Reinforcement Learning." *International Conference on Machine Learning (ICML)*.
- Ngiam, Jiquan, Zhenghao Chen, Pang Wei Koh, and Andrew Y. Ng. 2011. "Learning Deep Energy Models." *International Conference on Machine Learning (ICML)*, 1105–12.
- Nguyen, Quan, and Adji Bousso Dieng. 2024. "Quality-Weighted Vendi Scores and Their Application to Diverse Experimental Design." *Proceedings of the 41st International Conference on Machine Learning* 235: 37667–82. <https://proceedings.mlr.press/v235/nguyen24d.html>.
- Nijkamp, Erik, Mitch Hill, Tian Han, Song-Chun Zhu, and Ying Nian Wu. 2020. "On the Anatomy of MCMC-Based Maximum Likelihood Learning of Energy-Based Models." *AAAI Conference on Artificial Intelligence* 34: 5272–80. <https://doi.org/10.1609/aaai.v34i04.5973>.
- Nijkamp, Erik, Mitch Hill, Song-Chun Zhu, and Ying Nian Wu. 2019. "Learning Non-Convergent Non-Persistent Short-Run MCMC Toward Energy-Based Model." *Advances in Neural Information Processing Systems (NeurIPS)*.
- Nikitin, Alexander, Jannik Kossen, Yarin Gal, and Pekka Marttinen. 2024. "Kernel Language Entropy: Fine-Grained Uncertainty Quantification for LLMs from Semantic Similarities." *Advances in Neural Information Processing Systems (NeurIPS)*. <https://arxiv.org/abs/2405.20003>.
- Niu, Shuaicheng, Jiayang Wu, Yifan Zhang, et al. 2022. "Efficient Test-Time Model Adaptation Without Forgetting." *International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/2204.02610>.
- Niu, Shuaicheng, Jiayang Wu, Yifan Zhang, et al. 2023. "Towards Stable Test-Time Adaptation in Dynamic Wild World." *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2302.12400>.
- Ochs, W. 1975. "A New Axiomatic Characterization of the von Neumann Entropy." *Reports on Mathematical Physics* 8 (1): 109–20. [https://doi.org/10.1016/0034-4877\(75\)90022-1](https://doi.org/10.1016/0034-4877(75)90022-1).

References

- Ospanov, Azim, Jingwei Zhang, Mohammad Jalali, Xuenan Cao, Andrej Bogdanov, and Farzan Farnia. 2024. "Towards a Scalable Reference-Free Evaluation of Generative Models." *Advances in Neural Information Processing Systems (NeurIPS)* 37: 120892–927. <https://doi.org/10.52202/079017-3841>.
- Ou, Zijing, Tingyang Xu, Qinliang Su, Yingzhen Li, Peilin Zhao, and Yatao Bian. 2022. "Learning Neural Set Functions Under the Optimal Subset Oracle." *Advances in Neural Information Processing Systems (NeurIPS)* 35: 35021–34.
- Ouyang, Long, Jeff Wu, Xu Jiang, et al. 2022. "Training Language Models to Follow Instructions with Human Feedback." *Advances in Neural Information Processing Systems* 35 (NeurIPS).
- Owen, Guillermo. 1972. "Multilinear Extensions of Games." *Management Science* 18 (5): P64–79. <https://doi.org/10.1287/mnsc.18.5.P64>.
- Owen, Guillermo. 1975. "Multilinear Extensions and the Banzhaf Value." *Naval Research Logistics Quarterly* 22 (4): 741–50. <https://doi.org/10.1002/nav.3800220409>.
- Özcan, Gözde, Chengzhi Shi, and Stratis Ioannidis. 2025. "Learning Set Functions with Implicit Differentiation." *AAAI Conference on Artificial Intelligence*, 19777–85.
- Paltridge, Garth W. 1975. "Global Dynamics and Climate — a System of Minimum Entropy Exchange." *Quarterly Journal of the Royal Meteorological Society* 101 (429): 475–84. <https://doi.org/10.1002/qj.49710142906>.
- Paninski, Liam. 2003. "Estimation of Entropy and Mutual Information." *Neural Computation* 15 (6): 1191–253. <https://doi.org/10.1162/089976603321780272>.
- Park, Mincheol, Heeji Won, Won Woo Ro, and Suhyun Kim. 2025. "Rethinking Entropy in Test-Time Adaptation: The Missing Piece from Energy Duality." *Advances in Neural Information Processing Systems* 38 (NeurIPS).
- Parry, Matthew, A. Philip Dawid, and Steffen Lauritzen. 2012. "Proper Local Scoring Rules." *Annals of Statistics* 40 (1): 561–92. <https://doi.org/10.1214/12-AOS971>.
- Parshakova, Tetiana, Jean-Marc Andreoli, and Marc Dymetman. 2019. "Distributional Reinforcement Learning for Energy-Based Sequential Models." *Optimization Foundations for Reinforcement Learning Workshop at NeurIPS 2019*.
- Pasarkar, Amey P., and Adji Bousso Dieng. 2024. "Cousins of the Vendi Score: A Family of Similarity-Based Diversity Metrics for Science and Machine Learning." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, vol. 238. <https://arxiv.org/abs/2310.12952>.

References

- Patil, G. P., and C. Taillie. 1982. "Diversity as a Concept and Its Measurement." *Journal of the American Statistical Association* 77 (379): 548–61. <https://doi.org/10.1080/01621459.1982.10477845>.
- Pearl, Judea. 1988. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann.
- Prigogine, Ilya. 1945. "Modération Et Transformations Irréversibles Des Systèmes Ouverts." *Bulletin de La Classe Des Sciences, Académie Royale de Belgique* 31: 600–606.
- Prigogine, Ilya. 1947. *Étude Thermodynamique Des Phénomènes Irréversibles*. Dunod, Paris; Desoer, Liège.
- Prigogine, Ilya. 1978. "Time, Structure, and Fluctuations." *Science* 201 (4358): 777–85. <https://doi.org/10.1126/science.201.4358.777>.
- Prigogine, Ilya, and Jean-Marie Wiame. 1946. "Biologie Et Thermodynamique Des Phénomènes Irréversibles." *Experientia* 2 (11): 451–53. <https://doi.org/10.1007/BF02153597>.
- Qin, Lianhui, Sean Welleck, Daniel Khashabi, and Yejin Choi. 2022. "COLD Decoding: Energy-Based Constrained Text Generation with Langevin Dynamics." *Advances in Neural Information Processing Systems (NeurIPS)*.
- Rafailov, Rafael, Joey Hejna, Ryan Park, and Chelsea Finn. 2024. "From r to Q^* : Your Language Model Is Secretly a Q-Function." *arXiv Preprint arXiv:2404.12358*.
- Rafailov, Rafael, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. "Direct Preference Optimization: Your Language Model Is Secretly a Reward Model." *Advances in Neural Information Processing Systems* 36 (NeurIPS).
- Rao, C. Radhakrishna. 1982. "Diversity and Dissimilarity Coefficients: A Unified Approach." *Theoretical Population Biology* 21 (1): 24–43. [https://doi.org/10.1016/0040-5809\(82\)90004-1](https://doi.org/10.1016/0040-5809(82)90004-1).
- Ratliff, Nathan D., J. Andrew Bagnell, and Martin A. Zinkevich. 2006. "Maximum Margin Planning." *International Conference on Machine Learning (ICML)*, 729–36.
- Rényi, Alfréd. 1961. "On Measures of Entropy and Information." *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*, 547–61.
- Rose, Kenneth, Eitan Gurewitz, and Geoffrey C. Fox. 1990. "Statistical Mechanics and Phase Transitions in Clustering." *Physical Review Letters* 65 (8): 945–48. <https://doi.org/10.1103/PhysRevLett.65.945>.

References

- Sahin, Aytunc, Yatao Bian, Joachim M. Buhmann, and Andreas Krause. 2020. "From Sets to Multisets: Provable Variational Inference for Probabilistic Integer Submodular Models." *International Conference on Machine Learning (ICML)*, 8388–97.
- Sánchez Giraldo, Luis Gonzalo, Murali Rao, and José C. Príncipe. 2015. "Measures of Entropy from Data Using Infinitely Divisible Kernels." *IEEE Transactions on Information Theory* 61 (1): 535–48. <https://doi.org/10.1109/TIT.2014.2370058>.
- Santos, Roberto J. V. dos. 1997. "Generalization of Shannon's Theorem for Tsallis Entropy." *Journal of Mathematical Physics* 38 (8): 4104–7. <https://doi.org/10.1063/1.532107>.
- Schrödinger, Erwin. 1944. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge University Press.
- Schulman, John, Sergey Levine, Pieter Abbeel, Michael I. Jordan, and Philipp Moritz. 2015. "Trust Region Policy Optimization." *International Conference on Machine Learning (ICML)*.
- Schulman, John, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. "Proximal Policy Optimization Algorithms." *arXiv Preprint arXiv:1707.06347*.
- Shannon, Claude E. 1948. "A Mathematical Theory of Communication." *Bell System Technical Journal* 27 (3): 379–423.
- Shao, Zhihong, Peiyi Wang, Qihao Zhu, et al. 2024. "DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models." *arXiv Preprint arXiv:2402.03300*. <https://arxiv.org/abs/2402.03300>.
- Shi, Chence, Shitong Luo, Minkai Xu, and Jian Tang. 2021. "Learning Gradient Fields for Molecular Conformation Generation." *International Conference on Machine Learning (ICML)*.
- Shi, Yongquan, Zijing Ou, Shiping Wang, and Yatao Bian. 2026. "Advancing Optimal Subset Oracle via Learning Relaxation of Neural Set Functions." *arXiv Preprint arXiv:2607.11555*.
- Shirts, Michael R., Eric Bair, Giles Hooker, and Vijay S. Pande. 2003. "Equilibrium Free Energies from Nonequilibrium Measurements Using Maximum-Likelihood Methods." *Physical Review Letters* 91: 140601. <https://doi.org/10.1103/PhysRevLett.91.140601>.
- Shore, John E., and Rodney W. Johnson. 1980. "Axiomatic Derivation of the Principle of Maximum Entropy and the Principle of Minimum Cross-Entropy." *IEEE Transactions on Information Theory* 26 (1): 26–37.
- Simpson, E. H. 1949. "Measurement of Diversity." *Nature* 163 (4148): 688. <https://doi.org/10.1038/163688a0>.

References

- Sipos, Ruben, Pannaga Shivaswamy, and Thorsten Joachims. 2012. "Large-Margin Learning of Submodular Summarization Models." *Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 224–33.
- Sohl-Dickstein, Jascha, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. 2015. "Deep Unsupervised Learning Using Nonequilibrium Thermodynamics." *International Conference on Machine Learning (ICML)*, 2256–65.
- Solow, Andrew R., and Stephen Polasky. 1994. "Measuring Biological Diversity." *Environmental and Ecological Statistics* 1 (2): 95–103. <https://doi.org/10.1007/BF02426650>.
- Song, Yang, Conor Durkan, Iain Murray, and Stefano Ermon. 2021. "Maximum Likelihood Training of Score-Based Diffusion Models." *Advances in Neural Information Processing Systems (NeurIPS)* 34.
- Song, Yang, and Stefano Ermon. 2019. "Generative Modeling by Estimating Gradients of the Data Distribution." *Advances in Neural Information Processing Systems (NeurIPS)*.
- Song, Yang, Sahaj Garg, Jiaxin Shi, and Stefano Ermon. 2019. "Sliced Score Matching: A Scalable Approach to Density and Score Estimation." *Uncertainty in Artificial Intelligence (UAI)*.
- Song, Yang, and Diederik P. Kingma. 2021. "How to Train Your Energy-Based Models." *arXiv Preprint arXiv:2101.03288*.
- Song, Yang, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. 2021. "Score-Based Generative Modeling Through Stochastic Differential Equations." *International Conference on Learning Representations (ICLR)*.
- Stirling, Andy. 2007. "A General Framework for Analysing Diversity in Science, Technology and Society." *Journal of the Royal Society Interface* 4 (15): 707–19. <https://doi.org/10.1098/rsif.2007.0213>.
- Stoyanov, Veselin, Alexander Ropson, and Jason Eisner. 2011. "Empirical Risk Minimization of Graphical Model Parameters Given Approximate Inference, Decoding, and Model Structure." *International Conference on Artificial Intelligence and Statistics (AISTATS)*.
- Sutskever, Ilya, and Tijmen Tieleman. 2010. "On the Convergence Properties of Contrastive Divergence." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, Proceedings of machine learning research, vol. 9: 789–95.
- Tanaka, Daiki, Daiki Ikami, Toshihiko Yamasaki, and Kiyoharu Aizawa. 2018. "Joint Optimization Framework for Learning with Noisy Labels." *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. <https://arxiv.org/abs/1803.11364>.

- Taskar, Ben, Carlos Guestrin, and Daphne Koller. 2003. "Max-Margin Markov Networks." *Advances in Neural Information Processing Systems (NeurIPS)* 16.
- Tempesta, Piergiulio. 2016. "Beyond the Shannon–Khinchin Formulation: The Composability Axiom and the Universal-Group Entropy." *Annals of Physics* 365: 180–97. <https://doi.org/10.1016/j.aop.2015.08.013>.
- The Royal Swedish Academy of Sciences. 2024. *Scientific Background to the Nobel Prize in Physics 2024: For Foundational Discoveries and Inventions That Enable Machine Learning with Artificial Neural Networks*. <https://www.nobelprize.org/prizes/physics/2024/advanced-information/>.
- Tieleman, Tijmen. 2008. "Training Restricted Boltzmann Machines Using Approximations to the Likelihood Gradient." *International Conference on Machine Learning (ICML)*, 1064–71.
- Todorov, Emanuel. 2006. "Linearly-Solvable Markov Decision Problems." *Advances in Neural Information Processing Systems (NeurIPS)* 19: 1369–76.
- Todorov, Emanuel. 2009. "Efficient Computation of Optimal Actions." *Proceedings of the National Academy of Sciences* 106 (28): 11478–83.
- Topsøe, Flemming. 1979. "Information-Theoretical Optimization Techniques." *Kybernetika* 15 (1): 8–27.
- Toussaint, Marc. 2009. "Robot Trajectory Optimization Using Approximate Inference." *International Conference on Machine Learning (ICML)*.
- Tsallis, Constantino. 1988. "Possible Generalization of Boltzmann–Gibbs Statistics." *Journal of Statistical Physics* 52 (1–2): 479–87.
- Tschiatschek, Sebastian, Aytunc Sahin, and Andreas Krause. 2018. "Differentiable Submodular Maximization." *International Joint Conference on Artificial Intelligence (IJCAI)*, 2731–38.
- Tsochantaridis, Ioannis, Thorsten Joachims, Thomas Hofmann, and Yasemin Altun. 2005. "Large Margin Methods for Structured and Interdependent Output Variables." *Journal of Machine Learning Research* 6: 1453–84.
- Uehara, Masatoshi, Yulai Zhao, Kevin Black, et al. 2024. "Fine-Tuning of Continuous-Time Diffusion Models as Entropy-Regularized Control." *arXiv Preprint arXiv:2402.15194*.
- Velikonivtsev, Fedor, Mikhail Mironov, and Liudmila Prokhorenkova. 2024. "Challenges of Generating Structurally Diverse Graphs." *Advances in Neural Information Processing Systems (NeurIPS)* 37: 57993–8022. <https://doi.org/10.52202/079017-1849>.

References

- Verschaffelt, Jules-Émile. 1954. "Sur Les Minima de Production d'entropie Et de Dissipation d'énergie." *Bulletin de La Classe Des Sciences, Académie Royale de Belgique* 40: 779–83.
- Vincent, Pascal. 2011. "A Connection Between Score Matching and Denoising Autoencoders." *Neural Computation* 23 (7): 1661–74.
- Wainwright, Martin J., and Michael I. Jordan. 2008. "Graphical Models, Exponential Families, and Variational Inference." *Foundations and Trends in Machine Learning* 1 (1–2): 1–305. <https://doi.org/10.1561/22000000001>.
- Wallace, Bram, Meihua Dang, Rafael Rafailov, et al. 2024. "Diffusion Model Alignment Using Direct Preference Optimization." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 8228–38.
- Wang, Dequan, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. 2021. "Tent: Fully Test-Time Adaptation by Entropy Minimization." *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2006.10726>.
- Wang, Xudong, Zhirong Wu, Long Lian, and Stella X. Yu. 2022. "Debiased Learning from Naturally Imbalanced Pseudo-Labels." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 14647–57. <https://doi.org/10.1109/CVPR52688.2022.01424>.
- Weilenmann, Mirjam, Lea Kraemer, Philippe Faist, and Renato Renner. 2016. "Axiomatic Relation Between Thermodynamic and Information-Theoretic Entropies." *Physical Review Letters* 117 (26): 260601. <https://doi.org/10.1103/PhysRevLett.117.260601>.
- Weitzman, Martin L. 1992. "On Diversity." *The Quarterly Journal of Economics* 107 (2): 363–405. <https://doi.org/10.2307/2118476>.
- Williams, Ronald J. 1992. "Simple Statistical Gradient-Following Algorithms for Connectionist Reinforcement Learning." *Machine Learning* 8 (3–4): 229–56.
- Wu, Fa-Yueh. 1982. "The Potts Model." *Reviews of Modern Physics* 54 (1): 235–68. <https://doi.org/10.1103/RevModPhys.54.235>.
- Wu, Jiaxiang, Tao Shen, Haidong Lan, Yatao Bian, and Junzhou Huang. 2021. "SE(3)-Equivariant Energy-Based Models for End-to-End Protein Folding." *bioRxiv Preprint*.
- Wu, Tailin, and Ian Fischer. 2020. "Phase Transitions for the Information Bottleneck in Representation Learning." *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2001.01878>.
- Wu, Tailin, Ian Fischer, Isaac L. Chuang, and Max Tegmark. 2019. "Learnability for the Information Bottleneck." *Uncertainty in Artificial Intelligence (UAI)*. <https://arxiv.org/abs/1907.0>

7331.

- Xie, Binghui, Yatao Bian, Kaiwen Zhou, et al. 2024. "Enhancing Neural Subset Selection: Integrating Background Information into Set Representations." *International Conference on Learning Representations (ICLR)*.
- Xie, Binghui, Yixuan Wang, Yongqiang Chen, et al. 2024. "HORSE: Hierarchical Representation for Large-Scale Neural Subset Selection." *Advances in Neural Information Processing Systems (NeurIPS)*, 4852–77.
- Xie, Jianwen, Yang Lu, Ruiqi Gao, and Ying Nian Wu. 2018. "Cooperative Learning of Energy-Based Model and Latent Variable Model via MCMC Teaching." *AAAI Conference on Artificial Intelligence*.
- Xie, Jianwen, Yang Lu, Song-Chun Zhu, and Ying Nian Wu. 2016. "A Theory of Generative ConvNet." *International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 48: 2635–44.
- Xie, Yutong, Ziqiao Xu, Jiaqi Ma, and Qiaozhu Mei. 2023. "How Much Space Has Been Explored? Measuring the Chemical Space Covered by Databases and Machine-Generated Molecules." *International Conference on Learning Representations (ICLR)*. <https://openreview.net/forum?id=Yo06F8kfMa1>.
- Xu, Minkai, Tomas Geffner, Karsten Kreis, et al. 2025. "Energy-Based Diffusion Language Models for Text Generation." *International Conference on Learning Representations (ICLR)*.
- Yan, Yuchen, Minkai Xu, Zaiquan Yang, and Yatao Bian. 2026. "Unified Energy for Invariant and Independent Decoding in Diffusion Language Models." *arXiv Preprint arXiv:2606.09159*.
- Yuan, Suqin, Jinkun Chen, Jiyang Zheng, et al. 2026. "Understanding Diversity Collapse in RLVR via the Lens of Overtraining." *arXiv Preprint arXiv:2606.15455*. <https://arxiv.org/abs/2606.15455>.
- Zaheer, Manzil, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander J. Smola. 2017. "Deep Sets." *Advances in Neural Information Processing Systems (NeurIPS)* 30: 3391–401.
- Zhang, Qingyang, Haitao Wu, Changqing Zhang, Peilin Zhao, and Yatao Bian. 2025. "Right Question Is Already Half the Answer: Fully Unsupervised LLM Reasoning Incentivization." *Advances in Neural Information Processing Systems (NeurIPS)*. <https://arxiv.org/abs/2504.05812>.
- Zhao, Dora, Jerone T. A. Andrews, Orestis Papakyriakopoulos, and Alice Xiang. 2024. "Position: Measure Dataset Diversity, Don't Just Claim It." *Proceedings of the 41st International Conference*

- on *Machine Learning* 235: 60644–73. <https://proceedings.mlr.press/v235/zhao24a.html>.
- Zhao, Stephen, Rob Brekelmans, Alireza Makhzani, and Roger Grosse. 2024. “Probabilistic Inference in Language Models via Twisted Sequential Monte Carlo.” *Proceedings of the 41st International Conference on Machine Learning (ICML)*, Proceedings of machine learning research, vol. 235.
- Zhao, Xuandong, Zhewei Kang, Aosong Feng, Sergey Levine, and Dawn Song. 2025. “Learning to Reason Without External Rewards.” *arXiv Preprint arXiv:2505.19590*. <https://arxiv.org/abs/2505.19590>.
- Zheng, Shuai, Sadeep Jayasumana, Bernardino Romera-Paredes, et al. 2015. “Conditional Random Fields as Recurrent Neural Networks.” *IEEE International Conference on Computer Vision (ICCV)*, 1529–37.
- Zhu, Song Chun, Ying Nian Wu, and David Mumford. 1997. “Minimax Entropy Principle and Its Application to Texture Modeling.” *Neural Computation* 9 (8): 1627–60. <https://doi.org/10.1162/neco.1997.9.8.1627>.
- Zhu, Song Chun, Ying Nian Wu, and David Mumford. 1998. “Filters, Random Fields and Maximum Entropy (FRAME): Towards a Unified Theory for Texture Modeling.” *International Journal of Computer Vision* 27 (2): 107–26.
- Zhu, Yuchang, Huizhe Zhang, Bingzhe Wu, et al. 2025. “Measuring Diversity in Synthetic Datasets.” *Proceedings of the 42nd International Conference on Machine Learning* 267: 80373–97. <https://proceedings.mlr.press/v267/zhu25ac.html>.
- Ziebart, Brian D., J. Andrew Bagnell, and Anind K. Dey. 2010. “Modeling Interaction via the Principle of Maximum Causal Entropy.” *International Conference on Machine Learning (ICML)*.
- Ziebart, Brian D., Andrew Maas, J. Andrew Bagnell, and Anind K. Dey. 2008. “Maximum Entropy Inverse Reinforcement Learning.” *AAAI Conference on Artificial Intelligence*, 1433–38.
- Ziegler, Daniel M., Nisan Stiennon, Jeffrey Wu, et al. 2019. “Fine-Tuning Language Models from Human Preferences.” *arXiv Preprint arXiv:1909.08593*.
- Zuo, Yuxin, Kaiyan Zhang, Li Sheng, et al. 2025. “TTRL: Test-Time Reinforcement Learning.” *Advances in Neural Information Processing Systems* 38 (NeurIPS).